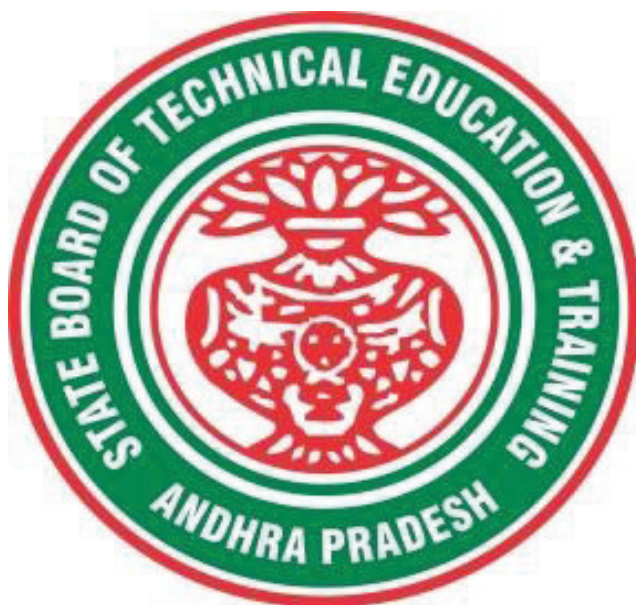


CURRICULUM - 2023

C -23

DIPLOMA IN COMMUNICATION & COMPUTER NETWORKING



**STATE BOARD OF TECHNICAL EDUCATION & TRAINING
ANDHRA PRADESH**

**DIPLOMA IN COMMUNICATION AND COMPUTER NETWORKING
CURRICULUM- 2023 (C-23)**

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PREAMBLE

Technical Education is a key driver of economic development and plays a crucial role in providing individuals with the skills and knowledge necessary to thrive in the workplace. As technological advancements continue to reshape industries and create new opportunities, it is critical that technical education curricula remain relevant and up-to-date.

The curriculum has been designed with this in mind, with a focus on practical skills, critical thinking, and problem-solving. We believe that these skills are essential for success in both academic and professional spheres. The revamping of the technical education curriculum is made with collaborative effort from educators, industry experts, policymakers, and students.

At the heart of the curriculum, is the belief that the technical education should be **student-centered**, empowering learners to take ownership of their learning and pursue their passions. We aim to create a learning environment that is safe, supportive, and nurturing, where every student has the opportunity to reach their fullest potential. We acknowledge that learning is a lifelong journey, and our curriculum is designed to provide a solid foundation for continued growth and development. We hope that our students will not only leave with a diploma but with employability and passion for learning.

The State Board of Technical Education and Training, (SBTET) AP, has been offering Diploma programmes to meet the above said aspirations of the stake holders: industries, students, academia, parents and the society at large. **The Curriculum should be flexible, adaptable, and responsive to the changing needs of the industry and society.** As such, it has been the practice of SBTET, A.P., to keep the curriculum abreast with the advances in technology through systematic and scientific analysis of current curriculum and bring out an updated revised version at regular intervals.

The design of Curriculum C-23 was started in the month of January - 2023. Feedback was collected from all stake holders: Students, Lecturers, Senior Lecturers, Head of Sections and Principals for all programmes for this purpose. Accordingly, a workshop was convened on 15th February 2023 by Smt. C. Naga Rani, I.A.S, Director of Technical Education & Chairperson, SBTET, AP to discuss on revamping of C-20 curriculum to meet the needs of industries and for improvement of placements.

The meeting was attended by Sri. Saurab Gaur, I.A.S, Principal Secretary, Skill Development & Training, Smt. Lavanya Veni, I.A.S, Director, Employment & Training. Thirteen Representatives from Industries and Fourteen Academicians from Higher Level Institutions and officials of ITI, Skill Development, CTE & SBTET attended the workshop.

Smt. C Naga Rani, I.A.S., Commissioner of Technical Education while addressing in the workshop, emphasized the necessity of industrial training and on-hand experience, that the students need to undergo to support the industries and the Gaps in the Curriculum need to be fixed to make the students passionate to work in the industry in order to support economy of the country.

The committees of each branch consisting of experts from Industries, Higher Level Institutions and Faculty of Polytechnics are informed to study the possibility of incorporating the following aspects while preparation of the curriculum so as to improve employability.

- **To bring out industry oriented Diploma Engineers.**
- **Internet of Things (IoT) for all branches**
- **Theoretical & Practical subjects 50: 50 Ratio**
- **Industry 4.0 concepts.**
- **5G Technology.**
- **Critical Thinking (Quantitative Aptitude, Data Interpretation, Quantitative reasoning etc) to face the written tests conducted by the industries during placements.**

In continuation, series of workshops with subject experts followed in the subsequent weeks for thorough perusal for preparation of draft curriculum. Also, the suggestions received from representatives from various industries, academic experts from higher level institutions, subject experts from Polytechnics, have been recorded, validated for incorporation into the **Curriculum C-23**. Finally, the draft curriculum was sent to academicians of higher-level institutions, industrial experts for Vetting.

The design of new Curricula C-23 for different diploma programmes has thus been finalised with the active participation of the members of the faculty teaching in the Polytechnics of Andhra Pradesh, and duly reviewed by Expert Committee constituted of academicians and representatives from industries. Thus, the primary objective of the curriculum change is to produce employable diploma holders in the country by correlating the growing needs of the industries with relevant academic input.

The outcome-based approach as given by NBA guidelines has been followed throughout the design of this curriculum and designed to meet the requirements of NBA Accreditation, too.

The Revised Curriculum i.e., Curriculum-2023 (C-23) is approved by 45th Academic Committee of SBTET, A.P for its implementation with effect from Academic Year 2023-24. Also, the SBTET, A.P under the aegis of the Department of Technical Education, Andhra Pradesh in it's 62nd Board Meeting held on 13-07-2023 (vide item no: 17) Approved to update the Polytechnic Curriculum C-23 with effect from the academic year 2023-2024 onwards after revamping the present C-20 curriculum, to meet the latest industrial technological developments including Industry 4.0 concepts.

2. HIGHLIGHTS OF CURRICULUM C-23

The following Courses/ Topics are incorporated in this curriculum C-23 as per the suggestions received from Industrial Experts, Faculty of Higher Level Institutions and Polytechnics to improve the Employability Skills of the Polytechnic Students.

1. Duration of course for regular Diploma and for sandwich Diploma is 3 years and 3½ years respectively.
2. The Curriculum is prepared in Semester Pattern. However, First Year is maintained as Year-wise pattern.
3. 6 Months Industrial training has been introduced for 3 years Diploma Courses and 6 months Industrial Training is introduced for 3 ½ years Sandwich Diploma courses.
4. Updated subjects relevant to the industry are introduced in all the Diploma courses.
5. The policy decisions taken at the State and Central level with regard to environmental science are implemented by including relevant topics in Chemistry. This is also in accordance with the Supreme Court guidelines issued in Sri Mehta's case.
6. Keeping in view the increased need of communication skills which is playing a major role in the success of Diploma Level students in the industries, emphasis is given for learning and acquiring listening, speaking, reading and writing skills in English. Further as emphasized in the meetings, Communication Skills lab and Life Skills lab are continuing for all the branches.
7. Upon reviewing the existing C-20 curriculum, it is found that the theory content is found to have more weightage than the Practical content. In C-23 curriculum, more emphasis is given to the practical content in Laboratories and Workshops, thus strengthening the practical skills.
8. With increased emphasis for the student to acquire Practical skills, the course content in all the subjects is thoroughly reviewed and structured as outcome based than the conventional procedure based.
9. Curricula of Laboratory and Workshops have been thoroughly revised based on the suggestions received from the industry and faculty, for better utilization of the equipment available in the Polytechnics. The experiments /exercises that are chosen for the practical sessions are identified to confirm to the field requirements of industry.
10. Student Centric activities(Library, Mini project, Sports, Preparation for placements, Reporting etc) are introduced in each year/semester with a weightage of 3 periods per week.
11. A new concept "Study on societal problems" is introduced to resolve the village level issue during Summer Vacation after completion of I Year at nearby village. A mini project report is to be submitted by the student.
12. Industrial Visits for students at least once in a year.
13. Industrial seminars by industrial experts in each year/semester.

3. ACKNOWLEDGEMENTS

The Members of the working group are grateful to Smt C. Naga Rani I.A.S., Commissioner of Technical Education & Chairman of SBTET, for continuous guidance and valuable inputs during process of revising, modifying and updating the Curriculum C-20 to Curriculum C-23.

We are grateful to Sri. S. Suresh Kumar, I.A.S, Principal Secretary, Skills Development & Training for his valuable suggestions to bring the revamped curriculum C-23 in to a final form to meet latest Industry 4.0 concepts.

We are grateful to Sri. Saurab Gaur, I.A.S, former Principal Secretary, Skills Development & Training who actively participated in the Industry-Academia workshop conducted on 15th February, 2023 and offered valuable suggestions and insights into the learning needs and preferences so that the curriculum is engaging, inclusive, and effective.

It is pertinent to acknowledge the support of the following in the making of Curriculum C-23. A series of workshops in different phases were conducted by SBTET, AP, Guntur involving faculty from Polytechnics, Premier Engineering Colleges & representatives from various Industries and Dr. C. R. Nagendra Rao, Professor & Head, NITTTR-ECV to analyse the Previous C-20 Curriculum and in designing of C-23 Curriculum, is highly appreciated and gratefully acknowledged.

We also extend our sincere thanks to Sri. V. Padma Rao, Joint Director of Technical Education, Sri K.V. Ramana Babu, Secretary, SBTE&T, Andhra Pradesh, Sri K. Vijaya Bhaskar, Deputy Director (Academic) , Andhra Pradesh, officials of Directorate of Technical Education and the State Board of Technical Education, Andhra Pradesh and all teaching fraternity from the Polytechnics who are directly or indirectly involved in preparation of the curricula.

4. RULES AND REGULATIONS OF C-23 CURRICULUM

4.1 Duration and pattern of the courses

All the Diploma programs run at various institutions are of AICTE approved 3 years or 3½ years duration of academic instruction. All the Diploma courses are run on year wise pattern in the first year, and the remaining two or two & half years are run in the semester pattern. In respect of few courses like Diploma in Bio-Medical course, the training will be in the seventh semester. **Run-through system is adopted for all the Diploma Courses, subject to eligibility conditions.**

4.2 Procedure for Admission into the Diploma Courses:

Selection of candidates is governed by the Rules and Regulations laid down in this regard from time to time.

- a) Candidates who wish to seek admission in any of the Diploma courses will have to appear for the Common Entrance Test for admissions into Polytechnics (POLYCET) conducted by the State Board of Technical Education and Training, Andhra Pradesh, Vijayawada. Only the candidates satisfying the following requirements will be eligible to appear for the Common Entrance Test for admissions into Polytechnics (POLYCET).
 - a. The candidates seeking admission should have appeared for S.S.C examination, conducted by the Board of Secondary Education, Andhra Pradesh or equivalent examination thereto, at the time of applying for the Common Entrance Test for admissions into Polytechnics (POLYCET). In case of candidates whose results of their

Qualifying Examinations is pending, their selection shall be subject to production of proof of their passing the qualifying examination in one attempt or compartmentally at the time of admission.

- b. Admissions are made based on the merit obtained in the Common Entrance Test (POLYCET) and the reservation rules stipulated by the Government of Andhra Pradesh from time to time.
- c. For admission into the following Diploma Courses for which entry qualification is 10+2, candidates need not appear for POLYCET. A separate notification will be issued for admission into these courses.
 - i). D.HMCT ii).D. Pharmacy

4.3 Medium of Instruction

The medium of instruction and examination shall be English.

4.4 Permanent Identification Number (PIN)

A cumulative / academic record is to be maintained of the Marks secured in sessional work and end examination of each year for determining the eligibility for promotion etc., A Permanent Identification Number (PIN) will be allotted to each admitted candidate to maintain academic records.

4.5 Number of Working Days Per Semester / Year:

- a) The Academic year for all the Courses shall be in accordance with the Academic Calendar.
- b) The Working days in a week shall be from Monday to Saturday
- c) There shall be 7 periods of 50 minutes duration each on all working days.
- d) The minimum number of working days for each semester / year shall be 90 / 180 days excluding examination days. If this prescribed minimum is not achieved due to any reason, special arrangements shall be made to conduct classes to complete the syllabus.

4.6 Eligibility (Attendance to Appear for the End Examination)

- a) A candidate shall be permitted to appear for the end examination in all subjects, if he or she has attended a minimum of 75% of working days during the year/Semester.
- b) Condonation of shortage of attendance in aggregate up to 10% (65% and above and below 75%) in each semester or 1st year may be granted on medical grounds.
- c) A stipulated fee shall be payable towards condonation for shortage of attendance.
- d) Candidates having less than 65% attendance shall be detained.
- e) Students whose shortage of attendance is not condoned in any semester / 1st year and not paid the condonation fee in time are not eligible to take their end examination of that class and their admissions shall stand cancelled. They may seek re-admission for that semester / 1st year when offered in the next subsequent academic semester/year.

For INDUSTRIAL TRAINING:

- i) During Industrial Training the candidate shall put in a minimum of 90% attendance.
- ii) If the student fails to secure 90% attendance during industrial training, the student shall reappear for 6 months industrial training at his own expenses.

4.7 Readmission

Readmission shall be granted to eligible candidates by the respective Principal/ Regional Joint Director.

- a) (i) Within 15 days after commencement of class work in any semester (Except Industrial Training).
- (ii) For Industrial Training: before commencement of the Industrial training.
- b) Within 30 days after commencement of class work in any year (including D. Pharmacy course or first year course in Engineering and Non-Engineering Diploma streams). Otherwise, such cases shall not be considered for readmission for that semester / year and are advised to seek readmission in the next subsequent eligible academic year.
- c) The percentage of attendance of the readmitted candidates shall be calculated from the first day of beginning of the regular class work for that year / Semester, as officially announced by CTE/SBTET but not from the day on which he/she has actually reported to the class work.

4.8 Scheme of Evaluation

a) First Year

Theory Courses: Each Course carries Maximum marks of 80 with an end examination of 3 hours duration, along with internal assessment for Maximum of 20 marks. (Sessional marks). However, there are no minimum marks prescribed for sessionals.

Laboratory Courses: There shall be 40/20 Marks for internal assessment i.e. sessional marks for each practical Course with an end examination of 3 hours duration carrying 60/30 marks. However, there are no minimum marks prescribed for sessional.

b) III, IV, V, VI and VII Semesters:

Theory Courses: End semester evaluation shall be of 3 hours duration and for a maximum of 80 marks.

Laboratory Courses: Each Course carry 60/30 marks of 3 hours duration 40/20 sessional marks.

4.9 Internal Assessment Scheme

- a) **Theory Courses:** Internal assessment shall be conducted for awarding Sessional marks on the dates specified. **Three-unit tests shall be conducted for I year students and two Unit Tests for semesters. The details are presented below.**

S. No.	Type of Assessment	Weightage Assigned
(i)	Testing of knowledge through mid-examination for year/sem as (Mid-1+Mid-2+Mid3) or (Mid-1 + Mid-2)	40
(ii)	Assignments	5
(iii)	<i>Dynamic Learning activities : Project Work/ Seminar/Tech-fest/Group Discussion, Quizzes etc./Extra-curricular activities/NSS/NCC/ IPSGM/Cleaning & Greening of Campus etc.</i>	5
	TOTAL	50

Internal Assessment shall be of 90 minutes duration and for a maximum of 40 marks for each test.

At least one assignment should be completed for each unit which carries 10 marks. The total assignment marks should be reduced to 5.

The dynamic learning activity is to be conducted which carries 10 marks. The total marks should be reduced to 5.

The total 50 marks assigned to internal assignment is to be scaled down to 20 marks.

b) Practical Courses:

(i) Drawing Courses:

The award of Sessional marks for internal Assessment shall be as given in the following table:

Distribution of Marks for the Internal Assessment Marks			
First Year (Total:40 Marks)		Semesters (Total:40 Marks)	
Max:20 Marks	Max:20 Marks	Max:20 Marks	Max:20 Marks
From the Average of THREE Unit Tests.	From the Average of Assessment of Regular Class work Exercises.	From the Average of TWO Unit Tests.	From the Average of Assessment of Regular Class work Exercises.

- For first year engineering drawing each unit test will be conducted for a duration of 2 hours with maximum marks of 40.
- (Part - A: 4 questions x 5 marks = 20 Marks; Part -B: 2 questions x 10 marks = 20 marks).
- For the semester drawing examinations, Two Unit tests shall be conducted as per the Board End Examination Question Paper Pattern.
- All Drawing exercises are to be filed in serial order and secured for further scrutiny by a competent authority

(ii) Laboratory Courses:

- (a) Student's performance in Laboratories / Workshop shall be assessed during the year/ semester of study for 40 marks in each practical Course.
- (b) Evaluation for Laboratory Courses, other than Drawing courses:
 - i. Instruction (teaching) in laboratory courses (except for the course on Drawing) here after shall be task/competency based as delineated in the Laboratory sheets, prepared by SBTET, AP & NITTTR- ECV and posted in SBTET website.
 - ii. Internal assessment for Laboratory shall be done on the basis of task/s performed by the student as delineated in the laboratory sheets, prepared by SBTET, AP & NITTTR- ECV and posted in AP, SBTET website.
 - iii. Question paper for End semester Evaluation shall also be task/s based and shall be prepared and distributed by SBTET as done in case of theory courses be prepared as per SBTET rules in vogue.
- c) Internal assessment in Labs / workshops / Survey field work etc., during the course of study shall be done and sessional marks shall be awarded by the concerned Teacher.

- d) For practical examinations, except in drawing, there shall be two examiners. External examiner shall be appointed by the Principal in consultation with respective Head of Section preferably choosing a qualified person from in the order of preference.
- i) Nearby Industry
 - ii) Govt / Semi Govt organization like R & B, PWD, PR, Railways, BSNL, APSRTC, APSEB etc.
 - iii) Govt / University Engg College.
 - iv) HoD/Senior Lecture (Selection Grade-II) from the Govt. Polytechnic
- Internal examiner shall be the person concerned with internal assessment as in (c) above. The end examination shall be held along with all theory papers in respect of drawing.
- e) Question Paper for Practicals: Question paper should cover (the experiments / exercise prescribed to test various) skills like handling, manipulating, testing, trouble shooting, repair, assembling and dismantling etc., from more than one experiment / exercise
- f) Records pertaining to internal assessment marks of both theory and practical Courses are to be maintained for official inspection.
- g) In case of Diploma programs having Industrial Training, Internal Assessment and Summative Evaluation, shall be done as illustrated in the following table:

Assessment no	Upon completion of	By	Based on	Max Marks
1	12 weeks	1.The faculty concerned (Guide) and 2. Training in charge (Mentor) of the industry	Learning outcomes as given in the scheme of assessment ,for Industrial Training	120
2	22 weeks			120
3. Final summative Evaluation	24 week	1.The faculty member concerned, 2.HoD concerned and 3.An external examiner	1.Demonstration of any one of the skills listed in learning outcomes	30
			2.Training Report	20
			3.Viva Voce	10
TOTAL				300

- h) Each staff member including Head of Section shall be assigned a batch of students 10 to 15 for making assessment during industrial training.

4.10 Minimum Pass Marks

a) Theory Examination:

For passing a theory Course, a candidate has to secure a minimum of 35% in end examination and a combined minimum of 35% of both Sessional and end examination marks put together.

b) Practical Examination:

For passing a practical Course, a candidate has to secure a minimum of 50% in end examination and a combined minimum of 50% of both sessional and practical end examination marks put together. In case of D.C.C.P., the pass mark for typewriting and short hand is 45% in the end examination. There are no sessional marks for typewriting and Shorthand Courses of D.C.C.P course.

C) Industrial Training:

- I. Monitoring: Similar to project work each teacher may be assigned a batch of 10-15 students irrespective of the placement of the students to facilitate effective monitoring of students learning during industrial training.
- II. Assessment: The Industrial training shall carry 300 marks and pass marks is 50% in assessments at industry (first and second assessment) and final summative assessment at institution level put together i.e. 150 marks out of 300 marks. And also student has to secure 50% marks in final summative assessment at institution level.
- III. **In-Plant Industrial Training for 3-Year Diploma (C-23) Courses is scheduled as per the Academic Calendar of the SBTET every year.**

4.11. Provision for Improvement

Improvement is allowed only after he / she has completed all the Courses from First Year to Final semester of the Diploma.

- a) Improvement is allowed in any 4 (Four) Courses of the Diploma.
- b) The student can avail of this improvement chance **ONLY ONCE**, that too within the succeeding two examinations after the completion of Diploma. However, the duration including Improvement examination shall not exceed **FIVE** years from the year of first admission.
- c) No improvement is allowed in Practical / Lab Courses or Project work or Industrial Training assessment. However, improvement in drawing Course(s) is allowed.
- d) If improvement is not achieved, the marks obtained in previous Examinations hold good.
- e) Improvement is not allowed in respect of the candidates who are punished under Mal-practice in any Examination.
- f) Examination fee for improvement shall be paid as per the notification issued by State Board of Technical Education and Training from time to time.
- g) All the candidates who wish to appear for improvement of performance shall deposit the original Marks Memos of all the years / Semesters and also original Diploma Certificate to the Board. If there is improvement in performance of the current examination, the revised Memorandum of marks and Original Diploma Certificate will be issued, else the submitted originals will be returned.

4.12. Rules of Promotion From 1ST YEAR TO 3rd, 4th, 5th, 6th and 7th Semesters:

A) For Diploma Courses of 3 Years duration

- i). A candidate shall be permitted to appear for first year examination provided he / she puts in 75% attendance (which can be condoned on Medical grounds up to 10%) and pay the examination fee.

- ii) A candidate shall be promoted to 3rd semester if he/she puts the required percentage of attendance in the first year and pays the examination fee. A candidate who could not pay the first year examination fee has to pay the promotion fee as prescribed by State Board of Technical Education and Training, AP from time to time before commencement of 3rd semester.

A candidate is eligible to appear for the 3rd semester examination if he/she puts the required percentage of attendance in the 3rd semester and pays the examination fee.

- iii) A candidate shall be promoted to 4th semester provided he/she puts the required percentage of attendance in the 3rd semester and pay the examination fee. A candidate, who could not pay the 3rd semester exam fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training AP from time to time before commencement of 4th semester. A candidate is eligible to appear for the 4th semester examination if he/she puts the required percentage of attendance in the 4th semester and pays the examination fee.

- iv) A candidate shall be promoted to 5th semester provided he / she puts the required percentage of attendance in the 4th semester and pays the examination fee. A candidate, who could not pay the 4th semester examination fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 5th semester.

A candidate is eligible to appear for the 5th semester examination if he/she puts the required percentage of attendance in the 5th semester and pays the examination fee.

- v) A candidate shall be sent to Industrial training / VI semester provided he/she puts in the required percentage of attendance in the 5th semester and pay the examination fee/ promotion fee as prescribed by SBTET.

A candidate is eligible to appear for Industrial Training assessment (Seminar/Viva-voce) puts the required percentage of attendance, i.e., 90% in 6th semester Industrial Training.

For IVC & ITI Lateral Entry students:

- i.) A candidate shall be permitted to appear for Third Semester examination provided he / she puts in 75% attendance (which can be condoned on Medical grounds up to 10%) and pay the examination fee for Third semester.
- ii) A candidate shall be promoted to 4th semester provided he/she puts the required percentage of attendance in the 3rd semester and pay the examination fee. A candidate, who could not pay the 3rd semester exam fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training AP from time to time before commencement of 4th semester.

A candidate is eligible to appear for the 4th semester examination if he/she puts the required percentage of attendance in the 4th semester and pays the examination fee.

- ii) A candidate shall be promoted to 5th semester provided he / she puts the required percentage of attendance in the 4th semester and pays the examination fee. A candidate, who could not pay the 4th semester examination fee, has to pay the promotion fee as

prescribed by State Board of Technical Education and Training from time to time before commencement of 5th semester.

A candidate is eligible to appear for the 5th semester examination if he/she puts the required percentage of attendance in the 5th semester and pays the examination fee.

- iii) A candidate shall be sent to Industrial training / VI semester provided he/she puts in the required percentage of attendance in the 5th semester and pay the examination fee/promotion fee as prescribed by SBTET.

A candidate is eligible to appear for Industrial Training assessment (Seminar/Viva-voce) puts the required percentage of attendance, i.e., 90% in 6th semester Industrial Training and pays the examination fee.

B) For Diploma Courses of 3 ½ Years duration (MET/ CH/ CHPP/ CHPC/ CHOT/ TT):

- i. A candidate shall be permitted to appear for 1st year examination provided he / she puts in 75% attendance (which can be condoned on Medical grounds upto 10%) i.e. attendance after condonation on Medical grounds should not be less than 65% and pay the examination fee.
- ii. A candidate shall be promoted to 3rd semester if he/she puts the required percentage of attendance in the 1st year and pays the examination fee. A candidate who could not pay the 1st year examination fee has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 3rd semester.
- iii. A candidate shall be promoted to 4th semester provided he/she puts the required percentage of attendance in the 3rd semester and pay the examination fee. A candidate, who could not pay the 3rd semester exam fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 4th semester.

A candidate is eligible to appear for the 4th semester exam if he/she puts the required percentage of attendance in the 4th semester

For IVC & ITI Lateral Entry students:

- a) Puts the required percentage of attendance in the 4th semester
- iv. A candidate shall be promoted to 5th semester industrial training provided he / she puts the required percentage of attendance in the 4th semester and pays the examination fee. A candidate, who could not pay the 4th semester examination fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 5th semester.
- v. Promotion from 5th to 6th semester is automatic (i.e., from 1st spell of Industrial Training to 2nd spell) provided he/she puts the required percentage of attendance, which in this case ie.,90 % of attendance and attends for the VIVA-VOCE examination at the end of training.
- vi. A candidate shall be promoted to 7th semester provided he / she puts the required percentage of attendance in the 6th semester and pays the examination fee. A candidate, who could not pay the 6th semester examination fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 7th semester.
- vii. A candidate shall be promoted to 7th semester of the course provided he/she has successfully completed both the spells of Industrial Training.
A candidate is eligible to appear for 7th semester examination if he/she

- a) Puts in the required percentage of attendance in the 7th semester

For IVC & ITI Lateral Entry students:

- a) Puts in the required percentage of attendance in the 7 th semester .

C) For Diploma Courses of 3 ½ Years duration (BM):

The same rules which are applicable for conventional courses also apply for this course. The industrial training in respect of this course is restricted to one semester (6 months) after the 6th semester (3 years) of the course.

- i. A candidate shall be permitted to appear for first year examination provided he / she puts in 75% attendance (which can be condoned on Medical grounds upto 10%) i.e. attendance after condonation on Medical grounds should not be less than 65% and pay the examination fee.
- ii. A candidate shall be promoted to 3rd semester if he/she puts the required percentage of attendance in the first year and pays the examination fee. A candidate who could not pay the first year examination fee has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 3rd semester.
- iii. A candidate shall be promoted to 4th semester provided he/she puts the required percentage of attendance in the 3rd semester and pay the examination fee. A candidate who could not pay the 3rd semester examination fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 4th semester.
A candidate is eligible to appear for the 4th semester examination if he/she
- a) Puts in the required percentage of attendance in the 4th semester

For IVC & ITI Lateral Entry Students:

A candidate is eligible to appear for the 4th semester examination if he/she puts the required percentage of attendance in the 4th semester

- iv. A candidate shall be promoted to 5th semester provided he / she puts the required percentage of attendance in the 4th semester and pays the examination fee. A candidate, who could not pay the 4th semester examination fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 5th semester.
A candidate is eligible to appear for the 5th semester exam if he/she
- a) Puts in the required percentage of attendance in the 5th semester.

For IVC & ITI Lateral Entry students:

- a) Puts in the required percentage of attendance in the 5th semester.
- v. A candidate shall be promoted to 6th semester provided he/she puts in the required percentage of attendance in the 5th semester and pays the examination fee.
A candidate who could not pay the 5th semester examination fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 6th semester.
A candidate is eligible to appear for 6th semester examination
- a) Puts in the required percentage of attendance in 6th semester

IVC & ITI Lateral Entry students:

- a) Puts in the required percentage of attendance in 6th semester.
- vi. A candidate shall be promoted to 7th semester provided he/she puts in the required percentage of attendance in 6th semester and pay the examination fee. A candidate, who could not pay the 6th semester examination fee, has to pay the promotion fee prescribed by SBTET from time to time before commencement of the 7th semester (Industrial Training).
A candidate is eligible to appear for 7th semester Industrial Training assessment (Seminar/Viva-voce) if he/she
 - a) Puts in the required percentage of attendance, i.e., 90% in 7th semester Industrial Training.

For IVC & ITI Lateral Entry students:

- a) Puts in the required percentage of attendance, i.e., 90% in 7th semester Industrial Training.

4.13. Students Performance Evaluation

Successful candidates shall be awarded the Diploma under the following divisions of pass.

- a) First Class with Distinction shall be awarded to the candidates who secure an overall aggregate of 75% marks and above.
- b) First Class shall be awarded to candidates who secure overall aggregate of 60% marks and above and below 75% marks.
- c) Second Class shall be awarded to candidates who secure a pass with an overall aggregate of below 60%.
 - i. The Weightage of marks for various year/Semesters which are taken for computing overall aggregate shall be 25% of I year marks + 100% of 3rd and subsequent Semesters.
 - ii. In respect IVC & ITI Lateral Entry candidates who are admitted directly into diploma course at the 3rd semester (i.e., second year) level the aggregate of (100%) marks secured at the 3rd and subsequent semesters of study shall be taken into consideration for determining the overall percentage of marks secured by the candidates for award of class/division.
- d) Second Class shall be awarded to all students, who fail to complete the Diploma in the regular 3 years/ 3 ½ years and four subsequent examinations from the year of first admission.

4.14. EXAMINATION FEE SCHEDULE:

The examination fee should be as per the notification issued by State Board of Technical Education and Training, AP from time to time.

4.15. Structure of Examination Question Paper:

I. Formative assessment (Internal examination)

a) For theory Courses:

Three-unit tests for first year and two-unit tests for semesters shall be conducted with a duration of 90 minutes for each test for maximum marks of 40. It consists of part A and Part B.

Part A contains five questions and carries 16 marks. Among these five questions first question consists of four objective items like one word or phrase answer/filling-in the blanks/true or false etc with one mark for each question. The other four questions are short answer questions and carry three marks each.

Part B carries 24 marks and consists of three questions with internal choice ie., Either/Or type , and each question carries 8 marks.

The sum of marks of 3 tests for I year and 2 tests for semesters including assignments and Dynamic learning activities (50 marks) shall be reduced to 20 marks in each Course for arriving at final sessional marks.

b) For drawing Courses:

For I year:

Three-unit tests with duration of 90 minutes and for maximum marks of 40 marks shall be conducted for first year. It consists of part A and Part B.

Part A consists four questions for maximum marks of 16 and each question carries four marks (4×4 marks=16 marks).

Part B carries maximum marks of 24 and consists of five questions while the student shall answer any three questions out of these five questions. Each question in this part carries a maximum mark of 8, (3×8 marks=24 marks).

The sum of marks obtained in 3-unit test marks shall be reduced to 20 marks for arriving at final sessional marks. Remaining 20 marks are awarded by the Course teacher based on the student's performance during regular class exercise.

For semester: Two-unit tests with duration of 90 minutes and for maximum marks of 40 marks shall be conducted. The sum of marks obtained in 2-unit test marks shall be reduced to 20 marks for arriving at final sessional marks. Remaining 20 marks are awarded by the Course teacher based on the student's performance during regular class exercise.

c) For Laboratory /workshop: 50% of total marks for the Course shall be awarded based on continuous assessment of the student in laboratory/workshop classes and the remaining 50% shall be based on the sum of the marks obtained by the students in two tests.

II. Summative assessment (End examination)

The question paper for theory examination is patterned in such a manner that the Weightage of periods/marks allotted for each of the topics for a particular Course be considered. End Examination paper is of 3 hours duration.

a) Each theory paper consists of Section 'A' and 'B'

Section 'A' with Max marks of 30, contains 10 short answer questions. All questions are to be answered and each carry 3 marks, i.e., $10 \times 3 = 30$.

Section 'B' with Max marks of 50 contains 8 essay type questions. Only 5 questions are to be answered and each carry 10 marks, i.e., Max. Marks: $5 \times 10 = 50$.
Thus, the total marks for theory examination shall be: 80.

b) **For Engineering Drawing Course (107) consist of section 'A' and section 'B'.**

Section 'A' with max marks of 20, contains four (4) questions. All questions in section 'A' are to be answered to the scale and each carries 5 marks, ie. $4 \times 5 = 20$.

Section 'B' with max marks of 40, contains six (6) questions. The student shall answer any four (4) questions out of the above six questions and each question carries 10 Marks, i.e., $4 \times 10 = 40$.

c) **Practical Examinations**

For Workshop practice and Laboratory Examinations, Each student has to pick up a question paper distributed by Lottery System.

Max. Marks for an experiment / exercise : 50

Max. Marks for VIVA-VOCE : 10

Total Max. Marks : 60

In case of practical examinations with 50 marks, the marks shall be distributed as

Max. Marks for an experiment / exercise : 25

Max. Marks for VIVA-VOCE : 05

Total Max. Marks : 30

In case of any change in the pattern of question paper, the same shall be informed sufficiently in advance to the candidates.

d) **Note: Evaluation for Laboratory Courses, other than Drawing courses:**

- I. Instruction (teaching) in laboratory courses (except for the course on Drawing) hereafter shall be task/competency based as delineated in the Laboratory sheets, prepared by SBTET, AP and posted in its website.
- II. Internal assessment for Laboratory shall be done on basis of task/s performed by the student as delineated in the laboratory sheets, prepared by SBTET, AP and posted in its website.
- III. Question paper for End semester Evaluation shall be prepared as per SBTET rules in vogue.

4.16. ISSUE OF MEMORANDUM OF MARKS

All candidates who appear for the end examination will be issued memorandum of marks without any payment of fee. However candidates who lose the original memorandum of marks have to pay the prescribed fee to the Secretary, State Board of Technical Education and Training, A.P. for each duplicate memo from time to time.

4.17. MAXIMUM PERIOD FOR COMPLETION OF DIPLOMA PROGRAMMES:

Maximum period for completion of the diploma courses is twice the duration of the course from the date of First admission (includes the period of detention and discontinuation of studies by student etc) failing which they will have to forfeit the claim for qualifying for the

award of Diploma (They will not be permitted to appear for examinations after that date). This rule applies for all Diploma courses of 3 years and 3 ½ years of engineering and non-engineering courses.

4.18. ELIGIBILITY FOR AWARD OF DIPLOMA

A candidate is eligible for award of Diploma Certificate if he / she fulfil the following academic regulations.

- i. He / She pursued a course of study for not less than 3 / 3 ½ academic years & not more than 6 / 7 academic years.
- ii. He / she have completed all the Courses.

Students who fail to fulfil all the academic requirements for the award of the Diploma within 6 / 7 academic years from the year of admission shall forfeit their seat in the course & their seat shall stand cancelled.

For IVC & ITI Lateral Entry students:

- i. He / She pursued a course of study for not less than 2 / 2 ½ academic years & not more than 4 / 5 academic years.
- ii. He / she has completed all the Courses.

Students who fail to fulfil all the academic requirements for the award of the Diploma within 4 / 5 academic years from the year of admission shall forfeit their seat in the course & their seat shall stand cancelled.

4.19. ISSUE OF PHOTO COPY OF VALUED ANSWER SCRIPT, RECOUNTING& REVERIFICATION:

A) FOR ISSUE OF PHOTO COPIES OF VALUED ANSWER SCRIPTS

- I. A candidate desirous of applying for Photo copy of valued answer script/s should apply within prescribed date from the date of the declaration of the result.
- II. Photo copies of valued answer scripts will be issued to all theory Courses and Drawing Course (s).
- III. The Photo copy of valued answer script will be dispatched to the concerned candidate's address as mentioned in the application form by post.
- IV. No application can be entertained from third parties.

B) FOR RE-COUNTING (RC) and RE-VERIFICATION(RV) OF THE VALUED ANSWER SCRIPT

- i. A candidate desirous of applying for Re-verification of valued answer script should apply within prescribed date from the date of the declaration of the result.
- ii. Re-verification of valued answer script shall be done for all theory Courses' and Drawing Course(s).
- iii. The Re-verification committee constituted by the Secretary, SBTETAP with Course experts shall re-verify the answer scripts.

I. RE-COUNTING

The Officer of SBTET will verify the marks posted and recount them in the already valued answer script. The variations if any will be recorded separately, without making any changes on the already valued answer script. The marks awarded in the original answer script are maintained (hidden).

II. RE-VERIFICATION

- (i) The Committee has to verify the intactness and genuineness of the answer script(s) placed for Re-verification.
- (ii) Initially single member shall carry out the re-verification.
- (iii) On re-verification by single member, if the variation is less than 12% of maximum marks, and if there is no change in the STATUS in the result of the candidate, such cases will not be referred to the next level ie., for 2-Tier evaluation.
- (iv) On re-verification by a single member, if the variation is more than 12% of maximum marks, it will be referred to 2-Tier evaluation.
- (v) If the 2-Tier evaluation confirms variation in marks as more than 12% of maximum marks, the variation is considered as follows:
 - a) If the candidate has already passed and obtains more than 12% of the maximum marks on Re-verification, then the variation is considered.
 - b) If the candidate is failed and obtains more than 12% of the maximum marks on Re-verification and secured pass marks on re-verification, then the status of the candidate changes to PASS.
 - c) If a candidate is failed and obtains more than 12% of the maximum marks on Re-verification and if the marks secured on re-verification are still less than the minimum pass marks, the status of the candidate remain FAIL only.
- (vii) After Re-verification of valued answer script the same or change if any therein on Re-verification, will be communicated to the candidate.
- (viii) On Re-verification of Valued Answer Script if the candidate's marks are revised, the fee paid by the candidate will be refunded or else the candidate has to forfeit the fee amount.

Note: No request for Photo copies/ Recounting /Re-verification of valued answer script would be entertained from a candidate who is reported to have resorted to Malpractice in that examination.

4.20. Mal Practice Cases:

If any candidate resorts to Mal Practice during examinations, he / she shall be booked and the Punishment shall be awarded as per SBTETAP rules and regulations in vogue.

4.21. Discrepancies/ Pleas:

Any Discrepancy /Pleas regarding results etc., shall be represented to the SBTETAP within one month from the date of issue of results. Thereafter, no such cases shall be entertained in any manner.

4.22. Issue of Duplicate Diploma

If a candidate loses his/her original Diploma Certificate and desires a duplicate to be issued he/she should produce written evidence to this effect. He / she may obtain a duplicate from the Secretary, State Board of Technical Education and Training, A.P., on payment of prescribed fee and on production of an affidavit signed before a First Class Magistrate (Judicial) and non-traceable certificate from the Department of Police. In case of damage of original Diploma Certificate, he / she may obtain a duplicate certificate by surrendering the original damaged certificate on payment of prescribed fee to the State Board of Technical Education and Training, A.P.

In case the candidate cannot collect the original Diploma within 1 year from the date of issue of the certificate, the candidate has to pay the penalty prescribed by the SBTET AP from time to time.

4.23. Issue of Migration Certificate and Transcripts:

The Board on payment of prescribed fee will issue these certificates for the candidates who intend to prosecute Higher Studies in India or Abroad.

4.24. General

- i. The Board may change or amend the academic rules and regulations or syllabi at any time and the changes or amendments made shall be applicable to all the students, for whom it is intended, with effect from the dates notified by the competent authority.
- ii. All legal matters pertaining to the State Board of Technical Education and Training, AP are within the jurisdiction of Mangalagiri.
- iii. In case of any ambiguity in the interpretation of the above rules, the decision of the Secretary, SBTET, A.P is final.

GUIDELINES FOR INDUSTRIAL TRAINING OF DIPLOMA IN COMMUNICATION AND COMPUTER NETWORKING ENGINEERING PROGRAMME

1. Duration of the training: 6 months.
2. Eligibility: The As per SBTET norms
3. Training Area: Students can be trained in either in In-house/Industry/ *TWO Online Certificate courses*
 - i) First 3 Months/12 weeks- Registration at either of Nptel/ Swayam/ Moocs/course era /lectera / caltech / oxford / hckerrank / udemy
for First Certificate Source
 - ii) Next 3 Months/12 Weeks- Registration at either of Nptel/ Swayam/ Moocs/course era /lectera / caltech / oxford / hckerrank / udemy
for Second Certificate Source.
4. In the areas of
Application Software Development / system software Development / firmware development/ Mobile application development/ Database applications / Web development/ IoT application development / smart technologies / Hardware interfacing/ Networking .
5. The candidate shall put a minimum of 90% attendance during Industrial Training.
6. If the student fails to secure 90% attendance during industrial training, the student shall reappear for 6 months industrial training.
7. Formative assessment at industry level shall be carried out by the Mentor from of the industry, where the student is undergoing training and the faculty in charge (Guide) from the concerned section in the institution.
8. The Industrial training shall carry 300 marks and pass marks is 50% in assessments at industry (first and second assessment) and final summative assessment at institution

level put together i.e. 150 marks out of 300 marks.

9. If the student fails to secure 50% marks in final summative assessment at institution level, the student should reappear for final summative assessment in the subsequent board examination.
10. Final summative assessment at institution level is done by a committee including
 1. Head of the section (of concerned discipline ONLY),
 2. External examiner from an industry and
 3. Faculty member who assessed the student during Industrial Training as members.

SPECIFIC CHANGES INCORPORATED IN CCNE C-23 WITH RESPECT TO CCNE C-20

Changes imposed in theory subjects

1. CCN-105 , Basics of Communication and Computer Networking Engineering(BCCNE) :
Introduced Concepts of i)Computer communication basics ii) Overview of Data Structures iii)Cloud Computing and removed Virtual Reality concepts.
2. CCN-302. Python Programming:
Retained the subject as it is but brought from CM-505
3. CCN-305, DBMS:
Syllabus changed
4. CCN-401, Network Management through TCP/IP:
A new Subject was introduced in place of CCN-401 – Engineering Mathematics – III
5. CCN-403, Removed Digital Electronics subject from 3rd Semester and placed as CCN-403 - Digital Electronics and Computer Organisation
6. CCN-404. OOPs Through Java
Introduced new subject to enable the students to learn Object Oriented Programming concept skills

Changes imposed in the Practical subjects

1. CCN-306: Python Programming Lab:
2. CCN-308: DBMS Lab:
Introduced the exercises on cursors and triggers in PL/SQL to enforce constraints and data validation and verification.
3. CCN-407: OOPs through Java Lab:
Introduced new Lab to learn Object Oriented Programming concepts thoroughly.
4. CCN-601 Industrial Training:
Online certificate courses on any two emerging computer technologies at NPTEL/SWAYAM/ MOOCS/Course Era//Lectera/Caltech/Oxford/Hackerrank/ Udemy/AP Skill Development courses etc. are introduced in Industrial training.

**DIPLOMA IN COMMUNICATION AND COMPUTER NETWORKING ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION**

(FIRST YEAR)

Sub Code	Name of the Subject	Instruction Periods/Week		Total Periods Per Year	Scheme Of Examinations			
		Theory	Practical		Duration (hrs)	Sessional Marks	End Exam Marks	Total Marks
THEORY SUBJECTS								
CCN-101	English-I	3	-	90	3	20	80	100
CCN-102	Engineering Mathematics - I	5	-	150	3	20	80	100
CCN-103	Engineering Physics	3	-	90	3	20	80	100
CCN-104	Engineering Chemistry and Environmental studies	3	-	90	3	20	80	100
CCN-105	Basics of Communication and Computer Networking Engineering	5	-	150	3	20	80	100
CCN-106	Programming in C	5	-	150	3	20	80	100
PRACTICAL SUBJECTS								
CCN-107	Engineering Drawing	-	3	180	3	40	60	100
CCN-108	Programming in C Lab		6	180	3	40	60	100
CCN-109	Physics Lab		1.5	45	1.5	20	30	50
CCN-110	Chemistry Lab		1.5	45	1.5	20	30	50
CCN-111	Computer Fundamentals Lab		3	90	3	40	60	100
	ACTIVITIES		3	90				
	Total	23	19		-			1000

CCN-101,102,103,104,107,109,110 common with all branches

**DIPLOMA IN COMMUNICATION AND COMPUTER NETWORKING ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION**

(III Semester)

Sub Code	Name of the Subject	Instruction Periods/Week		Total Periods Per Semester	Scheme Of Examinations			
		Theory	Practicals		Duration (hrs)	Sessional Marks	End Exam Marks	Total Marks
THEORY SUBJECTS								
CCN-301	Mathematics –II	4		60	3	20	80	100
CCN-302	Python Programming	5	-	75	3	20	80	100
CCN-303	Operating systems	4	-	60	3	20	80	100
CCN-304	Data Communication and Computer Networks	5	-	75	3	20	80	100
CCN-305	DBMS	5	-	75	3	20	80	100
PRACTICAL SUBJECTS								
CCN-306	Python Programming Lab	-	3	45	3	40	60	100
CCN-307	Computer Hardware and Network Maintenance Lab	-	6	90	3	40	60	100
CCN-308	DBMS Lab	-	4	60	3	40	60	100
CCN-309	Multimedia Lab		3	45	3	40	60	100
	ACTIVITIES		3	45				
	Total	23	19	630		260	640	900

CCN-301 Common to CME, AIML, AMG, AMT, CAI, CCB, CCN, WD

CCN-302 Common with CM-505, CCN-303 COMMON WITH CM-303, CCN-305 IS COMMON WITH CM-305

CCN-308 COMMON WITH CM-308, CCN – 309 COMMON WITH CM-309

**DIPLOMA IN COMMUNICATION AND COMPUTER NETWORKING ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION**

(IV Semester)

Sub Code	Name of the Subject	Instruction Periods / Week		Total Periods Per Semester	Scheme Of Examinations			
		Theory	Practicals		Duration (hrs)	Sessional Marks	End Exam Marks	Total Marks
THEORY SUBJECTS								
CCN-401	Network Management through TCP/IP	5	-	75	3	20	80	100
CCN-402	Web Technologies	5	-	75	3	20	80	100
CCN-403	Digital Electronics and Computer Organisation	5	-	75	3	20	80	100
CCN-404	OOP through Java	5	-	75	3	20	80	100
CCN-405	Cryptography and Network Security	5	-	75	3	20	80	100
PRACTICAL SUBJECTS								
CCN-406	Web Technologies Lab	-	4	60	3	40	60	100
CCN-407	Java Programming Lab	-	4	60	3	40	60	100
CCN-408	Communication Skills	-	3	45	3	40	60	100
CCN-409	Computer Networks and Security Lab	-	3	45	3	40	60	100
	ACTIVITIES		3	45				
	Total	25	17	630	-	260	640	900

CCN-402 COMMON WITH CM-402

CCN-403 COMMON WITH AIM-304 , CCN-404 COMMON WITH CM-404

CCN-406 COMMON WITH CM-406, CCN-407 COMMON WITH CM-407

CCN-408 COMMON WITH ALL BRANCHES

**DIPLOMA IN COMMUNICATION AND COMPUTER NETWORKING ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION**

(V Semester)

Sub Code	Name of the Subject	Instruction Periods/Week		Total Periods Per Semester	Scheme Of Examinations			
		Theory	Practicals		Duration (hrs)	Sessional Marks	End Exam Marks	Total Marks
THEORY SUBJECTS								
CCN-501	Industrial Management and Entrepreneurship	5	-	75	3	20	80	100
CCN-502	Wireless Networks	5	-	75	3	20	80	100
CCN-503	Software Engineering	5	-	75	3	20	80	100
CCN-504	Internet of Things	4	-	60	3	20	80	100
CCN-505	High Speed Networks	5	-	75	3	20	80	100
PRACTICAL SUBJECTS								
CCN-506	Wireless Networks Lab	-	3	45	3	40	60	100
CCN-507	Advanced Computer Networks Lab	-	3	45	3	40	60	100
CCN-508	Life Skills	-	3	45	3	40	60	100
CCN-509	Project work	-	6	90	3	40	60	100
	ACTIVITIES		3	45				
	Total	24	18	630	-	260	640	900

CCN-501 is common to all branches.

CCN-503 is common with CM-401, CCN-504 is common with CM-504

CCN-508 common with all branches

DIPLOMA IN COMMUNICATION AND COMPUTER NETWORKING ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION
Industrial Training

Sl. No.	Subject	Duration	Scheme of evaluation		
			Item	Nature	Max. Marks
1	Industrial Training	6 months	1.First Assessment at Industry (After 12 Weeks)	Assessment of learning outcomes by both the faculty and training mentor of the industry	120
			2.Second Assessment at the Industry (After 22 weeks))	Assessment of learning outcomes by both the faculty and training mentor of the industry	120
			Final Summative assessment at institution level	Training Report	20
				Demonstration of any one of the skills listed in learning outcomes	30
				Viva Voce	10
TOTAL MARKS					300

- The candidate shall put a minimum of 90% attendance during Industrial Training.
- If the student fails to secure 90% attendance during industrial training, the student shall reappear for 6 months industrial training.
- Formative assessment at industry level shall be carried out by the Mentor from of the industry, where the student is undergoing training and the faculty in charge (Guide) from the concerned section in the institution.
- The Industrial training shall carry 300 marks and pass marks is 50% in assessments at industry (first and second assessment) and final summative assessment at institution level put together i.e. 150 marks out of 300 marks.
- If the student fails to secure 50% marks in final summative assessment at institution level, the student should reappear for final summative assessment in the subsequent board examination.
- Final Summative assessment at institution level is done by a committee including Head of the section, External examiner and Faculty members who assessed the students during Industrial Training as members.

**DIPLOMA IN COMMUNICATION AND COMPUTER NETWORKING ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION**

CURRICULUM-2023

(FIRST YEAR)

Sub Code	Name of the Subject	Instruction Periods/Week		Total Periods Per Year	Scheme Of Examinations			
		Theory	Practical		Duration (hrs)	Sessional Marks	End Exam Marks	Total Marks
THEORY SUBJECTS								
CCN-101	English-I	3	-	90	3	20	80	100
CCN-102	Engineering Mathematics - I	5	-	150	3	20	80	100
CCN-103	Engineering Physics	3	-	90	3	20	80	100
CCN-104	Engineering Chemistry and Environmental studies	3	-	90	3	20	80	100
CCN-105	Basics of Communication and Computer Networking Engineering	5	-	150	3	20	80	100
CCN-106	Programming in C	5	-	150	3	20	80	100
PRACTICAL SUBJECTS								
CCN-107	Engineering Drawing	-	3	180	3	40	60	100
CCN-108	Programming in C Lab		6	180	3	40	60	100
CCN-109	Physics Lab		1.5	45	1.5	20	30	50
CCN-110	Chemistry Lab	-	1.5	45	1.5	20	30	50
CCN-111	Computer Fundamentals Lab		3	90	3	40	60	100
	ACTIVITIES		3	90				
	Total	23	19		-			1000

CCN-101,102,103,104,107,109,110 common with DCME branch.

Course Code	Course Title	No. of Periods/Week	Total No. of Periods	Marks for FA	Marks for SA
CCN-101	English	3	90	20	80

Time Schedule						
S.no.	Title of the Unit	Periods allotted	Weightage of Marks	No. of Short answer questions	No. of Long Answer questions	Mapping of COs
1	English for Employability	8	16	2	1	CO1, CO2, CO3, CO4, CO5
2	Living in Harmony	8			26	2
3	Connect with Care	8	CO1, CO2, CO3, CO4, CO5			
4	Humour for Happiness	8		CO1, CO2, CO3, CO4, CO5		
5	Never Ever Give Up!	8	10		1	1
6	Preserve or Perish	9	23	2		CO1, CO2, CO3, CO4, CO5
7	The Rainbow of Diversity	8				
8	New Challenges - Newer Ideas	8	1	1	CO1, CO2, CO3, CO4, CO5	
9	The End Point First	8				1
10	The Equal Halves	8	16	1	1	
11	Dealing with Disaster	9				30
	Total	90	110			

Course Objectives	- To improve grammatical knowledge and enrich vocabulary.
	- To develop effective reading, writing and speaking skills.
	- To comprehend themes related to Personality, Society, Environment to exhibit Universal Human Values.

CO No.	Course Outcomes
CO1	Learn and apply various grammatical concepts to communicate in academic, professional and everyday situations

CO2	Use appropriate vocabulary in various contexts
CO3	Read and comprehend different forms of academic, professional and general reading material
CO4	Communicate effectively in speaking and writing in academic, professional and everyday situations.
CO5	Display human values by applying the knowledge of themes related to Self, Society, Environment, Science and Technology for holistic development and harmonious living through communication.

CO-PO Matrix

CCN-101	Course Title: English Number of Course Outcomes: 5			No. of Periods: 90	
POs	Mapped CO No.	CO Periods Addressing PO in Column 1		Level of Mapping (1,2,3)	Remarks
		Number	Percentage		
PO1		Not directly Applicable for English course, however, the language activities make use of the content from Science and Technology relevant to the programme to enhance English communication skills.			
PO2					
PO3					
PO4					
PO5	CO5	16	18%	Level 1	Up to 20%: Level 1 21%-50%: Level 2 >50%: Level 3
PO6	CO1, CO2, CO3, CO4,	52	58%	Level 3	
PO7	CO1, CO2, CO3, CO4,CO5	22	24%	Level 2	

Level 3 – Strongly Mapped, Level 2- Moderately Mapped; Level 1- Slightly Mapped

Learning Outcomes

1. English for Employability

- 1.1. Perceive the need for improving communication in English for employability
- 1.2. Use adjectives and articles effectively while speaking and in writing
- 1.3. Write simple sentences

2. Living in Harmony

- 2.1. Develop positive self-esteem for harmonious relationships
- 2.2. Use affixation to form new words
- 2.3. Use prepositions and use a few phrasal verbs contextually

3. Connect with Care

- 3.1. Use social media with discretion
- 3.2. Speak about abilities and possibilities
- 3.3. Make requests and express obligations
- 3.4. Use modal verbs and main verbs in appropriate form
- 3.5. Write short dialogues about everyday situations

4. Humour for Happiness

- 4.1. Realize the importance of humour for a healthy living
- 4.2. Improve vocabulary related to the theme
- 4.3. Inculcate reading and speaking skills
- 4.4. Frame sentences with proper Subject – Verb agreement
- 4.5. Understand the features of a good paragraph and learn how to gather ideas as a preliminary step for writing a good paragraph.

5. Never Ever Give Up!

- 5.1. Learn to deal with failures in life
- 5.2. Use the present tense form for various every day communicative functions such as speaking and writing about routines, professions, scientific descriptions and sports commentary
- 5.3. Write paragraphs with coherence and other necessary skills

6. Preserve or Perish

- 6.1. Understand the ecological challenges that we face today and act to save the environment.
- 6.2. Narrate / Report past events and talk about future actions
- 6.3. Develop vocabulary related to environment
- 6.4. Write e-mails

7. The Rainbow of Diversity

- 7.1. Appraise and value other cultures for a happy living in multi-cultural workspace
- 7.2. Understand the usage of different types of sentences
- 7.3. Ask for or give directions, information, instructions
- 7.4. Use language to express emotions in various situations
- 7.5. Write letters in various real life situations

8. New Challenges – Newer Ideas

- 8.1. Understand the functional difference between Active Voice and Passive Voice
- 8.2. Use Passive Voice to speak and write in various contexts
- 8.3. Understand the major parts and salient features of an essay
- 8.4. Learn about latest innovations and get motivated

9. The End Point First!

- 9.1. Understand the importance of setting goals in life
- 9.2. Report about what others have said both in speaking and writing
- 9.3. Write an essay following the structure in a cohesive and comprehensive manner
- 9.4. Apply the words related to Goal Setting in conversations and in life

10. The Equal Halves

- 10.1. Value the other genders and develop a gender-balanced view towards life
- 10.2. Identify the use of different conjunctions in synthesising sentences
- 10.3. Write various types of sentences to compare and contrast the ideas
- 10.4. Apply the knowledge of sentence synthesis in revising and rewriting short essays

10.5. Develop discourses in speech and writing

11. Dealing with Disasters

11.1. be aware of different kinds of disasters and the concept of disaster management

11.2. Generate vocabulary relevant to disaster management and use it in sentences

11.3. Analyze an error in a sentence and correct it

11.4. Learn and write different kinds of reports

Textbook: **'INTERACT'** (A Text book of English for I Year Engineering Diploma Courses) - by SBTET, AP

Reference Books:

Martin Hewings: *Advanced Grammar in Use*, Cambridge University Press

Murphy, Raymond : *English Grammar in Use*, Cambridge University Press

Sidney Greenbaum : *Oxford English Grammar*, Oxford University Press

Wren and Martin (Revised by N.D.V. Prasad Rao) : *English Grammar and Composition*, Blackie ELT Books, S. Chand and Co.

Sarah Freeman: *Strengthen Your Writing*, Macmillan

Course Code	Course Title	No. of Periods/week	Total No. of periods	Marks for FA	Marks for SA
CCN-102	Engineering Mathematics-I	5	150	20	80

TIME SCHEDULE

S.No.	Chapter	No. of Periods	Marks Allotted	Short type	Essay type	COs mapped
Unit - I: Algebra						
1	Functions	6	3	1	0	CO1
2	Partial Fractions	5	3	1	0	CO1
3	Matrices and Determinants	20	16	2	1	CO1
Unit - II: Trigonometry						
4	Trigonometric Ratios	2	0	0	0	CO2
5	Compound Angles	5	3	1	0	CO2
6	Multiple and Submultiple angles	8	3	1	0	CO2
7	Transformations	6	5	0	1/2	CO2
8	Inverse Trigonometric Functions	6	5	0	1/2	CO2
9	Trigonometric Equations	6	5	0	1/2	CO2
10	Properties of triangles	5	5	0	1/2	CO2
11	Complex Numbers	6	3	1	0	CO2
Unit III: Co-ordinate Geometry						
12	Straight Lines	5	3	1	0	CO3
13	Circles	6	5	0	1/2	CO3
14	Conic Sections	12	5	0	1/2	CO3
Unit – IV: Differential Calculus						
15	Limits and Continuity	6	3	1	0	CO4
16	Differentiation	28	23	1	2	CO4
Unit – V: Applications of Derivatives						
17	Geometrical Applications	4	5	0	1/2	CO5
18	Physical Applications	6	5	0	1/2	CO5
19	Maxima and Minima	4	5	0	1/2	CO5
20	Errors and Approximations	4	5	0	1/2	CO5
Total		150	110	10	8	
Marks				30	80	

Course Objectives	(i) To apply the principles of Algebra, Trigonometry and Co-ordinate Geometry to real-time problems in engineering. (ii) To comprehend and apply the concept of Differential Calculus in engineering applications.
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Course Outcomes	CO1	Identify functions as special relations, resolve partial fractions and solve problems on matrices and determinants.
	CO2	Solve problems using the concept of trigonometric functions, their inverses and complex numbers.

	CO3	Find the equations and properties of straight lines, circles and conic sections in coordinate system.
	CO4	Evaluate the limits and derivatives of various functions.
	CO5	Find solutions for engineering problems using differentiation.

Learning Outcomes:

UNIT - I

C.O. 1 Identify functions, resolve partial fractions and solve problems on matrices and determinants.

- L.O.** 1.1 Define Set, Ordered pair and Cartesian product of two sets - examples.
 1.2 Explain Relations and Functions – examples
 1.3 Find Domain & Range of functions – simple examples.
 1.4 Define one-one and onto functions.
 1.5 Find the inverse of a function – simple examples.
 1.6 Define rational, proper and improper fractions of polynomials.
 1.7 Explain the procedure of resolving proper fractions of the types mentioned below into partial fractions

$$i) \frac{f(x)}{(ax+b)(cx+d)} \quad ii) \frac{f(x)}{(ax+b)^2(cx+d)}$$

- 1.8 Define a matrix and order of a matrix.
 1.9 State various types of matrices with examples (emphasis on 3rd order square matrices).
 1.10 Compute sum, difference, scalar multiplication and product of matrices. Illustrate the properties of these operations such as commutative, associative and distributive properties with examples and counter examples.
 1.11 Define the transpose of a matrix and state its properties – examples.
 1.12 Define symmetric and skew-symmetric matrices with examples. Resolve a square matrix into a sum of symmetric and skew-symmetric matrices and provide examples.
 1.13 Define determinant of a square matrix; minor, co-factor of an element of a 3x3 square matrix with examples. Expand the determinant of a 3 x 3 matrix using Laplace expansion formula. State and apply the properties of determinants to solve problems.
 1.14 Distinguish singular and non-singular matrices. Define multiplicative inverse of a matrix and list properties of adjoint and inverse. Compute adjoint and multiplicative inverse of a square matrix.
 1.15 Solve a system of 3 linear equations in 3 unknowns using Cramer's rule and matrix inversion method.

UNIT - II

C.O. 2 Solve problems using the concept of trigonometric functions, their inverses and complex numbers.

- L.O.** 2.1 Define trigonometric ratios of any angle - List the values of trigonometric ratios at specified values.
 2.2 Draw graphs of trigonometric functions - Explain periodicity of trigonometric functions.
 2.3 Define compound angles and state the formulae of $\sin(A \pm B)$, $\cos(A \pm B)$, $\tan(A \pm B)$ and $\cot(A \pm B)$.

- 2.4 Give simple examples on compound angles to derive the values of $\sin 15^\circ$, $\cos 15^\circ$, $\sin 75^\circ$, $\cos 75^\circ$, $\tan 15^\circ$, $\tan 75^\circ$ etc.
- 2.5 Derive identities like $\sin(A+B) \sin(A-B) = \sin^2 A - \sin^2 B$ etc.
- 2.6 Solve simple problems on compound angles.
- 2.7 Derive the formulae of multiple angles $2A$, $3A$ etc and sub multiple angle $A/2$ in terms of angle A of trigonometric functions.
- 2.8 Derive useful allied formulae like $\sin^2 A = (1 - \cos 2A)/2$ etc.
- 2.9 Solve simple problems using the multiple and submultiple formulae.

Syllabus for Unit test-I completed

- 2.10 Derive the formulae on transforming sum or difference of two trigonometric ratios in to a product and vice versa - examples on these formulae.
- 2.11 Solve problems by applying these formulae to sum or difference or product of two terms.
- 2.12 Explain the concept of the inverse of a trigonometric function by selecting an appropriate domain and range.
- 2.13 Define inverses of six trigonometric functions along with their domains and ranges.
- 2.14 Derive relations between inverse trigonometric functions so that the given inverse trigonometric function can be expressed in terms of other inverse trigonometric functions with examples.
- 2.15 State various properties of inverse trigonometric functions and identities like

$$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}, \text{ etc.}$$

- 2.16 Apply formulae like $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$, where $x \geq 0, y \geq 0, xy < 1$ etc.,

to solve Simple problems.

- 2.17 Explain what is meant by solution of trigonometric equations and find the general solutions of $\sin x = k$, $\cos x = k$ and $\tan x = k$ with appropriate examples.
- 2.18 Solve models of the type $a \sin^2 x + b \sin x + c = 0$ and $a \sin x + b \cos x = c$.
- 2.19 State sine rule, cosine rule, tangent rule and projection rule and solve a triangle using these formulae.
- 2.20 List various formulae for the area of a triangle with examples.
- 2.21 Define a complex number, its modulus, conjugate, amplitude and list their properties.
- 2.22 Define arithmetic operations on complex numbers with examples.
- 2.23 Represent the complex number in various forms like modulus-amplitude (polar) form, Exponential (Euler) form with examples.

UNIT - III

Coordinate Geometry

C.O. 3 Find the equations and properties of straight lines, circles and conic sections in coordinate system.

- L.O.** 3.1 Write different forms of a straight line – general form, point-slope form, slope-intercept form, two-point form, intercept form and normal form (or perpendicular form).
- 3.2 Find distance of a point from a line, acute angle between two lines, intersection of two non-parallel lines and distance between two parallel lines.
- 3.3 Define locus of a point and circle.
- 3.4 Write the general equation of a circle and find its centre and radius.
- 3.5 Find the equation of a circle, given (i) centre and radius, (ii) two ends of the diameter (iii) three non collinear points of type $(0,0)$, $(a,0)$, $(0, b)$.
- 3.6 Define a conic section - Explain the terms focus, directrix, eccentricity, axes and latus-rectum of a conic with illustrations.
- 3.7 Find the equation of a conic when focus, directrix and eccentricity are given.

- 3.8 Describe the properties of Parabola, Ellipse and Hyperbola in standard forms whose axes are along the co-ordinate axes and solve simple examples on these conics.

Syllabus for Unit test-II completed

C.O.4 Evaluate the limits and derivatives of various functions.

- L.O. 4.1 Explain the concept of limit and meaning of $\lim_{x \rightarrow a} f(x) = l$ and state the properties of limits.
- 4.2 Evaluate the limits of the type $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ and $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$
- 4.3 State the Standard limits $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a}$, $\lim_{x \rightarrow 0} \frac{\sin x}{x}$, $\lim_{x \rightarrow 0} \frac{\tan x}{x}$, $\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$, $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$, $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$, $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$ (without proof) and solve simple problems using these standard limits.
- 4.4 Explain the concept of continuity of a function at a point and on an interval
- 4.5 State the concept of derivative of a function $y = f(x)$ – definition, first principle as $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ and also provide standard notations to denote the derivative of a function.
- 4.6 Explain the significance of derivative in scientific and engineering applications.
- 4.7 Find the derivative of standard algebraic, logarithmic, exponential and trigonometric functions using the first principle.
- 4.8 Find the derivatives of inverse trigonometric, hyperbolic and inverse hyperbolic functions.
- 4.9 State the rules of differentiation of sum, difference, scalar multiplication, product and quotient of functions with simple illustrative examples.
- 4.10 Explain the method of differentiation of a function of a function (Chain rule) with illustrative examples.
- 4.11 Explain the method of differentiation of parametric functions with examples.
- 4.12 Explain the procedure for finding the derivatives of implicit functions with examples.
- 4.13 Explain the need of taking logarithms for differentiating some functions of $[f(x)]^{g(x)}$ type – examples on logarithmic differentiation.
- 4.14 Explain the concept of finding the second order derivatives with examples.
- 4.15 Explain the concept of functions of several variables, finding partial derivatives and difference between the ordinary and partial derivatives with simple examples.
- 4.16 Explain the concept of finding second order partial derivatives with simple problems.

C.O. 5 Evaluate solutions for engineering problems using differentiation

- L.O. 5.1 State the geometrical meaning of the derivative - Explain the concept of derivative to find the slopes of tangent and normal to a given curve at any point on it with examples.
- 5.2 Find the equations of tangent and normal to a given curve at any point on it – simple problems.
- 5.3 Explain the derivative as a rate of change in distance-time relations to find the velocity and acceleration of a moving particle with examples.
- 5.4 Explain the derivative as a rate measurer in the problems where the quantities like areas, volumes vary with respect to time- illustrative examples.

- 5.5 Define the concept of increasing and decreasing functions - Explain the conditions to find points where the given function is increasing or decreasing with illustrative examples.
- 5.6 Explain the procedure to find the extreme values (maxima or minima) of a function of single variable-simple problems for quadratic and cubic polynomials.
- 5.7 Apply the concept of derivatives to find the errors and approximations - simple problems.

Syllabus for Unit test-III completed

CO/PO – Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	2	2	3				3	2	2
CO2	3	3	2	2				3	2	2
CO3	3	3	2	2				3	2	2
CO4	3	3	3	3				3	3	3
CO5	3	3	3	3				3	3	3
Avg.	3	2.8	2.4	2.6				3	2.4	2.4

3 = Strongly mapped (High), **2** =moderately mapped (Medium), **1** =slightly mapped (Low)

Note: The gaps in CO/PO mapping can be met with appropriate activities as follows:

For PO5: Appropriate quiz programmes may be conducted at intervals and duration as decided by concerned faculty.

For PO6: Seminars on applications of mathematics in various engineering disciplines are to be planned and conducted.

For PO7: Plan activities in such a way that students can visit the Library to refer standard books on Mathematics and access the latest updates in reputed national and international journals. Additionally, encourage them to attend seminars and learn mathematical software tools.

PO- CO – Mapping strength

PO No	Mapped with CO no	CO periods addressing PO in column I		Level (1,2 or 3)	Remarks
		No	%		
1	CO1, CO2, CO3, CO4, CO5	150 (31+44+23+34+18)	100%	3	>40% Level 3 Highly addressed 25% to 40% Level 2 Moderately addressed 5% to 25% Level 1 Low addressed <5% Not addressed
2	CO1, CO2, CO3, CO4, CO5	80 (8+23+12+22+15)	53.3%	3	
3	CO1, CO2, CO3, CO4, CO5	61 (9+14+9+14+15)	40.6%	3	
4	CO1, CO2, CO3, CO4, CO5	61 (14+9+9+14+15)	40.6%	3	
PSO 1	CO1, CO2, CO3, CO4, CO5	150 (31+44+23+34+18)	100%	3	
PSO 2	CO1, CO2, CO3, CO4, CO5	62 (10+14+9+14+15)	41.3%	3	
PSO 3	CO1, CO2, CO3, CO4, CO5	62	41.3%	3	

		(10+14+9+14+15)			
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COURSE CONTENT

Unit-I Algebra

1. Functions:

Definitions of Set, Ordered pair, Cartesian product of two sets, Relations, Functions, Domain & Range of functions – One-one and onto functions, inverse of a function.

2. Partial Fractions:

Definitions of rational, proper and improper fractions of polynomials. Resolve rational fractions (proper fractions) into partial fractions covering the types mentioned below.

$$i) \frac{f(x)}{(ax+b)(cx+d)} \quad ii) \frac{f(x)}{(ax+b)^2(cx+d)}$$

3. Matrices:

Definition of a matrix, types of matrices - Algebra of matrices, equality of two matrices, sum, difference, scalar multiplication and product of matrices. Transpose of a matrix, Symmetric, skew-symmetric matrices - Determinant of a square matrix, minor and cofactor of an element, Laplace's expansion, properties of determinants - Singular and non-singular matrices, Adjoint and multiplicative inverse of a square matrix- System of linear equations in 3 variables-Solutions by Cramer's rule and Matrix inversion method.

Unit-II Trigonometry

4. Trigonometric ratios:

Definition of trigonometric ratios of any angle, values of trigonometric ratios at specified values, draw graphs of trigonometric functions, periodicity of trigonometric functions.

5. Compound angles:

Formulas of $\sin(A \pm B)$, $\cos(A \pm B)$, $\tan(A \pm B)$, $\cot(A \pm B)$, and related identities.

6. Multiple and sub multiple angles:

Formulae for trigonometric ratios of multiple angles $2A$, $3A$ and sub multiple angle $A/2$.

7. Transformations:

Transformations of products into sums or differences and vice versa.

8. Inverse trigonometric functions:

Definition, domains and ranges-basic properties.

9. Trigonometric equations:

Concept of a solution, principal value and general solution of trigonometric equations:

$\sin x = k$, $\cos x = k$, $\tan x = k$, where k is a constant. Solutions of simple quadratic equations and equations of type $a \sin x + b \cos x = c$.

10. Properties of triangles:

Relations between sides and angles of a triangle- sine rule, cosine rule, tangent rule and projection rule-area of a triangle.

11. Complex Numbers:

Definition of a complex number, modulus, conjugate and amplitude of a complex number - Arithmetic operations on complex numbers - Modulus-Amplitude (polar) form, Exponential form (Euler form) of a complex number.

UNIT-III

Coordinate geometry

12. **Straight lines:** Various forms of a straight line - Angle between two lines, perpendicular distance from a point, intersection of non-parallel lines and distance between parallel lines.
13. **Circle:** Locus of a point, Circle definition - Circle equation given (i) centre and radius, (ii) two ends of a diameter (iii) three non-collinear points of type $(0,0)$, $(a,0)$, $(0, b)$ - General equation of a circle –its centre and radius.
14. Definition of a conic section - Equation of a conic when focus, directrix and eccentricity are given - Properties of parabola, ellipse and hyperbola in standard forms.

UNIT-IV

Differential Calculus

15. **Concept of Limit-** Definition and Properties of Limits and Standard Limits - Continuity of a function at a point.
16. **Concept of derivative-** Definition (first principle)- different notations - Derivatives of standard algebraic, logarithmic, exponential, trigonometric, inverse trigonometric, hyperbolic and inverse hyperbolic functions - Derivatives of sum, difference, scalar multiplication, product, quotient of functions - Chain rule, derivatives of parametric functions, derivatives of implicit functions, logarithmic differentiation - Second order derivatives - Functions of several variables, first and second order partial derivatives.

UNIT-V

Applications of Derivatives

17. Geometrical meaning of the derivative, equations of tangent and normal to a curve at any point.
18. Physical applications of derivatives – Velocity, acceleration, derivative as a rate measurer.
19. Applications of the derivative to find the extreme values – Increasing and decreasing functions, maxima and minima for quadratic and cubic polynomials.
20. Absolute, relative and percentage errors - Approximate values due to errors in measurements.

Textbook:

Engineering Mathematics-I, a textbook for first year diploma courses, prepared & prescribed by SBTET, AP.

Reference Books:

1. Shanti Narayan, A Textbook of matrices, S.Chand&Co.
2. Robert E. Moyer & Frank Ayers Jr., Schaum's Outline of Trigonometry, 4th Edition, Schaum's Series.
3. G.B.Thomas, R.L.Finney, Calculus and Analytic Geometry, Addison Wesley, 9th Edition, 1995.
4. Frank Ayers & Elliott Mendelson, Schaum's Outline of Calculus, Schaum's Series.
5. M.Vygodsky, Mathematical Handbook, Mir Publishers, Moscow.

Unit Test Syllabus

Unit Test	Syllabus
Unit Test-I	From L.O. 1.1 to L.O. 2.9
Unit Test-II	From L.O. 2.10 to L.O. 3.8
Unit Test-III	From L.O.4.1 to L.O. 5.7

CCN-103	Engineering Physics	3	90	20	80
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TIME SCHEDULE

S.No	Major topics	No. of Periods	Weightage of Marks	Short Answer type (3 marks)	Essay type (10 marks)	COs mapped
1.	Units and measurements	09	03	1		CO1
2.	Statics	11	13	1	1	
3.	Gravitation	12	20	1	2	CO2
4.	Concepts of energy	10	13	1	1	
5.	Thermal physics	10	13	1	1	CO3
6.	Sound	12	16	2	1	
7.	Electricity & Magnetism	13	16	2	1	CO4
8.	Modern physics	13	16	2	1	
	Total:	90	110	10	8	

Course title : Engineering Physics	
Course objectives	(1) To understand the basic concepts of physics for various Engineering applications as required for industries. (2) To equip the students with the scientific advances in technology and make the student suitable for any industrial or scientific organization.

COURSE OUTCOMES	CO1	Familiarize with various physical quantities, their SI units and errors in measurements; understand the concepts of vectors and various forces in statics.
	CO2	Understand the concepts of gravitation with reference to applications in satellites, provide the knowledge of various forms of energy and their working principles.
	CO3	Familiarize with the knowledge of transmission of heat and gas laws; provide the knowledge on musical sound and noise as pollution and also the concepts of echo and reverberation.
	CO4	Provide basic knowledge of electricity and concepts of magnetism and magnetic materials; familiarize with the advances in Physics such as photoelectric effect, optical fibers, semiconductors, superconductors and nanotechnology.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	2	2	1			2	2		2
CO2	3	2	2	2	2		2	1		2
CO3	2		1		2		1		1	1
CO4	3	2	3	2	2		3	2		2

MATRIX SHOWING MAPPING OF COURSE OUTCOMES WITH PROGRAMME OUTCOMES

CO-PO Mapping Strength					No of periods 90
Course code Common - 103	Engineering Physics No of Course Objectives : 4				
POs	Mapped with CO No	POs addressing PO in Col 1 NO	%	Level 1,2,3	remarks
PO1	CO1,CO2,CO3,CO4	44	48.9 %	3	>40% level 3 (highly addressed) 25% to 40% level 2 (moderately addressed) 25% level 1 (Low addressed) < 5% (not addressed)
PO2	CO1,CO2, CO4	11	12.2%	1	
PO3	CO1, CO2,CO3, CO4	10	11.1%	1	
PO4	CO1, CO2,CO4	8	8.9%	1	
PO5	CO2,CO3, CO4	8	8.9%	1	
PO6					
PO7	CO1, CO2, CO3, CO4	9	10.0%	1	

3 = strongly mapped, 2 = moderately mapped,

1 = slightly mapped

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following.

- | | | | |
|------------------------|------------------------|--------------------------------|---------------------|
| (i) Seminars | (ii) Tutorials | (iii) Guest Lecturers | (iv) Assignments |
| (v) Quiz competitions | (vi) Industrial visits | (vii) Tech fest | (viii) Mini project |
| (ix) Group discussions | (x) Virtual classes | (xi) Library visit for e-books | |

Learning outcomes

Upon completion of the course the student shall be able to

1.0 Understand the concept of units and measurements

- 1.1 Explain the concept of units
- 1.2 Define the terms
 - a) Physical quantity
 - b) Fundamental physical quantities and
 - c) Derived physical quantities
- 1.3 Define unit
- 1.4 Define fundamental units and derived units
- 1.5 State SI units with symbols for fundamental and some derived quantities
- 1.6 State Multiples and Submultiples in SI system
- 1.7 State rules of writing S.I units
- 1.8 State advantages of SI units
- 1.9 What are direct and indirect measurements
- 1.10 Define accuracy and least count
- 1.11 Define error in measurement
- 1.12 Define absolute, relative and percentage errors with their formulae
- 1.13 Solve simple problems on absolute, relative and percentage errors

2.0 Understand the concepts of statics

- 2.1 Explain the concept of Vectors
- 2.2 Define scalar and vector quantities with examples
- 2.3 Represent vectors geometrically
- 2.4 Define the types of vectors (equal, negative, unit, co-initial, co-planar, position vector)
- 2.5 Resolve the vector into rectangular components
- 2.6 State and explain triangle law of addition of vectors
- 2.7 Define concurrent forces, co-planar forces and equilibrant.
- 2.8 State and explain Lami's theorem
- 2.9 State the parallelogram law of addition of forces with diagram.
- 2.10 Write the expressions for magnitude and direction of resultant (no derivation)
- 2.11 Illustrate parallelogram law with examples (i) flying of bird and (ii) working of sling.
- 2.12 Define moment of force and couple.
- 2.13 Write the formulae and S.I units of moment of force and couple.
- 2.14 Solve simple problems on (i) Resolution of force and (ii) Parallelogram law of forces (finding R , α and θ).

3.0 Understand the concepts of Gravitation

- 3.1 State and explain Newton's universal law of gravitation.
- 3.2 Define G and mention its value.
- 3.3 Explain the acceleration due to gravity (g)
- 3.4 Explain the factors affecting the value of g

- 3.5 Derive the relationship between g and G .
- 3.6 State and explain the Kepler's laws of planetary motion
- 3.7 Define a satellite.
- 3.8 What are natural and artificial satellites, Give examples.
- 3.9 Define orbital velocity and write its formula.
- 3.10 Define escape velocity and write its formula.
- 3.11 Write a brief note on Polar satellites.
- 3.12 Write a brief note on Geo-stationary satellites.
- 3.13 Mention the applications of artificial satellites.
- 3.14 Solve simple problems on (i) Newton's law of gravitation and (ii) calculation of orbital and escape velocities.
- 4.0 Understand the concepts of Energy.**
- 4.1 Define work done and energy. Mention their SI units.
- 4.2 List various types of energy.
- 4.3 Define P.E with examples. Write its equation.
- 4.4 Define K.E with examples. Write its equation.
- 4.5 Derive relationship between K.E and momentum.
- 4.6 State the law of conservation of energy. Give various examples.
- 4.7 Write a brief note on solar energy.
- 4.8 Explain the principle of solar thermal conversion.
- 4.9 Explain the principle of photo voltaic effect
- 4.10 Solve simple problems on (i) work done (ii) P.E & K.E and (iii) Relation between K.E & momentum.
- 5.0 Understand the concepts of thermal physics**
- 5.1 Define the concepts of heat and temperature
- 5.2 State different modes of transmission of heat
- 5.3 Explain conduction, convection and radiation with two examples each.
- 5.4 State and explain Boyle's law
- 5.5 Define absolute zero temperature
- 5.6 Explain absolute scale of temperature
- 5.7 State the relationship between degree Celsius, Kelvin and Fahrenheit temperatures
- 5.8 State Charle's law and write its equation
- 5.9 State Gay-Lussac's law and write its equation
- 5.10 Define ideal gas
- 5.11 Derive ideal gas equation
- 5.12 Explain why universal gas constant (R) is same for all gases in nature
- 5.13 Calculate the value of R for 1 gram mole of gas.
- 5.14 Solve simple problems on (i) Inter conversion of temperatures between $^{\circ}\text{C}$, K and F
(ii) Gas laws and (iii) Ideal gas equation.
- 6.0 Understand the concepts of Sound**
- 6.1 Define the term sound
- 6.2 Define longitudinal and transverse waves with one example each
- 6.3 Explain the factors which affect the velocity of sound in air
- 6.4 Distinguish between musical sound and noise
- 6.5 Explain noise pollution and state SI unit for intensity of sound
- 6.6 Explain source s of noise pollution
- 6.7 Explain effects of noise pollution
- 6.8 Explain methods of minimizing noise pollution
- 6.9 Define Doppler effect.
- 6.10 List the Applications of Doppler effect

- 6.11 Define reverberation and reverberation time
- 6.12 Write Sabine's formula and name the physical quantities in it.
- 6.13 Define echoes and explain the condition to hear an echo.
- 6.14 Mention the methods of reducing an echo
- 6.15 Mention the applications of an echo
- 6.16 What are ultrasonics
- 6.17 Mention the applications of ultra-sonics, SONAR
- 6.18 Solve simple problems on echo
- 7.0 Understand the concepts of Electricity and Magnetism**
- 7.1 Explain the concept of P.D and EMF
- 7.2 State Ohm's law and write the formula
- 7.3 Explain Ohm's law
- 7.4 Define resistance and specific resistance. Write their S.I units.
- 7.5 State and explain Kichoff's first law.
- 7.6 State and explain Kirchoff's second law.
- 7.7 Describe Wheatstone bridge with legible sketch.
- 7.8 Derive an expression for balancing condition of Wheatstone bridge.
- 7.9 Describe Meter Bridge experiment with necessary circuit diagram.
- 7.10 Write the formulae to find resistance and specific resistance in meter bridge
- 7.11 Explain the concept of magnetism
- 7.12 What are natural and artificial magnets (mention some types)
- 7.13 Define magnetic field and magnetic lines of force.
- 7.14 Write the properties of magnetic lines of force
- 7.15 State and explain the Coulomb's inverse square law of magnetism
- 7.16 Define magnetic permeability
- 7.17 Define para, dia, ferro magnetic materials with examples
- 7.18 Solve simple problems on (i) Ohm's law (ii) Kirchoff's first law (iii) Wheatstone bridge (iv) meter bridge and (v) Coulomb's inverse square law
- 8.0 Understand the concepts of Modern physics**
- 8.1 State and explain Photo-electric effect.
- 8.2 Write Einstein's Photo electric equation and name the physical quantities in it.
- 8.3 State laws of photo electric effect
- 8.4 Explain the Working of photo electric cell
- 8.5 List the Applications of photoelectric effect
- 8.6 Recapitulate refraction of light and its laws
- 8.7 Define critical angle
- 8.8 Explain the Total Internal Reflection
- 8.9 Explain the principle and working of Optical Fiber
- 8.10 List the applications of Optical Fiber
- 8.11 Explain the energy gap based on band structure
- 8.12 Distinguish between conductors, semiconductors and insulators based on energy gap
- 8.13 Define doping
- 8.14 Explain the concept of hole
- 8.15 Explain the types of semiconductors, Intrinsic and extrinsic
- 8.16 Explain n-type and p-type semiconductors
- 8.17 Mention the applications of semiconductors
- 8.18 Define superconductor and superconductivity
- 8.19 List the applications of superconductors
- 8.20 Nanotechnology definition, nano materials and application

COURSECONTENT

1. Units and measurements

Introduction – Physical quantity – Fundamental and Derived quantities – Fundamental and derived units - SI units – Multiples and Sub multiples – Rules for writing S.I. units-Advantages of SI units – Direct and indirect measurements – Accuracy and least count – Errors : Absolute, relative and percentage errors –Problems.

2. Statics

Scalars and Vectors– Representation of a vector - Types of vectors - Resolution of vector into rectangular components – Triangle law of vectors – Concurrent forces - Lami's theorem - Parallelogram law of forces : Statement, equations for magnitude and direction of resultant, examples – Moment of force and couple – Problems.

3. Gravitation

Newton's law of gravitation and G – Concept of acceleration due to gravity (g) – Factors affecting the value of g – Relation between g and G - Kepler's laws – Satellites : Natural and artificial – Orbital velocity and escape velocity – Polar and geostationary satellites – Applications of artificial satellites – Problems.

4. Concepts of energy

Work done & Energy-Definition and types of energy - potential energy - kinetic energy-- K.E and momentum relation – Law of Conservation of energy, examples - Solar energy, principles of thermal and photo conversion – Problems.

5. Thermal physics

Modes of transmission of heat – Expansion of Gases - Boyle's law – Absolute scale of temperature - Thermometric scales and their inter conversion - Charle's law - Gay-Lussac's law - Ideal gas equation - Universal gas constant (R) - Problems.

6. Sound

Sound - Nature of sound - Types of wave motion, Longitudinal and transverse – Factors affecting the velocity of sound in air - musical sound and noise - Noise pollution – Causes & effects - Methods of reducing noise pollution - Doppler effect - Echo- Reverberation -Reverberation time-Sabine 's formula – Ultrasonics & applications – SONAR - Problems.

7. Electricity & Magnetism

Concept of P.D and EMF - Ohm's law and explanation-Specific resistance - Kirchoff's laws – Wheat stone's bridge - Meter bridge.

Natural and artificial magnets – magnetic field and magnetic lines of force – Coulomb's inverse square law – Permeability – Magnetic materials – Para, dia, ferro – Examples – Problems.

8. Modern Physics

Photoelectric effect – laws of photoelectric effect – photoelectric cell – Applications of photoelectric cell - Total internal reflection - Fiber optics - Principle and working of an optical fiber - Applications of optical fibers – Semiconductors : Based on Energy gap – Doping – Hole - Intrinsic and extrinsic semiconductors (n-type & p-type) – Applications of semiconductors – Superconductivity – applications – Nanotechnology definition, nano materials, applications.

REFERENCES

- | | |
|--|-----------------------------------|
| 1. Intermediate physics - Volume - I & 2 | Telugu Academy (English version) |
| 2. Unified physics Volume 1, 2, 3 and 4 | Dr. S.L Guptha and Sanjeev Guptha |
| 3. Concepts of Physics, Vol 1 & 2 | H.C. Verma |
| 4. Text book of physics Volume I & 2 | Resnick & Halliday |
| 5. Fundamentals of physics | Brijlal & Subramanyam |
| 6. Text book of applied physics | Dhanpath Roy |

NCERT Text Books of physics

Class XI & XII Standard

1. e-books/e-tools/websites/Learning Physics software/eLMS

Course Code	Course Title	No. of Periods/week	Total No. of periods	Marks for FA	Marks for SA
CCN-104	Engineering Chemistry and Environmental Studies	3	90	20	80

Time Schedule

S.No	Unit Title/Chapter	No of Periods	Weightage of marks	Question wise distribution		Mapped with CO
				Essay	Short	
1	Fundamentals of Chemistry	14	21	1½*	2	CO1
2	Solutions, Acids and Bases	16	21	1½*	2	CO1
3	Electrochemistry	12	13	1	1	CO2
4	Corrosion	8	13	1	1	CO2
5	Water Treatment	8	13	1	1	CO3
6	Polymers & Engineering materials.	12	13	1	1	CO4
7	Fuels	6	3	0	1	CO4
8	Environmental Studies	14	13	1	1	CO5
Total		90	110	8	10	

Course Objectives	1. To familiarize with the concepts of chemistry involved in the process of various Engineering Industrial Applications.
	2. To know the various natural and man-made environmental issues and concerns with an interdisciplinary approach that include physical, chemical, biological and socio cultural aspects of environment.
	3. to reinforce theoretical concepts by conducting relevant experiments/exercises

Course Outcomes	CO1	Explain Bohr's atomic model, chemical bonding, mole concept, acids and bases, P ^H and Buffer solutions.
	CO2	Explain electrolysis, Galvanic cell, batteries and corrosion
	CO3	Explain the chemistry involved in the treatment of hardness in water.
	CO4	Explain the methods of preparation and applications of Polymers and Elastomers, chemical composition and applications of Alloys, Composite Materials, Liquid Crystals , Nano Materials and Fuels.
	CO5	Explain Global impacts due to air pollution, causes , effects and controlling methods of water pollution and understand the environment, forest resources, e-Pollution and Green Chemistry Principles.

Engineering. Chemistry and Environmental studies No of Course Outcomes:5					No Of periods 90
POs	Mapped with CO No	CO periods addressing PO in Col NO. 1	%	Level 1,2,3	remarks
PO1	CO1,CO2,CO3	42	46.7 %	3	>40% level 3 (highly addressed) 25% to 40% level 2(moderately addressed) 5% to 25% level 1 (Low addressed) < 5%(not addressed)
PO2	CO2,CO3	16	17.8%	1	
PO3	CO4	12	13.3%	1	
PO4	CO4	6	6.7%	1	
PO5	CO5	14	15.5%	1	
PO6	-	-	-	-	
PO7	-	-	-	-	

COs-POs mapping strength (as per given table)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-
CO2	3	1	-	-	-	-	-	-	-	-
CO3	3	1	-	-	-	-	-	-	-	-
CO4	-	-	1	1	-	-	-	-	-	-
CO5	-	-	-	-	1	-	-	-	-	-
Average	3	1	1	1	1		-	-	-	-

3=strongly mapped 2= moderately mapped 1= slightly mapped

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following i) Seminars ii) Tutorials iii) Guest Lectures iv) Assignments v) Quiz competitions vi) Industrial visit vii) Tech Fest

viii) Mini project ix) Group discussions x) Virtual classes xi) Library visit for e-books

Model Blue Print with Weightage for Blooms category and questions for each chapter and COs mapped

***One question of 10 marks should be given with 50% weightage from unit title 1 and 2**

Upon completion of the course, the student shall be able to learn out

ENGINEERING CHEMISTRY AND ENVIRONMENTAL STUDIES

1.0 Atomic structure

- 1.1 Explain the charge, mass of fundamental particles of an atom (electron, proton and neutron) and the concept of atomic number and mass number.
- 1.2 State the Postulates of Bohr's atomic theory and its limitations.
- 1.3 Explain the significance of four Quantum numbers and draw the atomic structures of Silicon and Germanium.
- 1.4 Define Orbital of an atom and draw the shapes of s,p and d-orbitals.
- 1.5 Explain 1. Aufbau principle, 2. Pauli's exclusion principle 3. Hund's principle.
- 1.6 Write the electronic configuration of elements up to atomic number 30.
- 1.7 Explain the significance of chemical bonding.
- 1.8 Explain the Postulates of Electronic theory of valency.
- 1.9 Define and explain Ionic and Covalent bonds with examples of NaCl , H_2 , O_2 and N_2 .(* Lewis dot method).
- 1.10 List out the Properties of Ionic compounds and covalent compounds and distinguish between their properties.

2.0 Solutions, Acids and Bases

- 2.1 Define the terms 1. Solution, 2. Solute and 3. Solvent.
- 2.2 Classify solutions based on solubility.
- 2.3 Define the terms 1. Atomic weight, 2. Molecular weight and 3. Equivalent weight.
Calculate Molecular weight and Equivalent weight of the given acids (HCl, H_2SO_4 , H_3PO_4) Bases (NaOH, Ca(OH)_2 , Al(OH)_3 and Salts (NaCl, Na_2CO_3 , CaCO_3).
- 2.4 Define mole and solve numerical problems on mole concept.
- 2.5 Define molarity, normality and solve numerical problems on molarity and normality.
 - a) Calculate the Molarity or Normality, if weight of solute and volume of solution are given.
 - b) Calculate the weight of solute, if Molarity or Normality with volume of solution are given.
 - c) Problems on dilution to convert high concentrated solutions to low concentrated solutions.
- 2.6 Explain Arrhenius theory of Acids and Bases and give its limitations.
- 2.7 Define ionic product of water, pH and solve numerical problems on pH (Strong Acids and Bases).
- 2.8 Define buffer solution and classify buffer solutions with examples. Give its applications.

3.0 Electrochemistry

- 3.1 Define the terms 1. Conductor 2. Semiconductor 3. Insulator, 4. Electrolyte 5. Non-electrolyte. Give two examples each.
- 3.2 Distinguish between Metallic conduction and Electrolytic conduction.
- 3.3 Explain electrolysis by taking an example of used NaCl and list out the applications of electrolysis.
- 3.4 Define Galvanic cell. Explain the construction and working of Galvanic cell.
- 3.5 Distinguish between electrolytic cell and galvanic cell.
- 3.6 Define battery and list the types of batteries with examples.
- 3.7 Explain the construction, working and applications of i) Dry cell (Leclanche cell), ii) Lead storage battery, iii) Lithium-Ion battery and iv) Hydrogen-Oxygen fuel cell.

4.0 Corrosion

- 4.1 Define the term corrosion.
- 4.2 state the Factors influencing the rate of corrosion.
- 4.3 Describe the formation of (a) composition cell (b) stress cell (c) concentration cell during corrosion.
- 4.4 Define rusting of iron and explain the mechanism of rusting of iron.
- 4.5 Explain the methods of prevention of corrosion by
 - (a) Protective coatings (anodic and cathodic coatings).
 - (b) Cathodic protection (Sacrificial anode process and Impressed-voltage process).

5.0 Water Treatment

- 5.1 Define soft water and hard water with respect to soap action.
- 5.2 Define and classify the hardness of water.
- 5.3 List out the salts that causing hardness of water (with Formulae).
- 5.4 State the disadvantages of using hard water in industries.
- 5.5 Define Degree of hardness and units of hardness (mg/L and ppm).
- 5.6 Solve numerical problems on hardness.

5.7 Explain the methods of softening of hard water by (i) Ion-exchange process and (ii) Reverse Osmosis process.

6.0 Polymers & Engineering materials.

A) Polymers

6.1 Explain the concept of polymerization.

6.2 Describe the methods of polymerization (a) addition polymerization of ethylene (b) condensation polymerization of Bakelite (Only flowchart).

6.3 Define plastic. Explain a method of preparation and uses of the following plastics:

1. PVC 2. Teflon 3. Polystyrene 4. Nylon 6,6.

6.4 Define elastomers. Explain a method of preparation and applications of the following:

1. Buna-S 2. Neoprene.

B) Engineering Materials

6.5 Define an alloy. Write the composition and applications of the following:

1. Nichrome 2. Duralumin 3. Stainless Steel.

6.6 Define Composite Materials and give any two examples. State their Properties and applications.

6.7 Define Liquid Crystals and give any two examples. State their Properties and applications.

6.8 Define Nano Materials and give any two examples. State their Properties and applications.

7.0 Fuels

7.1 Define the term fuel.

7.2 Classify the fuels based on occurrence.

7.3 Write the composition and uses of the following:

1. LPG 2. CNG 3. Biogas 4. Power alcohol

7.4 Write the commercial production of Hydrogen as future fuel. Give its advantages and disadvantages.

8.0 ENVIRONMENTAL STUDIES

8.1 Explain the scope and importance of environmental studies.

8.2 Define environment. Explain the different segments of environment.

1. Lithosphere 2. Hydrosphere 3. Atmosphere 4. Biosphere

8.3 Define the following terms:

1. Pollutant 2. Pollution 3. Contaminant 4. Receptor 5. Sink 6. Particulates 7. Dissolved oxygen (DO) 8. Threshold Limit Value (TLV) 9. BOD 10. COD 11. Eco system 12. Producers 13. Consumers 14. Decomposers with examples.

8.4 State the renewable and non-renewable energy sources with examples.

8.5 State the uses of forest resources.

8.6 Explain the causes and effects of deforestation.

8.7 Define air pollution and explain its Global impacts 1. Greenhouse effect, 2. Ozone layer depletion and 3. Acid rain.

8.8 Define Water pollution. Explain the causes, effects and controlling methods of Water pollution.

8.9 Define e-Pollution, State the sources of e-waste. Explain its health effects and control methods.

8.10 Define Green Chemistry. Write the Principles and benefits of Green Chemistry.

COURSE CONTENT

ENGINEERING CHEMISTRY AND ENVIRONMENTAL STUDIES

1. Fundamentals of Chemistry

Atomic Structure: Introduction - Fundamental particles – Bohr's theory – Quantum numbers – Atomic structure of Silicon and Germanium - Orbitals, shapes of s, p and d orbitals - Aufbau's principle - Hund's rule - Pauli's exclusion Principle - Electronic configuration of elements.

Chemical Bonding: significance – Electronic theory of valency - Types of chemical bonds – Ionic and covalent bond with examples – Properties of Ionic and Covalent compounds.

2. Solutions, Acids and Bases

Solutions: Types of solutions - Mole concept – Numerical problems on mole concept - Methods of expressing concentration of a solution – Molarity and Normality – Numerical problems on molarity and normality.

Acids and Bases: Arrhenius theory of acids and bases – Ionic product of water - pH – Numerical problems on pH – Buffer solutions – Classification - applications.

3. Electrochemistry

Conductors, semiconductors, insulators, electrolytes and non-electrolytes – Electrolysis of fused NaCl– Applications of electrolysis - Galvanic cell – Battery-Types- Dry Cell (Leclanche Cell), Lead- Storage battery- Lithium-Ion battery -Hydrogen-Oxygen Fuel cell.

4. Corrosion

Introduction - Factors influencing corrosion - Composition, Stress and Concentration Cells– Rusting of iron and its mechanism – Prevention of corrosion by Protective Coating methods, Cathodic Protection methods.

5. Water treatment

Introduction– Soft and Hard water– Causes of hardness– Types of hardness– Disadvantages of hard water – Degree of hardness (ppm and mg/lit) – Numerical problems on hardness - Softening methods – Ion-Exchange process– Reverse Osmosis process.

6. Polymers & Engineering materials

Polymers:

Concept of polymerization – Types of polymerization – Addition, condensation with examples – Plastics - Preparation and uses of i).PVC ii) Teflon iii) Polystyrene and iv) Nylon 6,6.

Elastomers: Preparation and application of i)Buna-s and ii) Neoprene.

Engineering Materials:

Alloys- Composition and applications of i) Nichrome, ii) Duralumin and iii) Stainless Steel.

Composite Materials- Properties and applications.

Liquid Crystals- Properties and applications.

Nano Materials- Properties and applications.

7. Fuels

Definition and classification of fuels – Composition and uses of i) LPG ii) CNG iii) Biogas and iv) Power alcohol – Hydrogen as a future fuel-production- advantages and disadvantages.

8. ENVIRONMENTAL STUDIES

Scope and importance of environmental studies – Environment - Important terms related to environment– Renewable and non-renewable energy sources–Forest resources – Deforestation –Air pollution–Global impacts on environment –Water pollution – causes – effects – control measures- e-Pollution –Sources of e-waste - Health effects - Control methods - Green Chemistry- Principles -Benefits.

Table specifying the scope of syllabus to be covered for Unit Test- 1, Unit Test- 2 and Unit Test -3

Unit Test	Learning outcomes to be covered
Unit Test – 1	From 1.1 to 2.8
Unit Test – 2	From 3.1 to 5.7
Unit Test – 3	From 6.1 to 8.10

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-105	Basics of Communication and Computer Networking Engineering	4	120	20	80

Time Schedule

S.No	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1	Fundamentals of Computers	15	16	2	1	CO1, CO3, CO4
2	Programming Methodology	15	21	2	1 ½	CO2
3	Operating System basics	25	21	2	1 ½	CO1, CO3
4	Computer Hardware and Networking Basics	35	26	2	2	CO1, CO4, CO5
5	Basics of Data structures and Emerging Trends in Computer Technologies	30	26	2	2	CO2, CO4
Total		120	110	10	8	
Marks				30	80	

Course Objectives	Upon completion of the course the student shall be able to
	i) know the fundamentals of Computers. ii) familiarize programming methodologies like algorithms and flowcharts iii) understand Operating system basics iv) To familiarize basics of data structures and Emerging Technologies

Course Outcomes	At the end of the course the student able to learn following:		
	CO1	CCN-105.1	Explain computer fundamentals
	CO2	CCN-105.2	Explain various flowchart, algorithm
	CO3	CCN-105.3	Apply the features of Basic Computer operating systems
	CO4	CCN-105.4	Explain functioning of various Hardware components and Network communication in computers
	CO5	CCN-105.5	Explain basics of data structures and emerging technologies in communication and computer network engineering.

Learning Outcomes:

1.0 Fundamentals of Digital Computer

- 1.1. Define various terms related to computers – Computer, Hardware, Software, Firmware, High Level Language , Low Level Language
- 1.2. Draw and explain block diagram of a computer in detail
- 1.3. Describe the current family of CPUs used in Computers.
- 1.4. State the use of storage devices used in a computer.
- 1.5. List the two types of memory used in a computer.
- 1.6. State the importance of cache memory.
- 1.7. Explain the generations of computers.
- 1.8. Classification of computers - based on a) size, b) processor.
- 1.9. State the importance of binary number system for use in Digital Computers

2.0 Implement Programming Methodology.

- 2.1. State the different steps involved in problem solving.
- 2.2. Define algorithm.
- 2.3. List four characteristics of algorithm.
- 2.4. Define a program
- 2.5. Differentiate between program and algorithm.
- 2.6. State the steps involved in algorithm development.
- 2.7. Differentiate between algorithm and flowchart.
- 2.8. Develop algorithms for simple problems.
- 2.9. Draw the symbols used in flowcharts.
- 2.10. Draw flowcharts for simple problems.

3.0 Operating Systems basics

- 3.1. Describe the need for an operating system.
- 3.2. List the various operating systems used presently.
- 3.3. List and explain
 - 3.3.1.Types of DOS commands
 - 3.3.2.Any 10 Internal Commands
 - 3.3.3.Any 5 External Commands
 - 3.3.4.Features of Windows desktop.
 - 3.3.5.Components of a Window.
- 3.4. State the function of each component of a Window.
- 3.5. Describe the Method of starting a program using start button
- 3.6. Explain usage of maximize, minimize, restore down and close buttons.
- 3.7. State the meaning of a file, folder.
- 3.8. Describe the Method of viewing the contents of hard disk drive using Explorer

- 3.9. Describe the Method of finding a file using search option.
- 3.10. Use control panel for
 - 3.10.1. installing and uninstalling software
 - 3.10.2. installing and uninstalling hardware
 - 3.10.3. Changing the system date and time
 - 3.10.4. Installing a printer
- 3.11. Explain Drive space using system tool option of Accessories group
- 3.12. Explain Disk defragmentation using System tools
- 3.13. Explain the procedure for changing resolution, color, appearance, screensaver options of the display

4.0 Computer Hardware, communication and Networking Basics

4.1 Hardware Basics

- 4.1.1 Identify hardware used for I/P, O/P & inside computer case, system board components used for communication among devices
- 4.1.2 Software - 3 types of Software: ROM BIOS, OS, application software
- 4.1.3 Explain Functions of BIOS
- 4.1.4 Explain boot process
- 4.1.5 Explain POST and important beep codes
- 4.1.6 Describe about different connectors.

4.2 Computer Communication Basics

- 4.2.1. Define Data communications
- 4.2.2. Explain the four fundamental characteristics of Data communication.
- 4.2.3. List the components of the Data communications.
- 4.2.4. Explain different ways of representing Data.
- 4.2.5. Explain Modes of communication (simplex, half-duplex, and full-duplex)

4.3 Networking Basics

- 4.3.1 Explain meaning of a computer network.
- 4.3.2 Describe the concept of a Local Area Network, Wide Area Network
- 4.3.3 Compare Internet and Intranet
- 4.3.4 Describe about internet service provider.
- 4.3.5 Explain the role of a modem in accessing the Internet.
- 4.3.6 Describe address format and IP address
- 4.3.7 What is browser and List various browsers
- 4.3.8 Explain the role of search engines with examples.
- 4.3.9 Explain Internet Security.

5.0 Basics of data structures and Emerging Computer Technologies.

5.1. Overview of Data Structures

- 5.1.1. Define Data structures.
- 5.1.2. Classify Data Structures.
- 5.1.3. Stack Concepts (Definition, operations, Applications)
- 5.1.4. Queue Concepts (Definition, operations, Applications)
- 5.1.5. List (Definition, Types-Single, Double, Circular, Operations-Insertion, Deletion, Search, Sort)
- 5.1.6. Define the terms Searching and sorting
- 5.1.7. Illustrate Linear search
- 5.1.8. List various sorting techniques
- 5.1.9. List the applications of searching and sorting.
- 5.2.1. Define the terms Tree, Binary Tree, Graph, Root, Internal node, Sibling node, LeafNode, Degree of a Node and Height of Tree
- 5.1.10. List the application of trees and graphs.

5.2. Introduction to Cloud fundamentals in Network Administration

5.2.1. Introduction to AWS

- 5.2.1.1. AWS IAM – Identity and Access Management
- 5.2.1.2. AWS VPC – Virtual Private Cloud
- 5.2.1.3. AWS EC2 - Elastic Cloud Computing
- 5.2.1.4. AWS EC2 – Load balancing

5.2.2. Introduction to Azure

- 5.2.2.1. Azure Cloud services Configuration
- 5.2.2.2. List Azure services
- 5.2.2.3. List Core Azure services
- 5.2.2.4. List Azure Identities and Governance

5.3. Basics of Ethical Hacking

- 5.3.1. Define Ethical Hacking and List the categories of Hackers
- 5.3.2. Describe Roles and responsibilities of Ethical Hackers
- 5.3.3. List and explain the phases in Ethical Hacking and Explain Penetrate testing

5.4. Introduction about Security threats in Networking

- 5.4.1 Backdoor
- 5.4.2 Logic Bomb
- 5.4.3 Trojan Horse
- 5.4.4 Virus
- 5.4.5 Worms

COURSE CONTENT

1.0 Fundamentals of Digital Computer

Block diagram of a digital computer, functional parameters of CPU, Clock speed and word length, Functional blocks of a CPU: ALU and Control unit, types of memory RAM, ROM, purpose of cache memory

2.0 Programming Methodology.

Steps involved in problem solving - Define algorithm, Program - Characteristics of algorithm - Differentiate between program and algorithm- Steps involved in algorithm development - Differentiate algorithm and flowchart - Algorithms for simple problems - Symbols used in flowcharts -Flowcharts for simple problems.

3.0 Understand Operating Systems

Need for an operating system - List the various operating systems - Types of commands, Internal & External Commands Features of Windows desktop - Components of a Window - Function of each component of a Window - Method of starting a program using start button -Maximize, minimize, restore down and close buttons- Meaning of a file and folder -Viewing the contents of hard disk drive using explorer -Finding a file - Formatting a floppy disk using explore option - Installing and uninstalling new software using control panel - installing and un installing a new hardware using control panel - Drive space - disk defragmentation - Installing a printer - Changing resolution, colour, appearance and screensaver options of the display - Changing the system date and time

4.0 Computer Hardware and Networking Basics

Hardware Basics- I/P, O/P - inside computer case- system board components - 3 types of Software - BIOS- boot process - POST - different connectors.

Data Communication -characteristics-components-Data Representation-Modes of communication.

Networking Basics - computer network - Local Area Network - Wide Area Network - Compare Internet and Intranet - internet service provider - role of a modem - address format and IP address - browser - search engines with examples -Describe Internet Security.

5.0 Basics of Data structures and Emerging Computer Technology

Overview of Data Structures-Definition-Classification- concepts of Stacks, Queues, Lists, Trees and Graphs.

Introduction to Cloud fundamentals in Network Administration- AWS-Azure

Basics of Ethical Hacking - categories of Hackers - Penetrate testing -Roles and responsibilities of Ethical Hackers-
phases in Ethical Hacking

REFERENCE BOOKS

1. Information Technology - Curtin.
2. Computer Science Theory & Application - E. Balaguruswamy, B. Sushila
3. Introduction to Computers (Special Indian Edition) - Peter Norton
4. <https://aws.amazon.com/>
5. <https://learn.microsoft.com/en-us/certifications/azure-fundamentals/>
8. Data Structures using 'C' - TanenbaumLangsam and Augenstein (PHI).

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.3
Unit test-2	From 3.4 to 4.1
Unit test-3	From 4.2 to 5.4

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-106	Programming in C	5	150	20	80

Time Schedule

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1	Introduction to C Language	20	16	2	1	CO1, CO2
2	Input and output statements, Operators and Expressions In C.	25	21	2	1 ½	CO1, CO2, CO3
3	Decision making, iterative and other control statements	40	21	2	1 ½	CO1, CO2, CO3
4	Arrays and strings, Structures and Unions	30	26	2	2	CO1, CO2, CO3
5	User defined functions, pointers, Dynamic memory allocation	35	26	2	2	CO1, CO2, CO3, CO4, CO5
Total		150	110	10	8	
Marks				30	80	

Course Objectives	Upon completion of the course the student shall be able to
	1. Relate basics of programming language constructs using C Language 2. Classify and implement data types, derived data types, pointers. 3. Analyse and develop effective modular programming 4. Construct mathematical, logical and scientific problems and real time applications using C-language

Course Outcomes	Upon completion of the course the student shall be able to		
	CO1	CCN-106.1	Develop, compile and debug programs using C-fundamentals and different programming statements in C language.
	CO2	CCN-106.2	Use various operations using primary and derived data types in C.
	CO3	CCN-106.3	Analyse programs using predefined functions, modules and recursive techniques
	CO4	CCN-106.4	Write scientific and logical programs using pointers, file pointers
	CO5	CCN-106.5	Develop programs using information passing

Learning Outcomes

1.0 Introduction to C-Language

- 1.1 Describe the history of C-language, structure of C-language program
- 1.2 Describe the programming style of C language
- 1.3 Explain the steps involved in Editing, compiling, executing and debugging of C program
- 1.4 Describe character set, C-Tokens, Keywords, Identifiers, Constants, Variables
- 1.5 Define Data Type
- 1.6 Classify Data Types and explain them with examples.
- 1.7 Explain declaration of constants and variables
- 1.8 Explain initializing values to variables in declaration
- 1.9 Explain about user defined data types with a simple program
- 1.10 Explain the usage of type qualifiers

2.0 Input and output statements, Operators and Expressions in C

- 2.1 Explain the importance of Pre-processor Directive #include.
- 2.2 Define an operator, an expression
- 2.3 Explain
 - 2.3.1 Various arithmetic operators and explain the evaluation of arithmetic expressions with example.
 - 2.3.2 Various relational operators and discuss evaluation of relational expressions.
 - 2.3.3 Various logical operators and discuss evaluation of logical expressions.
- 2.4 Explain the difference between unary and binary operators.
- 2.5 Describe various assignment operators, increment and decrement operators.
- 2.6 Illustrate nested assignment.
- 2.7 Explain conditional operators with an example.
- 2.8 Explain
 - 2.8.1 Bit-wise operators and explain each with an example.
 - 2.8.2 Special operators with examples.
 - 2.8.3 Precedence and Associativity of operators.
- 2.9 Describe evaluation of compound expression.
- 2.10 Illustrate type conversion techniques.
- 2.11 Write sample programs by using all the operators.
- 2.12 Illustrate
 - 2.12.1 Reading a character using getch(),getche() and getchar().

- 2.12.2 writing a character using `putch()`, `putchar()`.
- 2.12.3 Formatted input using `scanf()` & write sample programs using it.
- 2.12.4 Formatted output using `printf()` & write sample programs using it.
- 2.13 Explain character functions.

3.0 Decision making, iterative and other control statements

- 3.1 Explain decision making statements and its need in programming
 - 3.2 Explain
 - 3.2.1 Simple if and if-else statement with syntax and sample program
 - 3.2.2 Nested if..else statements with syntax and sample program
 - 3.2.3 if-else-if ladder with syntax and sample program
 - 3.2.4 switch statement with syntax and sample program
 - 3.3 State the importance of break statement with switch and illustrate
 - 3.4 Compare
 - 3.4.1 Conditional operator with if-else statement
 - 3.4.2 if-else with switch statement
 - 3.5 Define looping or iteration
 - 3.6 List and explain iterative statements with syntax and examples
 - 3.7 Compare different loop statements
 - 3.8 What is nested loop and illustrate.
 - 3.9 Explain the usage of goto, break and continue statements with loop statements
 - 3.10 Differentiate break and continue statements.
 - 3.11 Define structured programming.

4.0 Arrays, strings, Structures and Unions

- 4.1 Define Array
- 4.2 Explain
 - 4.2.1 Declaration and initialization of One Dimensional (1D) Array with syntax and sample programs.
 - 4.2.2 Accessing the elements in 1D-Array with sample programs.
 - 4.2.3 Reordering an array in ascending order.
- 4.3 Explain declaration and initialization and usage of two Dimensional (2D) Arrays.
- 4.4 Illustrate the concept of arrays with sample programs on matrix addition, subtraction and matrix multiplication
- 4.5 Define String
- 4.6 Explain
 - 4.6.1 Declare and initialize of String variables.
 - 4.6.2 `gets()` and `puts()`
 - 4.6.3 Reading and displaying of strings from terminal with sample programs.
 - 4.6.4 Explain about various String handling functions with sample programs.
- 4.7 Explain Character arithmetic.
- 4.8 Define a structure.
- 4.9 Explain
 - 4.10 Initializing structure, declaring structure, Declaring Structure Variables.
 - 4.11 Accessing of the structure members
 - 4.12 Structure assignment.
 - 4.13 How to find size of a structure.
 - 4.14 Nested structure concept.
 - 4.15 Structures containing arrays
 - 4.16 Array of structures
 - 4.17 Define Union, declare, initialize and use of union.
 - 4.18 Distinguish between Structures and Unions
 - 4.19 Write sample programs for all the concepts of structures and unions

5.0 User defined functions, pointers, file management and pre-processor directives

5.1 Explain

5.1.1 Need of user defined functions

5.1.2 Advantages of the functions

5.1.3 Elements of function

5.1.4 Return values and their types

5.2 Define a function call, function prototype

5.3 Explain

5.3.1 Function declaration in programs

5.3.2 Functions with no arguments and no return values with sample programs

5.3.3 Functions with arguments with no return values with sample programs

5.3.4 Functions with arguments with return values with sample programs

5.3.5 Functions with no arguments with return values with sample programs

5.3.6 Functions that return multiple values with sample programs

5.3.7 Recursion with sample programs

5.3.8 Passing arrays to functions with sample programs

5.3.9 Structure as function arguments and structures as function values.

5.3.10 Structures containing pointers.

5.3.11 Self-referential structures with examples.

5.3.12 Storage classes-auto, register, static, extern

5.3.13 Scope, visibility and lifetime of variables in functions

5.4 Differentiate Local and External variables

5.5 Define Global variable

5.6 Discuss passing the global variables as parameters using sample programs

5.7 Explain

5.7.1 Declaration and initialization of Pointers.

5.7.2 Accessing the address of a variable using &operator

5.7.3 Accessing the value of a variable through pointer

5.7.4 Pointer Arithmetic

5.7.5 Precedence of address and de-referencing operators.

5.7.6 Relationship between arrays and pointers.

5.7.7 Accessing array elements using pointers

5.7.8 Pointers as function arguments

5.7.9 Pointer arrays with examples.

5.8 Differentiate between address and de-referencing operators.

5.9 Explain

5.9.1 Dynamic memory management functions with examples.

5.9.2 Structures containing pointers.

5.9.3 Pointer to structure.

5.9.4 Self-referential structures with examples.

5.10 State Need of pre-processor directives.

COURSE CONTENT

- 1. Introduction to C Language:** History of C language - importance of C language - structure of C language - programming style of C language - steps involved in executing the C program-Character set - C Tokens - Keywords and Identifiers- Constants and Variables - Data Types and classification - declaration of constants and variables-initializing values to variables-user defined data types-usage of type qualifiers.
- 2. Input and output statements, Operators and Expressions in C:** Importance of Pre-processor #include-reading and writing a single character function- formatted input and output statements-operators-

classification of operators-operator precedence and associativity- expressions and expression evaluation-type conversion techniques.

3. **Understand Decision making, iterative and other control statements:** Simple if, if-else, if else ladder, nested if-else-switch statement - else if, nested if , else if ladder, switch statements- Classification of various loop statements- while statement – do.. while statement ram - for loop statement - nesting of loops- Comparisons of different loop statements –goto statement-break and continue statements – concept of structured programming
4. **Understand Arrays and strings, basics of Structures and Unions:** Arrays -One Dimensional Arrays – array programs -two Dimensional Arrays- programs on matrix - Strings – String handling functions - Structure- Array of structures - Nested structures- pointer to structure Self-referential structures - Union and illustrate use of a union – difference between Structures and Union
5. **Understand User defined functions, basics of pointers, file management and pre-processor directives:** Function – user defined functions – Advantages - Recursion concept - parameter passing –storage classes - scope, visibility and lifetime of variables in functions- Local and External variables -Global variable- - Pointer - Differentiate address and de-referencing operators - Pointer Arithmetic- precedence of address and de- referencing operators -
6. Relationship between Arrays and Pointers - Pointers as Function Arguments - Dynamic memory management-

REFERENCE BOOKS

- | | | | |
|---|--------------------------|------------------|------------------|
| 1 | Programming in ANSI C | E. Balaguruswamy | Tata McGrawHill |
| 2 | Programming with C | Gottfried | Tata McGraw Hill |
| 3 | C The complete Reference | Schildt | Tata McGrawHill |

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 2.13
Unit test-2	From 3.1 to 4.6
Unit test-3	From 4.7 to 5.12

Course code	Course Title	No. of periods /week	Total No. of periods	Marks for FA	Marks for SA
CCN-107	ENGINEERING DRAWING	3	90	40	60

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1	Use of Drawing Instruments, Free Hand Lettering and Dimensioning Practice	10	10	2	0	CO1, CO3, CO4
2	Principles of Geometric Constructions	15	15	1	1	CO2
3	Projections of points, lines, planes and solids	20	25	1	2	CO1, CO3
4	Sectional Views	20	10	0	1	CO1, CO4, CO5
5	Orthographic projection	25	20	0	2	CO2, CO4
Total		90	80	4	6	
Marks				20	40	

Course Objectives and Course Outcomes

Course Objectives	Upon completion of the course the student shall be able to understand the basic graphic skills and use them in preparation, reading and interpretation of engineering drawings.
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Course Outcomes	CO1	CN-107.1	Practice the use of engineering drawing instruments and Familiarise with the conventions to be followed in engineering drawing as per BIS
	CO2	CN-107.2	Construct the i) basic geometrical constructions ii) engineering curves
	CO3	CN-107.3	Visualise and draw the projections of i) Points ii) Lines iii) Regular Planes iv) Regular Solids
	CO4	CN-107.4	Visualise and draw the sectional views of components
	CO5	CN-107.5	Visualise and draw the orthographic projections of components

LEARNING OUTCOMES

Upon completion of the course the student shall able to

1.0 Use of Drawing Instruments, Free Hand Lettering and Dimensioning Practice

- 1.1 State the importance of drawing as an engineering communication medium
- 1.2 Select the correct instruments to draw the different lines / curves.
- 1.3 Use correct grade of pencil and other instruments to draw different types of lines and for different purposes
- 1.4 Identify the steps to be taken to keep the drawing clean and tidy.
- 1.5 Write titles using vertical and slopping (inclined) lettering and numerals of 7mm, 10mm and 14mm height.
- 1.6 Acquaint with the conventions, notations, rules and methods of dimensioning in engineering drawing as per the B.I.S.
- 1.7 Dimension a given drawing using standard notations and desired system of dimensioning.

2.0 Principles of Geometric Constructions

- 2.1 Practice the basic geometric constructions like i) dividing a line into equal parts
i) Exterior and interior tangents to the given two circles
ii) Tangent arcs to two given lines and arcs
- 2.2 Draw any regular polygon using general method when i) side length is given
i) Inscribing circle radius is given ii) describing circle radius is given
- 2.3 Draw the engineering curves like i) involute ii) cycloid

3.0 Projections of points, lines, planes and solids (All in first quadrant only)

- 3.1 Explain the basic principles of the orthographic projections
- 3.2 Visualise and draw the projection of a point with respect to reference planes (HP & VP)
- 3.3 Visualise and draw the projections of straight lines with respect to two reference Planes (up to lines parallel to one plane and inclined to other plane)
- 3.4 Visualise and draw the projections of planes (up to planes perpendicular to one plane and inclined to other plane)
- 3.5 Visualise and draw the projections of regular solids like Prisms, Pyramids, Cylinder, Cone (up to axis of solids parallel to one plane and inclined to other plane)

4.0 Sectional Views

- 4.1 Identify the need to draw sectional views.
- 4.2 Draw sectional views of regular solids by applying the principles of hatching.

5.0 Orthographic projection

- 5.1 Draw the orthographic views of an object from its pictorial drawing.
- 5.2 Draw the minimum number of views needed to represent a given object fully.

Competencies and Key competencies to be achieved by the student

S.No	Major topic	Key Competency
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1.	Use of Drawing Instruments, Free Hand Lettering and Dimensioning Practice	Explain the linkages between Engineering drawing and other subjects of study in Diploma course.
		Select the correct instruments to draw various entities in different orientation
		Write titles using sloping and vertical lettering and numerals as per B.I.S (Bureau of Indian standards)
		Dimension a given drawing using standard notations and desired system of dimensioning
2.	Geometrical construction	Dividing a line into equal parts, tangents to circles, Construct involute, cycloid from the given data.
3.	Projection of points, Lines, Planes & Solids	Draw the projections of points, straight lines, planes & solids with respect to reference planes (HP& VP)
4.	Sectional Views	- Differentiate between true shape and apparent shape of section - Apply principles of hatching. - Draw simple sections of regular solids
5.	Orthographic Projection	Draw the minimum number of views needed to represent a given object fully.

COURSE CONTENTS:

- NOTES:
1. B.I.S Specification should invariably be followed in all the topics.
 2. A-3 Size Drawing Sheets are to be used for all Drawing Practice Exercises.

1.0 Use of Drawing Instruments, Free Hand Lettering and Dimensioning Practice

Explanation of the scope and objectives of the subject of Engineering Drawing . Its importance as a graphic communication -Need for preparing drawing as per standards – SP-46 –1988 – Mention B.I.S - Role of drawing in -engineering education - Basic Tools, tools for drawing– Mentioning of names under each classification and their brief description -Scales: Recommended scales reduced & enlarged -Lines: Types of lines, selection of line thickness - Selection of Pencils -Sheet Sizes: A0, A1, A2, A3, A4, A5, Layout of drawing sheets in respect of A0, A1, A3 sizes, Sizes of the Title block and its contents - Care and maintenance of Drawing Sheet, Importance of lettering – Types of lettering -Guide Lines for Lettering Practicing of letters & numbers of given sizes (7mm, 10mm and 14mm)-Advantages of single stroke or simple style of lettering - Use of lettering stencils- Purpose of engineering Drawing, Need of B.I.S code in dimensioning -Shape description of an Engineering object -Definition of Dimensioning size description -Location of features, surface finish, fully dimensioned Drawing -Notations or tools of dimensioning, dimension line extension line, leader line, arrows, symbols, number and notes, rules to be observed in the use of above tools -Placing dimensions: Aligned system and unidirectional system (SP-46-1988)-Arrangement of dimensions Chain, parallel, combined progressive, and dimensioning by co-ordinate methods-The rules for dimensioning standard, features “Circles (holes) arcs, angles, tapers, chamfers, and dimension of narrow spaces.

2.0 Geometric Constructions

Division of a straight line into given number of equal parts –Drawing interior and exterior tangents to two circles of given radii and centre distance-Drawing tangent arc of given radius to touch two lines inclined at given angle (acute, right and obtuse angles), Tangent arc of given radius touching a circle or an arc and a given

line, Tangent arcs of radius R, touching two given circles internally and externally-Construction of any regular polygon by general method for given side length, inscribing circle radius and describing/superscripting circle radius - Involute, Cycloid, explanations as locus of a moving point, their engineering application, viz., Gear tooth profile, screw threads, springs etc. – their construction

3.0 Projection of points, lines and planes and Solids (All in first quadrant only)

Classification of projections, Observer, Object, Projectors, Projection, Reference Planes, Reference Line, Various angles of projections –Differences between first angle and third angle projections

Projections of points -Projections of straight line –(a) Parallel to both the planes, (b)Perpendicular to one of the planes and (c) Inclined to one plane and parallel to other planes-Projections of regular planes-(a) Plane parallel to one of the reference planes, (b) Plane perpendicular to HP and inclined to VP and vice versa-Projections of regular solids- (a) Axis perpendicular to one of the planes, (b) Axis parallel to VP and inclined to HP and vice versa.

4.0 Sectional Views

Need for drawing sectional views – what is a sectional view - Hatching – Section of regular solids inclined to one plane and parallel to other plane

5.0 Orthographic Projections

Meaning of orthographic projection - Using a viewing box and a model – Number of views obtained on the six faces of the box, - Legible sketches of only 3 views for describing object -Concept of front view, top view, and side view sketching these views for a number of engineering objects - Explanation of first angle projection. – Positioning of three views in First angle projection -Projection of points as a means of locating the corners of the surfaces of an object – Use of meter line in drawing a third view when other two views are given -Method of representing hidden lines -Selection of minimum number of views to describe an object fully.

Unit Test	Learning Outcomes to be Covered
Unit Test – I	From 1.1 to 2.3
Unit Test – II	From 3.1 to 3.5
Unit Test – III	From 4.1 to 5.2

REFERENCE BOOKS

- 1 Engineering Graphics by P I Varghese – (McGraw-hill)
- 2 Engineering Drawing by Basant Agarwal & C.M Agarwal - (McGraw-hill)
- 3 Engineering Drawing by N.D.Bhatt.
- 4 T.S.M. & S.S.M on “ Technical Drawing” prepared by T.T.T.I., Madras.
- 5 SP-46-1998 – Bureau of Indian Standards.

Course Code	Course title	No of periods/week	Total no of periods	Marks for FA	Marks for SA
CCN-108	Programming in C Lab	06	180	40	60

S No	Chapter/ Unit Title	No. of Periods	COs Mapped
1.	Fundamentals and Input / Output statements	15	CO1, CO2
2.	Control statements	45	CO1, CO2, CO4
3.	Arrays, structures and unions	60	CO1, CO2, CO3, CO4
4.	User defined functions, storage classes, pointers	60	CO1,CO2,CO3,CO4,CO5, CO6
	Total	180	

COURSE OBJECTIVES	Upon completion of the course the student shall be able to 1. Edit, compile and debug execution of C-Programs. 2. Learn the syntax of all the statements, keywords, user defined identifiers and usage of writing statements in C-Program. 3. Evaluate all the expressions using different primary types of data, derived data, operators and with their precedence. 4. Write C-programs using I/O statements, decision making statements. 5. Write structured and modular C-programs. 6. Write C-programs to implement dynamic memory allocation using pointer concepts.
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Course Outcomes		Upon completion of the course the student shall be able to
CO 1	CCN-108.1	Perform Edit, compile and debug and execution of C-Programs.
CO 2	CCN-108.2	Develop programs using different predefined functions, keywords, user defined identifiers.
CO 3	CCN-108.3	Write different expressions using available C-operators and valid data supported by C-language.
CO 4	CCN-108.4	Develop C-programs using control statements, array's, structures, unions.
CO 5	CCN-108.5	Develop C-programs using user defined functions and recursion.
CO 6	CCN-108.6	Develop C-programs to implement dynamic memory concept.

1. Fundamentals and Input / Output statements

1. Exercise on structure of C Program.
2. Exercise on Keywords and identifiers.
3. Exercise on constants and variables.
4. Execution of simple C program.
5. Exercise on operators and expressions.
6. Exercise on special operators.
7. Exercise on input and output of characters.
8. Exercise on formatted input and output.
9. Exercise on escape sequence characters.

2. Control statements

(Note: Every statement must be repeated with at least 5 different applications)

1. Exercise on simple if statement.
2. Exercise on if..else statement.
3. Exercise on if..else..if ladder statement.
4. Exercise on switch statement.
5. Exercise on conditional operator comparing with if-else statement.
6. Exercise on while statement.
7. Exercise on for statement.
8. Exercise on do. While statement.

3. Arrays, structures and unions

1. Exercise on one dimensional arrays
2. Exercise on two dimensional arrays
3. Exercise on strings
4. Exercise on structure
5. Exercise on union
6. Exercise on array of structures

4. User defined functions, storage classes, pointers.

1. Exercise on user-defined function
2. Exercise on storage classes
3. Exercise on parameter passing techniques
4. Exercise on recursion
5. Exercise on pointers

The competencies and key competencies to be achieved by the student

S.N o.	Name of the experiment	Objectives	Key Competencies
1	Exercise on structure of C program	For a given C program, identify the different building blocks	❖ Identify different building block in a C program
2	Exercise on Keywords and identifiers	For a given C program identify the keywords and identifiers	❖ Identify different keywords ❖ Check whether the keywords are in lowercase ❖ Differentiate identifiers and keywords
3	Exercise on constants and variables	For a given C program identify the constants and variables	❖ Identify the constants ❖ Identify the variables ❖ Declare variables with proper names ❖ Know the assignment of values to variables

4	Execution of simple C program	Execute a simple C program	❖ Acquaint with C program editing ❖ Compile the program ❖ Rectify the syntactical errors ❖ Execute the program
5	Exercise on operators and expressions	Write a C program that uses different arithmetic operators	❖ Identify different arithmetic operators ❖ Build arithmetic expressions ❖ Identify the priorities of operators ❖ Evaluate arithmetic expression ❖ Compile the program ❖ Rectify the syntactical errors ❖ Execute the program ❖ Check the output for its correctness
6	Exercise on special operators	Write a C program that uses special operators	❖ Identify different special operators ❖ Build expressions using special operators ❖ Compile the program ❖ Rectify the syntactical errors ❖ Execute the program ❖ Check the output for its correctness
7	Exercise on input and output of characters	Write a C program for reading and writing characters	❖ Know the use of get char() function ❖ Know the use of put char() function ❖ Compile the program ❖ Rectify the syntactical errors ❖ Execute the program ❖ Check whether the correct output is printed for the given input
8	Exercise on formatted input and output	Write a C program using formatted input and formatted output	❖ Know the use of format string for different types of data in scan f() function ❖ Know the use of format string for different types of data in print f() function ❖ Check whether the data is read in correct format ❖ Check whether the data is printed in correct format
9	Exercise on Escape Sequence Characters	Write a C program using Escape Sequence Characters	❖ Know the use of Escape sequence characters ❖ Use the Escape sequence characters ❖ Check whether the data is read in correct format ❖ Rectify the syntax errors ❖ Check the output for correctness
10	Exercise on simple if statement	Write a C program using simple if statement	❖ Build a relational expression ❖ Use the if statement for decision making ❖ Rectify the syntax errors ❖ Check the output for correctness
11	Exercise on if..else statement	Write a C program using if..else statement	❖ Build a relational expression ❖ Use the if..else statement for decision making ❖ Rectify the syntax errors ❖ Check the output for correctness
12	Exercise on else..if ladder statement	Write a C program using else..if ladder statement	❖ Use else..if ladder statements with correct syntax ❖ Rectify the syntax errors ❖ Debug logical errors

			❖ Check the output for correctness
13	Exercise on switch statement	Write a C program using switch statement	❖ Use switch statement with correct syntax ❖ Identify the differences between switch and else..if ladder ❖ Rectify the syntax errors ❖ Debug logical errors ❖ Check the output for correctness
14	Exercise on conditional operator	Write a C program using (? :) conditional operator	❖ Build the three expressions for conditional operator ❖ Use conditional operator with correct syntax ❖ Rectify the syntax errors ❖ Debug logical errors ❖ Differentiate conditional operator and if..else statement
15	Exercise on while statement	Write a C program using while statement	❖ Build the termination condition for looping ❖ Use while statement with correct syntax ❖ Check whether correct number of iterations are performed by the while loop ❖ Rectify the syntax errors ❖ Debug logical errors
16	Exercise on for statement	Write a C program using for statement	❖ Build the initial, increment and termination conditions for looping ❖ Use for statement with correct syntax ❖ Rectify the syntax errors ❖ Debug logical errors ❖ Check whether correct number of iterations are performed by the for loop ❖ Differentiate for and while statements
17	Exercise on do..while statement	Write a C program using do statement	❖ Build the termination condition for looping ❖ Use do statement with correct syntax ❖ Rectify the syntax errors ❖ Debug logical errors ❖ Check whether correct number of iterations are performed by the while loop ❖ Differentiate do..while , while and for statements
18	Exercise on one dimensional arrays	Write a C program to create and access one dimensional array	❖ Create a one-dimensional array with correct syntax ❖ Store elements into array ❖ Read elements from array ❖ Validate boundary conditions while accessing elements of array ❖ Rectify the syntax errors ❖ Debug logical errors ❖ Check for the correctness of output for the given input
19	Exercise on two dimensional arrays	Write a C program to create and access two-dimensional array	❖ Create a two-dimensional array with correct syntax ❖ Store elements into array ❖ Read elements from array ❖ Validate boundary conditions while accessing elements of array

			❖ Rectify the syntax errors ❖ Debug logical errors ❖ Check for the correctness of output for the given input
20	Exercise on strings	Write a C program for reading and writing strings	❖ Declare and initialize string variables ❖ Read strings from keyboard ❖ Print strings to screen
21	Exercise on structure	Write a C program using structure	❖ Define a structure with correct syntax ❖ Identify different members of a structure ❖ Declare a structure variable ❖ Access different members of structure ❖ Observe the size of the structure ❖ Rectify the syntax errors ❖ Debug logical errors ❖ Check for the correctness of output for the given input
22	Exercise on union	Write a C program using union	❖ Define a union with correct syntax ❖ Identify different members of a union ❖ Declare a union variable ❖ Access different members of union ❖ Observe the size of the union ❖ Rectify the syntax errors ❖ Debug logical errors ❖ Check for the correctness of output for the given input
23	Exercise on array of structures	Write a C program to create an array of structures and store and retrieve data from that array	❖ Define a structure with correct syntax ❖ Identify different members of a structure ❖ Declare a structure variable ❖ Create an array of structure ❖ Access individual element of the array of structure ❖ Access different members of structure ❖ Rectify the syntax errors ❖ Debug logical errors ❖ Check for the correctness of output for the given input
24	Exercise on user-defined function	Write a C program to define and call user-defined functions	❖ Identify the different parts of function declaration ❖ Define function with correct syntax ❖ Classify functions based on it parameters and return types ❖ Identify parameters passed ❖ Identify parameter passing method used ❖ Identify return value ❖ Rectify the syntax errors ❖ Debug logical errors ❖ Check for the correctness of output for the given input
25	Exercise on storage classes	Write a C program using different storage classes	❖ Know the use of different storage classes ❖ Use the different storage classes

			❖ Check whether the scope of variables is correctly defined or not. ❖ Rectify the syntax errors ❖ Check the output for correctness
26	Exercise on parameter passing techniques	Write a C program using parameter passing techniques	❖ Know the use of parameter passing ❖ Use the different parameter passing techniques ❖ Check whether the parameters passed correctly or not. ❖ Rectify the syntax errors ❖ Check the output for correctness
27	Exercise on recursion	Write a C program using recursion	❖ Identify where recursive call is made in the function ❖ Validate the termination condition ❖ Rectify the syntax errors ❖ Debug logical errors ❖ Check for the correctness of output for the given input
28	Exercise on pointers	Write a C program using pointer data type	❖ Declare pointer variable ❖ Initialize pointer variable ❖ Access a variable through its pointer ❖ Rectify the syntax errors ❖ Debug logical errors ❖ Check for the correctness of output for the given input

Course Code	Course title	No of periods/week	Total no of periods	Marks for FA	Marks for SA
CCN-109	Engineering Physics Lab	03	45	40	60

Course objectives	(1) To provide strong practical knowledge of Physics to serve as a tool for various device applications in Engineering. (2) To enhance scientific skills of the students by incorporating new experiments so as to enrich the technical expertise of the students as required for industries.
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COURSE OUTCOMES	CO1	Improving accuracy in various measurements; understanding the nature of the forces keeping the body in equilibrium.
	CO2	Estimating the acceleration caused by the gravity of earth; Practical study of the concepts of refraction of light at curved/plane surface
	CO3	Understanding the pressure of the gas as function of its volume; study of the combined magnetic field of the earth and an artificial magnet to estimate its pole strength; Estimating the velocity of sound in air through resonance phenomenon.
	CO4	Applying Kirch off's laws to evaluate the specific resistance of a wire; Study of exchange of heat from system to surrounding by graphical analysis; Conversion of light energy to micro currents as potential engineering application.

CO-PO MAPPING

CO	PO 1	PO2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3	2	2	2	2	1	2
CO2	3		1	1	1	1	1
CO3	3	2			1		
CO4	3	2	2			1	2

CO-PO Mapping Strength

Course code Common - 109	Engineering Physics No of Course Objectives : 4				No of periods 45
POs	Mapped with CO No	CO periods addressing PO in Col NO %		1,2,3	remarks
PO1	CO1,CO2,CO3,CO4	15	33.3 %	2	>40% level 3 (highly addressed) 25% to 40% level 2 (moderately addressed) 25% level 1 (Low addressed) < 5% (not addressed)
PO2	CO1,CO3, CO4	8	17.8%	1	
PO3	CO1, CO2, CO4	6	13.3%	1	
PO4	CO1, CO2	3	6.7%	1	
PO5	CO1,CO2, CO3	5	11.1%	1	
PO6	CO1, CO2, CO4	3	6.7%	1	
PO7	CO1, CO2, CO4	5	11.1%	1	

3 = strongly mapped, 2 = moderately mapped, 1 = slightly mapped

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following.

- (i) Seminars (ii) Viva-voce (iii) Assignments
 (iv) Quiz competitions (v) Industrial visits (vi) Tech fest (vii) Mini project
 (viii) Group discussions (ix) Virtual labs (x) Library visit for e-books

ENGINEERING PHYSICS LAB

Subject Title : Engineering Physics Lab
 Subject Code : Common-109
 Periods per week : 03
 Total periods per year : 45

TIMESCHEDULE

S.No	List of experiments	No.of Periods
1.	Vernier calipers	03
2.	Micrometer (Screw gauge)	03
3.	Verification of Lami's theorem using concurrent forces	03
4.	Determination of 'g' using simple pendulum	03
5.	Focal length and focal power of convex lens	03

6.	Refractive index of solid using travelling microscope	03
7.	Verification of Boyle's law using Quill tube	03
8	Determination of pole strength of the bar magnet through magnetic field lines	03
9	Resonance apparatus – Determination of velocity of sound in air	03
	Experiments for demonstration	
10	Meter bridge – Determination of resistance and specific resistance of a wire	03
11	Verification of Newton's law of cooling	03
12	Photo electric cell – Study of its characteristics	03
	Revision	06
	Test	03
	Total:	45

Learning Outcomes

Upon completion of the course the student shall be able to

- 1.0 Practice with Vernier calipers to determine the volumes of cylinder and sphere.
- 2.0 Practice with Screw gauge to determine thickness of a glass plate and cross sectional area of a wire.
- 3.0 Verify the Lami's theorem using concurrent forces.
- 4.0 Determine the value of acceleration due to gravity (g) using Simple Pendulum. To verify the result from $l-T^2$ graph.
- 5.0 Calculate the Focal length and focal power of convex lens using distant object method and U-V method. To verify the result from U-V graph and $1/U - 1/V$ graph methods.
- 6.0 Determine the refractive index of a solid using travelling microscope
- 7.0 Verify the Boyle's law using Quill tube. To draw a graph between P and $1/l$.
- 8.0 Determination of magnetic pole strength of a bar magnet by drawing magnetic lines of force and locating null points (either N - N or N - S method)
- 9.0 Determine the velocity of sound in air at room temperature and its value at zero degree Centigrade using resonance apparatus.
- 10.0 Determine the resistance and specific resistance of material of a wire using Meter Bridge
- 11.0 To verify the Newton's law of cooling.
- 12..0 To study the characteristics of photo electric cell.

Course Outcomes

S.No	List of experiments	No.of Periods	COs
1.	Vernier calipers	03	

2.	Micrometer (Screw gauge)	03	CO1
3.	Verification of Lami's theorem using concurrent forces	03	
4.	Determination of g using simple pendulum	03	CO2
5.	Focal length and power of convex lens	03	
6.	Refractive index of solid using travelling microscope	03	
7.	Verification of Boyle's law using Quill tube	03	CO3
8.	Determination of pole strength of the bar magnet through magnetic field lines	03	
9.	Resonance apparatus – Determination of velocity of sound in air	03	
10.	Meter bridge – Determination of resistance and specific resistance of a wire	03	CO4
11.	Verification of Newton's law of cooling	03	
12.	Photo electric cell – Study of its characteristics	03	

Competencies and Key competencies to be achieved by the student

Name of the Experiment (Nu)	Competencies	Key competencies
1 . Practice on Vernier Calipers (03)	- Find the Least count - Fix the specimen in position - Read the scales - Calculate the physical quantities of given object	- Read the scales - Calculate the requisite physical quantities of given objects - Calculating volumes of the cylinder and sphere
2. Practice on Screw gauge(03)	- Find the Least count - Fix the specimen in position - Read the scales - Calculate thickness of glass plate and cross section of wire from radius	- Read the scales - Noting zero error - Calculate thickness of given glass plate - Calculate cross section of wire from radius

3. Verification of Lami's theorem forces(03)	• Making experimental set up • Fix suitable weights • Note the positions of threads on drawing sheet • Find the angles between the concurrent forces • Changing weights appropriately • Verify Lami's theorem	• Measuring angles between the forces • Marking the directions forces on a paper • Verifying Lami's theorem from the weights and measured angles between the forces.
4. Simple pendulum(03)	• Fix the simple pendulum to the stand • Adjust the length of pendulum • Find the time for number of oscillations (say 20) • Find the time period • Calculate the acceleration due to gravity • Draw $I-T^2$ graph	• Find the time for number of oscillations • Find the time period • Calculate the acceleration due to gravity • Verify form $I-T^2$ graph
5. Focal length and Focal power of convex lens (03)	• Fix the object distance • Find the Image distance • Calculate the focal length and power of convex lens • Draw u-v and $1/u - 1/v$ graphs	• Find focal length from distant object method. • Calculate the focal length and power of convex lens • Verify result from u-v and $1/u - 1/v$ graphs
6 Refractive index of solid using traveling microscope(03)	• Find the least count of Vernier on microscope • Place the graph paper below microscope • Read the scales	• Reading the scales on Microscope. • Finding real and apparent thickness of the slab • Calculate the refractive index
7 . Boyle's law verification (03)	• Note the atmospheric pressure • Fix the Quill tube to retort stand • Find the length of air column • Find the pressure of enclosed air • Find and compare the calculated values of $P \times l$	• Fixing Quill tube in various positions on retort stand. • Find the length of air column • Find the pressure of enclosed air • Find the values of $P \times l$ • Verify Boyle's law.

8. Mapping of magnet lines of force (03)	• Draw magnetic meridian • Place the bar magnet in N-N or N-S directions • Draw magnetic lines of force • Locate the neutral points	• Draw the pattern of magnetic lines of force • Locate the neutral points • Calculating pole strength of the bar magnet
9. Velocity of sound in air – Resonance method (03)	• Arrange the resonance apparatus • Adjust the reservoir level for booming sound • Find the first and second resonating lengths • Calculate velocity of sound .	• Adjust the reservoir level • Find the first and second resonating lengths • Calculate velocity of sound at room temperature and at 0° C
10. Meter bridge(03)	• Make the circuit connections • Find the balancing length • Calculate unknown resistance • Find the radius of wire • Calculate the specific resistance	• Making connections as per circuit diagram. • Find the balancing length • Calculate unknown resistance • Calculate the specific resistance of the given wire
11. Verification of Newton's law of Cooling (03)	• Heating liquid in a beaker using a heating element • Inserting thermometer in liquid in calorimeter • Stirring liquid • Measuring temperatures as a function of time using thermometer • Plotting a cooling curve	• Measuring temperature of a liquid as function of time. • Plotting a cooling curve. • Verifying Newton's law of cooling.
12. Photo electric cell – Study of its Characteristics (03)	• Experimental set up and making connections • Verifying intensity of light by varying distances between light source and photocell. • Measuring Voltage and current values.	• Making connections for experimental set up. • Varying distances appropriately • Measuring Voltage and current values. • Study of V- I Characteristics form graph.

Scheme of Valuation for End Practical Examination :

Activity	Marks
For writing, Apparatus, formulae, least count (if applicable)	5
Procedure & precautions	5
Drawing Tables	3
Readings, calculations, graph (if applicable), reporting the findings	12
Viva-voce	5
Total marks	30

Course Code	Course title	No of periods/week	Total no of periods	Marks for FA	Marks for SA
CCN-110	Chemistry Laboratory	03	45	40	60

CO1	Operate and practice volumetric apparatus and preparation of standard solution.
CO2	Evaluate and judge the neutralization point in acid base titration.
CO3	Evaluate the end point of reduction and oxidation reaction.
CO4	Judge the stable end point of complex formation, stable precipitation.
CO5	Judge operate and demonstrate and perform precise operations with instrument for investigation of water pollution parameters.

PO- CO mapping

Course code Common-110	Chemistry Laboratory No. of CO's:5				o. of periods : 45
POs	Addressed with CO No.	No of periods addressing	%	Level 1,2,3	Remarks

		PO in Col. No. 1			
PO1	CO1,CO2,CO3,CO4,CO5	12	.66	2	3 (highly addressed) 40% 2 (moderately addressed) 25% (Low addressed) not addressed)
PO2	CO1,CO2,CO3,CO4,CO5	9	0	1	
PO3					
PO4	CO1,CO2,CO3,CO4,CO5	12	.66	2	
PO5	CO2,CO3,CO4,CO5	12	.66	2	
PO6	-	-	-	-	
PO7	-	-	-	-	

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	2	1	-	2	-	-	-	-	-	-
CO2	2	1	-	2	2	-	-	-	-	-
CO3	2	1	-	2	2	-	-	-	-	-
CO4	2	1	-	2	2	-	-	-	-	-
CO5	2	1	-	2	2	-	-	-	-	-

COs-POs mapping strength (as per given table)

3=strongly mapped 2= moderately mapped 1= slightly mapped

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following: i) Seminars ii) Tutorials iii) Guest Lectures iv) Assignments v) Quiz competitions vi) Industrial visit vii) Tech Fest viii) Mini project ix) Group discussions x) Virtual classes xi) Library visit for e-books

TIME SCHEDULE

S.No	Name of the Experiment	No. of Periods	Mapped with COs
1.	a) Recognition of chemical substances and solutions used in the laboratory by senses. b) Familiarization of methods for Volumetric analysis.	03	CO1
2.	Preparation of Std. Na_2CO_3 solution and making solutions of different dilutions	03	CO1
3.	Estimation of HCl solution using Std. Na_2CO_3 solution.	03	CO2
4.	Estimation of NaOH using Std. HCl solution.	03	CO2
5.	Determination of acidity of water sample.	03	CO2
6.	Determination of alkalinity of water sample.	03	CO2
7.	Estimation of Mohr's Salt using Std. KMnO_4 Solution.	03	CO3
8.	Estimation of Ferrous ion by using Std. $\text{K}_2\text{Cr}_2\text{O}_7$ solution.	03	CO3
9.	Determination of total hardness of water sample using Std. EDTA solution.	03	CO4
10.	Estimation of Chlorides present in water sample by using Std. AgNO_3 solution.	03	CO4
11.	Estimation of Dissolved Oxygen(D.O) in water sample by using Std. hypo solution.	03	CO5
12.	Determination of pH using pH meter..	03	CO5
13.	Determination of conductivity of water and adjusting ionic strength required	03	CO5
14.	Determination of turbidity of water.	03	CO5
15.	Estimation of total solids present in water sample.	03	CO5
	Total:	45	

Objectives:

Upon completion of the course the student shall be able to

- 1.0 To identify the chemical compounds and solutions by senses.
Practice volumetric measurements (using pipettes, measuring jars, volumetric flask, burettes) and gravimetric measurements (using different types of balances), making dilutions, etc.
- 2.0 Practice making standard solutions with pre weighed salts and to make solutions of desired dilutions using appropriate techniques.
- 3.0 Conduct titrations adopting standard procedures and using Std. Na_2CO_3 solution for estimation of HCl.
- 4.0 Conduct titrations adopting standard procedures and using Std. HCl solution for estimation of NaOH.
- 5.0 Conduct titrations adopting standard procedures to determine the acidity of given samples of water (One ground water and one surface / tap water, and rain water if available).
- 6.0 Conduct titrations adopting standard procedures to determine the alkalinity of given samples of water (One ground water and one surface / tap water).
- 7.0 Conduct titrations adopting standard procedures and using Std. KMnO_4 solution for estimation of Mohr's Salt.
- 8.0 Conduct titrations adopting standard procedures and using Std. $\text{K}_2\text{Cr}_2\text{O}_7$ solution for estimation of Ferrous ion.
- 9.0 Conduct titrations adopting standard procedures to determine the total hardness of given samples of water (One ground water and one surface / tap water) using Std. EDTA solution.
10. Conduct titrations adopting standard procedures to determine the chlorides present in the given samples of water (One ground water and one surface / tap water) using Std. AgNO_3 solution.
11. Conduct the test using titrimetric / electrometric method to determine. Dissolved Oxygen (D.O) in the given water samples (One sample from closed container and one from open container / tap water) by Std. Hypo solution.
12. Conduct the test on given samples of water / solutions (like soft drinks, sewage, etc.) to determine their pH using standard pH meter.
13. Conduct the test on given samples of water / solutions.
 - a) to determine conductivity.
 - b) to adjust the ionic strength of the sample to the desired value.
14. Conduct the test on given samples of solutions (coloured and non-coloured) to determine their turbidity in NTU.
15. Determine the total solids present in given samples of water (One ground water and one surface / tap water).

Competencies and Key competencies to be achieved by the student

Name of the Experiment (No of Periods)	Competencies	Key competencies
Recognition of chemical substances and solutions. Familiarization of methods for Volumetric analysis. (03)	-	--
Preparation of Std. Na_2CO_3 solution and making solutions of different dilutions. (03)	▪ Weighing the salt to the accuracy of .01 mg. ▪ Measuring the water with volumetric flask, measuring jar, volumetric pipette and graduated pipette. ▪ Making appropriate dilutions.	▪ Weighing the salt to the accuracy of 0.01 mg. ▪ Measuring the water with volumetric flask, measuring jar, volumetric pipette and graduated pipette. ▪ Making appropriate dilutions.
Estimation of HCl solution using Std. Na_2CO_3 solution. (03)	▪ Cleaning the glassware and rinsing with appropriate solutions. ▪ Making standard solutions. ▪ Measuring accurately the standard solutions and titrants. ▪ Filling the burette with titrant. ▪ Fixing the burette to the stand. ▪ Effectively Controlling the flow of the titrant. ▪ Identifying the end point. ▪ Making accurate observations. ▪ Calculating the results.	▪ Making standard solutions. ▪ Measuring accurately the standard solutions and titrants. ▪ Effectively Controlling the flow of the titrant. ▪ Identifying the end point. ▪ Making accurate observations.
Estimation of NaOH using Std. HCl solution. (03)		
Determination of acidity of water sample. (03)		
Determination of alkalinity of water sample. (03)		
Estimation of Mohr's Salt using Std. KMnO_4 solution. (03)		
Estimation of Ferrous ion by using Std. $\text{K}_2\text{Cr}_2\text{O}_7$ solution (03)		
Determination of total hardness of water using Std. EDTA solution. (03)		
Estimation of Chlorides present in water sample using Std. AgNO_3 solution (03)		
Estimation of Dissolved Oxygen(D.O) in water sample (By titration method) (03)		
Determination of pH using pH meter. (03)	▪ Familiarize with instrument. ▪ Choose appropriate 'Mode' / 'Unit'. ▪ Prepare standard solutions / buffers, etc. ▪ Standardize the instrument with	▪ Prepare standard solutions / buffers, etc. ▪ Standardize the instrument with appropriate standard solutions. ▪ Plot the standard curve.
Determination of conductivity of water and adjusting ionic strength to required level. (03)		

Determination of turbidity of water. (03)	appropriate standard solutions. ▪ Plot the standard curve. ▪ Make measurements accurately. ▪ Follow Safety precautions.	▪ Make measurements accurately.
Estimation of total solids present in water sample. (03)	▪ Measuring the accurate volume and weight of sample. ▪ Filtering and air drying without losing any filtrate. ▪ Accurately weighing the filter paper, crucible and filtrate. ▪ Drying the crucible in an oven.	▪ Measuring the accurate volume and weight of sample. ▪ Filtering and air drying without losing any filtrate. ▪ Accurately weighing the filter paper, crucible and filtrate.

SCHEME OF VALUATION

A) Writing Chemicals, apparatus, principle and procedure.	5M
B) Demonstrated competencies.	20M
Making standard solutions.	
Measuring accurately the standard solutions and titrants.	
Effectively controlling the flow of the titrant.	
Identifying the end point.	
Making accurate observations.	
C) Viva-voce.	5M
Total	30M

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-111	Computer Fundamentals Lab	4	120	40	60

Time schedule:

S.No.	Chapter/Unit Title	No. of sessions each of 3 periods duration	No. of Periods	CO's Mapped
1.	Computer hardware Basics	2	8	CO 1
2.	Windows Operating System	2	8	CO 1
3.	MS Word	8	32	CO 2
4.	MS Excel	7	28	CO 3
5.	MS PowerPoint	5	20	CO 4
6	Adobe Photoshop	6	24	CO 5
Total periods		30	120	

Course Objectives	i) To know Hardware Basics ii) To familiarize operating systems iii) To use MS Office effectively to enable to students use these skills in future courses iv) To use Adobe Photoshop in image editing.
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Course Outcomes	At the end of the course students will be able to		
	CO1	CCN-111.1	Identify hardware and software components
	CO2	CCN-111.2	Prepare documents with given specifications using word processing software
	CO3	CCN-111.3	Use Spread sheet software to make calculation and to draw various graphs / charts.

	CO4	CCN-111.4	Use Power point software to develop effective presentation for a given theme or topic.
	CO5	CCN-111.5	Edit digital or scanned images using Photoshop

Learning Outcomes:

I. Computer Hardware Basics

1. a)To Familiarize with Computer system and hardware connections
- b.To Start and Shut down Computer correctly
- c)To check the software details of the computer
- To check the hardware present in your computer

II. Windows's operating system

3. To Explore Windows Desktop
4. Working with Files and Folders
5. Windows Accessories: Calculator – Notepad – WordPad – MS Paint

III. Practice with MS-WORD

6. To familiarize with Ribbon layout of MS Word
- Home – Insert- Page layout – References – Review- View.
- 7.To practice Word Processing Basics
8. To practice Formatting techniques
9. To insert a table of required number of rows and columns
10. To insert Objects, Clipart and Hyperlinks
11. To use Mail Merge feature of MS Word
12. To use Equations and symbols features

IV. Practice with MS-EXCEL

13. To familiarize with MS-EXCEL layout
14. To access and enter data in the cells
15. To edit a spread sheet- Copy, Cut, Paste, and selecting Cells
16. To use built in functions and Formatting Data
17. To create Excel Functions, Filling Cells
18. To enter a Formula for automatic calculations
19. To sort and filter data in table.
20. To present data using Excel Graphs and Charts.
21. To develop lab reports of respective discipline.
22. To format a Worksheet in Excel, Page Setup and Print

V. Practice with MS-POWERPOINT

23. To familiarize with Ribbon layout features of PowerPoint 2007.
24. To create a simple PowerPoint Presentation
25. To set up a Master Slide in PowerPoint
26. To insert Text and Objects
27. To insert a Flow Charts
28. To insert a Table
29. To insert a Charts/Graphs

30. To insert video and audio
31. To practice Animating text and objects
32. To Review presentation

VI. Practice with Adobe Photoshop

33. To familiarize with standard toolbox
34. To edit a photograph.
35. To insert Borders around photograph.
36. To change Background of a Photograph.
37. To change colors of Photograph.
38. To prepare a cover page for the book in your subject area.
39. To adjust the brightness and contrast of the picture so that it gives an elegant look.
40. To type a word and apply the shadow emboss effects.

Key competencies:

Expt. No	Name of Experiment	Competencies	Key competencies
1 (a).	To familiarize with Computer system and hardware connections	a. Identify the parts of a computer system: i). CPU ii). Mother Board iii) Monitor iv) CD/DVD Drive v) Power Switch vi) Start Button vii) Reset Button b. Identify and connect various peripherals c. Identify and connect the cables used with computer system d. Identify various ports on CPU and connect Keyboard & Mouse	Connect cables to external hardware and operate the computer
1 (b).	To Start and Shut down Computer correctly	a. Log in using the password b. Start and shut down the computer c. Use Mouse and Key Board	a. Login and logout as per the standard procedure b. Operate mouse & Key Board
1 (c).	To Explore Windows Desktop	a. Familiarize with Start Menu, Taskbar, Icons and Shortcuts b. Access application programs using Start menu, Task manager c. Use Help support	a. Access application programs using Start menu b. Use taskbar and Task manager
2.	To check the software details of the computer	a. Find the details of Operating System being used b. Find the details of Service Pack installed	Access the properties of computer and find the details

3.	To check the hardware present in your computer	a. Find the CPU name and clock speed b. Find the details of RAM and Hard disk present c. Access Device manager using Control Panel and check the status of devices like mouse and key board d. Use My Computer to check the details of Hard drives and partitions e. Use the Taskbar	a. Access device manager and find the details b. Type /Navigate the correct path and Select icon related to the details required
4.	Working with Files and Folders	a. Create folders and organizing files in different folders b. Use copy / paste move commands to organize files and folders	a. Create files and folders Rename, arrange and search for the required folder/file
		c. Arrange icons – name wise, size, type, Modified d. Search a file or folder and find its path e. Create shortcut to files and folders (in other folders) on Desktop f. Familiarize with the use of My Documents g. Familiarize with the use of Recycle Bin	b. Restore deleted files from Recycle bin
5.	To use Windows Accessories: Calculator – Notepad – WordPad – MS Paint	a. Familiarize with the use of Calculator b. Access Calculator using Run command c. Create Text Files using Notepad and WordPad and observe the difference in file size d. Use MS paint and create .jpeg, .bmp files using MS Paint	a. Use windows accessories and select correct text editor based on the situation. b. Use MS pain to create /Edit pictures and save in the required format.
6.	To familiarize with Ribbon layout of MS word. – Home – Insert- page layout- References-Review- View	a. Create/Open a document b. Use Save and Save as features c. Work on two Word documents simultaneously d. Choose correct Paper size and Printing options	a. Create a Document and name appropriately and save b. Set paper size and print options
7.	To practice Word Processing Basics	a. Typing text b. Keyboard usage c. Use mouse (Left click / Right click / Scroll)	a. Use key board and mouse to enter/edit text in the document.

		d. Use Keyboard shortcuts e. Use Find and Replace features in MS- word f. Use Undo and Redo Features g. Use spell check to correct Spellings and Grammar	b. Use shortcuts c. Use spell check/ Grammar features for autocorrections.
8.	To practice Formatting techniques	a. Formatting Text b. Formatting Paragraphs c. Setting Tabs d. Formatting Pages e. The Styles of Word f. Insert bullets and numbers g. Themes and Templates h. Insert page numbers, header and footer	a. Format Text and paragraphs and use various text styles. b. Use bullets and numbers to create lists c. Use Templates / Themes d. Insert page numbers date, headers and footers
9.	To insert a table of required number of rows and columns	a. Edit the table by adding the fields – Deleting rows and columns –inserting sub table –marking borders. Merging and splitting of cells in a Table b. Changing the background colour of the table c. Use table design tools d. Use auto fit – fixed row/ column height/length – Even distribution of rows / columns features e. Convert Text to table and Table to Text f. Use Sort feature of the Table to arrange data in ascending/descending order	a. Insert table in the word document and edit b. Use sort option for arranging data.
10.	To Insert objects, clipart and Hyperlinks	a. Create a 2-page document. &Insert hyperlinks and t Bookmarks. b. Create an organization chart c. Practice examples like preparing an Examination schedule notice with a hyperlink to Exam schedule table.	a. Insert hyperlinks &Bookmarks b. Create organization charts/flow charts
11.	To Use Mail merge feature of MS Word	a. Use mail merge to prepare individually addressed letters b. Use mail merge to print envelopes.	Use Mail merge feature

12.	To use Equations and symbols features.	a. Explore various symbols available in MS Word b. Insert a symbol in the text c. Insert mathematical equations in the document	Enter Mathematical symbols and Equations in the word document
13.	To Practice with MS-EXCEL	a. Open /create an MS Excel spread sheet and familiarize with MS Excel 2007 layout like MS office Button b. Use Quick Access Toolbar- Title Bar- Ribbon- Worksheets- Formula Bar- Status Bar	a. Familiarize with excel layout and use b. Use various features available in toolbar
14.	To access and Enter data in the cells	a. Move Around a Worksheets- Quick access -Select Cells b. Enter Data-Edit a Cell-Wrap Text-Delete a Cell Entry-Save a File-Close Excel	a. Access and select the required cells by various addressing methods b. Enter data and edit
15.	To edit spread sheet Copy, Cut, Paste, and selecting cells	a. Insert and Delete Columns and Rows-Create Borders-Merge and Center b. Add Background Color-Change the Font, Font Size, and Font Color c. Format text with Bold, Italicize, and Underline-Work with Long Text-Change a Column's Width	Format the excel sheet
16.	To use built in functions and Formatting Data	a. Perform Mathematical Calculations verify -AutoSum b. Perform Automatic Calculations-Align Cell Entries	Use built in functions in Excel
17.	To enter a Formula for automatic calculations	a. Enter formula b. Use Cell References in Formulae c. Use Automatic updating function of Excel Formulae d. Use Mathematical Operators in Formulae e. Use Excel Error Message and Help	Enter formula for automatic calculations
18.	To Create Excel Functions, Filling Cells	a. Use Reference Operators b. Work with sum, Sum if, Count and Count If Functions c. Fill Cells Automatically	a. Create Excel sheets involving cross references and equations b. Use the advanced functions for conditional calculations

19.	To sort and filter data in table	a. Sort data in multiple columns b. Sort data in a row c. Sort data using Custom order d. Filter data in work sheet	a. Refine the data in a worksheet and keep it organized b. Narrow a worksheet by selecting specific choice
20.	To Practice Excel Graphs and Charts	a. Produce an Excel Pie Chart b. Produce c. Excel Column Chart	a. Use data in Excel sheet to Create technical charts and graphs Produce Excel Line Graph b. Produce a Pictograph in Excel
21.	To develop lab reports of respective discipline	Create Lab reports using MS Word and Excel	a. Insert Practical subject name in Header and page numbers in Footer
22.	To format a Worksheet in Excel, page setup and print	a. Shade alternate rows of data b. Add currency and percentage symbols c. Change height of a row and width of a column d. Change data alignment e. Insert Headers and Footers f. Set Print Options and Print	a. Format Excel sheet b. Insert headers & footers and print
23.	To familiarize with Ribbon layout & features of PowerPoint 2007.	Use various options in PowerPoint a. Home b. Insert c. Design d. Animation e. Slideshow f. View g. Review	Access required options in the tool bar
24.	To create a simple PowerPoint Presentation	a. Insert a New Slide into PowerPoint b. Change the Title of a PowerPoint Slide c. PowerPoint Bullets d. Add an Image to a PowerPoint Slide e. Add a Textbox to a PowerPoint slide	a. Create simple PowerPoint presentation with photographs/ClipArt and text boxes b. Use bullets option
25.	To Set up a Master Slide in PowerPoint and add notes	a. Create a PowerPoint Design Template b. Modify themes c. Switch between Slide master view and Normal view d. Format a Design Template Master Slide e. Add a Title Slide to a Design Template	a. Setup Master slide and format b. Add notes

		f. The Slide Show Footer in PowerPoint f. Add Notes to a PowerPoint Presentation	
26.	To Insert Text and Objects	a. Insert Text and objects b. Set Indents and line spacing c. Insert pictures/ clipart d. Format pictures e. Insert shapes and word art f. Use 3d features g. Arrange objects	Insert Text and Objects Use 3d features
27.	To insert a Flow Chart / Organizational Charts	a. Create a Flow Chart in PowerPoint b. Group and Ungroup Shapes c. Use smart art	Create organizational charts and flow charts using smart art
28.	To insert a Table	a. PowerPoint Tables b. Format the Table Data c. Change Table Background d. Format Series Legend	Insert tables and format
29.	To insert a Charts/Graphs	a. Create 3D Bar Graphs in PowerPoint b. Work with the PowerPoint Datasheet c. Format a PowerPoint Chart Axis d. Format the Bars of a Chart e. Create PowerPoint Pie Charts f. Use Pie Chart Segments g. Create 2D Bar Charts in PowerPoint h. Format the 2D Chart e. Format a Chart Background	Create charts and Bar graphs, Pie Charts and format.
30.	To Insert audio & video, Hyperlinks in a slide Add narration to the slide	a. Insert sounds in the slide and hide the audio symbol b. Adjust the volume in the settings c. Insert video file in the format supported by PowerPoint in a slide d. Use automatic and on click options e. Add narration to the slide f. Insert Hyperlinks	a. Insert Sounds and Video in appropriate format. b. Add narration to the slide c. Use hyperlinks to switch to different slides and files
31.	To Practice Animation effects	a. Apply transitions to slides b. To explore and practice special animation effects like <i>Entrance, Emphasis, Motion Paths & Exit</i>	Add animation effects
32.	Reviewing presentation	a. Checking spelling and grammar b. Previewing presentation c. Set up slide show	a. Use Spell check and Grammar feature b. Setup slide show

		d. Set up resolution e. Exercise with Rehearse Timings feature in PowerPoint f. Use PowerPoint Pen Tool during slide show g. Saving h. Printing presentation (a) Slides (b) Hand-out	c. Add timing to the slides d. Setup automatic slide show
33	To familiarize with standard toolbox	a. Open Adobe Photoshop b. Use various tools such as i. The Layer Tool ii. The Color & Swatches Tool iii. Custom Fonts & The Text Tool iv. Brush Tool v. The Select Tool vi. The Move Tool vii. The Zoom Tool viii. The Eraser ix. The Crop Tool x. The Fill Tool	photograph and save it in Photoshop
34	To edit a photograph	a. Use the Crop tool b. Trim edges c. Change the shape and size of a photo d. Remove the part of photograph including graphics and text	to edit image by using corresponding tools.
35	To insert Borders around photograph	1. Start with a single background layer 2. Bring the background forward 3. Enlarge the canvas 4. Create a border color 5. Send the border color to the back 6. Experiment with different colors	create a border or frame around an image to add visual interest to a photo
36	To change Background of a Photograph	a. open the foreground and background image b. Use different selection tools to paint over the image c. Copy background image and paste it on the foreground. d. Resize and/or drag the background image to reposition. e. In the Layers panel, drag the background layer below the foreground image layer.	swap background elements using the Select and Mask tool and layers.

37	To change colors of Photograph	a. Change colors using: Color Replacement tool ii) Hue/Saturation adjustment layer tool	Able to control color saturation
38	To prepare a cover page for the book in subject area	a. open a file with height 500 and width 400 for the cover page. b. apply two different colors to work area by dividing it into two parts using Rectangle tool. c. Copy any picture and place it on work area→ resize it using free transform tool. d. Type text and apply color and style e. Apply effects using blended options	Able to prepare cover page for the book
39	To adjust the brightness and contrast of picture to give an elegant look	a. open a file b. Go to image→ adjustments→ Brightness/Contrast. f. adjust the brightness and contrast g. save the image	Able to control brightness / contrast.
40	To type a word and apply the shadow emboss effects	a. open a file b. Select the text tool and type text. c. Select the typed text go to layer→ layer style→ blended option→ drop shadow, inner shadow, bevel and emboss→ contour→ satin→ gradient overlay d. Save the image.	Able to apply shadow emboss effects

**DIPLOMA IN COMMUNICATION AND COMPUTER NETWORKING ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION**

CURRICULUM-2023

(III Semester)

Sub Code	Name of the Subject	Instruction Periods/Week		Total Periods Per Semester	Scheme Of Examinations			
		Theory	Practicals		Duration (hrs)	Sessional Marks	End Exam Marks	Total Marks
THEORY SUBJECTS								
CCN-301	Mathematics –II	4		60	3	20	80	100
CCN-302	Python Programming	5	-	75	3	20	80	100
CCN-303	Operating systems	4	-	60	3	20	80	100
CCN-304	Data Communication and Computer Networks	5	-	75	3	20	80	100
CCN-305	DBMS	5	-	75	3	20	80	100
PRACTICAL SUBJECTS								
CCN-306	Python Programming Lab	-	3	45	3	40	60	100
CCN-307	Computer Hardware and Network Maintenance Lab	-	6	90	3	40	60	100
CCN-308	DBMS Lab	-	4	60	3	40	60	100
CCN-309	Multimedia Lab		3	45	3	40	60	100
	ACTIVITIES		3	45				
	Total	23	19	630		260	640	900

CCN-301 Common to CME, AIML,AMG,AMT,CAI,CCB,CCN, WD

CCN-302 Common with CM-505 ,

CCN-303 COMMON WITH CM-303 ,CCN-305 IS COMMON WITH CM-305

CCN-308 COMMON WITH CM-308, CCN – 309 COMMON WITH CM-309

Course Code	Course Title	No. of Periods/week	Total No. of periods	Marks for FA	Marks for SA
CCN-301	Engineering Mathematics-II	4	60	20	80

TIME SCHEDULE

S.No	Chapter/Unit title	No. of Periods	Weightage Allotted	Short Type	Essay Type	COs mapped
Unit – I: Integral Calculus						
1	Indefinite integration	17	26	2	2	CO1
2	Definite integrals	5	16	2	1	CO1
Unit – II: Differential Equations						
3	Introduction to Differential equations	2	3	1	0	CO2
4	Solutions of Differential equations of first order	6	10	0	1	CO2
Unit – III: Graph Theory and Probability						
5	Graph theory	5	6	2	0	CO3
6	Probability	12	26	2	2	CO3
Unit – IV : Statistics						
6	Measures of CentralTendency	1	0	0	0	CO4
7	Measures of Dispersion	3	3	1	0	CO4
8	Correlation	4	10	0	1	CO4
9	Simple linear regression	5	10	0	1	CO4
Total		60	110	10	8	
Marks				30	80	

Course Objectives	(iii) To understand the concepts of indefinite integration and definite integration. (iv) To understand the formation of differential equations and learn various methods of solving first order differential equations. (v) To comprehend the concepts of graph theory and probability. (vi) To learn different statistical techniques for data analysis.
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Course Outcomes	CO1	Integrate various functions using different methods and evaluate definite integrals.
	CO2	Obtain differential equations and solve differential equations of first order and first degree.
	CO3	Able to define the basic concepts of Graph Theory and use the principles of Probability in computational systems.
	CO4	Apply various statistical techniques for data analysis.

Unit-I

Integral Calculus

C.O. 1 Integrate various functions using different methods and evaluate definite integrals.

L.O.1.1. Explain the concept of Indefinite integral as an anti-derivative.

1.2. State the indefinite integral of standard functions and properties of $\int (u + v) dx$ and $\int k u dx$ where u, v are functions of x and k is constant.

1.3. Solve problems involving standard functions using these properties.

1.4. Evaluate integrals involving simple functions of the following type by the method of substitution.

i) $\int f(ax + b) dx$, where $f(x)$ is in standard form.

ii) $\int (f(x))^n f'(x) dx$, $n \neq -1$

iii) $\int \frac{1}{f(x)} dx$

iv) $\int [f(g(x))] g'(x) dx$

1.5. Find the integrals of $\tan x$, $\cot x$, $\sec x$ and $\operatorname{cosec} x$ w.r.t. x .

1.6. Evaluate the Standard integrals of the functions of the type

i) $\frac{1}{a^2 + x^2}$, $\frac{1}{a^2 - x^2}$, $\frac{1}{x^2 - a^2}$

ii) $\frac{1}{\sqrt{a^2 + x^2}}$, $\frac{1}{\sqrt{a^2 - x^2}}$, $\frac{1}{\sqrt{x^2 - a^2}}$

iii) $\sqrt{a^2 + x^2}$, $\sqrt{a^2 - x^2}$, $\sqrt{x^2 - a^2}$

1.7. Evaluate integrals using decomposition method.

1.8. Solve problems using integration by parts.

1.9. Use Bernoulli's rule for evaluating the integrals of the form $\int u.v dx$.

1.10. Evaluate the integrals of the form $\int e^x [f(x) + f'(x)] dx$

1.11. State the fundamental theorem of integral calculus

1.12. Explain the concept of definite integral.

1.13. Solve simple problems on definite integrals.

1.14. State various properties of definite integrals.

1.15. Evaluate simple problems on definite integrals using these properties.

Unit -II

Differential Equations

C.O. 2 Obtain differential equations and solve differential equations of first order and first degree.

L.O.2.1. Define a differential equation, its order and degree

2.2 Find order and degree of a given differential equation.

2.3 Form a differential equation by eliminating arbitrary constants.

2.4 Solve the first order and first degree differential equations by variables separable method.

2.5 Solve linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where P and Q are functions of x only or constants.

Syllabus for Unit test-I completed

Unit-III

Graph Theory and Probability

C.O. 3 Able to define the basic concepts of Graph Theory and use the principles of Probability in computational systems.

L.O. 3.1 Define a graph.

3.2 Explain the terminology of a graph, vertices, edges, parallel edges, adjacent vertices, self-loops.

3.3 State the significance of Graph Theory in Computer Science applications.

3.4 Explain incidence and degree of a graph.

3.5 Explain the relation between degree and edges of a graph.

3.6 Explain various types of graphs, null graph, trivial graph, simple graph, multigraph, directed graph, non-directed graph and cyclic graph.

3.7 Define walk, path, circuit, length of a graph, distance between two vertices.

3.8 Explain the formation of adjacency matrix of a graph.

3.9 Recall the basic probability principles.

3.10 Define permutations and combinations with examples.

3.11 State addition theorem of probability for two mutually exclusive and exhaustive events.

3.12 Solve simple problems on addition theorem.

3.13 Explain conditional event and conditional probability.

3.14 Solve simple problems on conditional probability.

3.15 Explain dependent, independent events and state multiplication theorem.

3.16 Solve simple problems on multiplication theorem.

3.17 Explain the concept of priori and posteriori probabilities.

3.18 State Bayes' theorem and solve simple problems.

Unit-IV

Statistics

C.O. 4 Apply various statistical techniques for data analysis.

L.O. 4.1 Recall the measures of central tendency.

4.2 Explain the significance of measures of dispersion to determine the degree of heterogeneity of the data.

4.3 Find the measures of dispersion, Range, Mean Deviation and Standard Deviation for ungrouped data.

4.4 Explain the merits and demerits of these measures of dispersion

4.5 Explain bivariate data.

4.6 Explain the concept of covariance and correlation between two variables.

4.7 Calculate Pearson's correlation coefficient between two variables.

4.8 Find Spearman's rank correlation coefficient.

4.9 Explain predictor variables, outcome variables and simple linear regression.

4.10 Calculate the regression coefficients and regression equations with simple problems.

Syllabus for Unit test-II completed

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	2	2	2				3	2	2
CO2	3	2	2	2				3	2	2
CO3	3	3	3	3				3	3	3
CO4	3	3	3	3				3	3	3
Avg.	3	2.5	2.5	2.5				3	2.5	2.5

3 = Strongly mapped (High), 2 = Moderately mapped (Medium), 1 = Slightly mapped (Low)

Note: The gaps in CO/PO mapping will be met with appropriate activities as follows:

- For PO5: Appropriate quiz programmes may be conducted at intervals and duration as decided by concerned faculty.
- For PO6: Seminars on applications of mathematics in various engineering disciplines are to be planned and conducted.
- For PO7: Plan activities in such a way that students can visit the Library to refer standard books on Mathematics and access the latest updates in reputed national and international journals. Additionally, encourage them to attend seminars and learn mathematical software tools.

PO no	Mapped with CO no	CO periods addressing PO in column I		Level (1,2 or 3)	Remarks
		Number	%		
1	CO1, CO2, CO3, CO4	60 (22+8+17+13)	100%	3	>40% Level 3 Highly addressed
2	CO1, CO2, CO3, CO4	38 (8+3+17+10)	63%	3	
3	CO1, CO2, CO3, CO4	38 (8+3+17+10)	63%	3	
4	CO1, CO2, CO3, CO4	38 (8+3+17+10)	63%	3	
5					25% to 40% Level 2 Moderately addressed
6					
7					
PSO 1	CO1, CO2, CO3, CO4	60 (22+8+17+13)	100%	3	5% to 25% Level 1 Low addressed
PSO 2	CO1, CO2, CO3, CO4	38 (8+3+17+10)	63%	3	
PSO 3	CO1, CO2, CO3, CO4	38 (8+3+17+10)	63%	3	

COURSE CONTENTS

Unit-I

Indefinite Integration:

- Integration regarded as anti-derivative – Indefinite integrals of standard functions - Properties of indefinite integrals - Integration by substitution or change of variable - Integrals of $\tan x$, $\cot x$, $\sec x$, $\operatorname{cosec} x$.

Evaluation of integrals which are of the following forms:

$$i) \frac{1}{a^2 + x^2}, \frac{1}{a^2 - x^2}, \frac{1}{x^2 - a^2}$$

$$ii) \frac{1}{\sqrt{a^2 + x^2}}, \frac{1}{\sqrt{a^2 - x^2}}, \frac{1}{\sqrt{x^2 - a^2}}$$

$$iii) \sqrt{a^2 + x^2}, \sqrt{a^2 - x^2}, \sqrt{x^2 - a^2}$$

Integration by decomposition of the integrand into simple rational, algebraic functions - Integration by parts, Bernoulli's rule and integrals of the form $\int e^x [f(x) + f'(x)] dx$.

Definite integral-fundamental theorem of integral calculus, properties of definite integrals, evaluation of simple definite integrals.

Unit -II

Differential Equations:

2. Definition of a differential equation, order and degree of a differential equation, formation of differential equations - Solutions of differential equations of first order and first degree using variables separable method and linear differential equation of the type $\frac{dy}{dx} + Py = Q$.

Unit-III

Graph Theory and Probability

3. Definition of a graph, terminology of a graph, significance in computer science applications - Incidence and degree, relationship between degree and edges - Various types of graphs, null graph, trivial graph, simple graph, multigraph, directed graph, non-directed graph and cyclic graph - Walk, path, circuit, length of a graph, distance between two vertices - Formation of adjacency matrix.
Permutations and Combinations - Addition theorem of probability, conditional probability, dependent and independent events with multiplication theorem - Priori and posteriori probability, Baye's theorem.

Unit III

Statistics

4. Measures of dispersion, range, mean deviation and standard deviation of ungrouped data, merits and demerits - Bivariate data, correlation, Pearson's correlation coefficient, Spearman's rank correlation coefficient - Predictor and outcome variables, simple linear regression coefficients and regression equations.

Textbook:

Engineering Mathematics-II, a textbook for second year third semester diploma courses, prepared & prescribed by SBTET, AP.

Reference Books:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers.
2. Schaum's Outlines Differential Equations, Richard Bronson & Gabriel B. Costa
3. Trembley and Manohar, Discrete Mathematical Structures with Applications to Computer Science, Tata-McGraw-Hill.
4. Narsing Deo, Graph Theory, PHI India.
5. Schaum's Outline: Introduction to Probability and Statistics, Seymour Lipschutz & John J. Schiller.

Unit Test Syllabus

Unit Test	Syllabus
Unit Test-I	From L.O 1.1 to L.O 2.5
Unit Test-II	From L.O 3.1 to L.O 4.10

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-302	Python Programming	5	75	20	80

TIME SCHEDULE

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1.	Introduction	10	26	2	2	CO1
2.	Control Flow and Loops	15	13	1	1	CO2
3.	Functions and Arrays	15	26	2	2	CO3
4.	Data Structures	15	26	2	2	CO4
5.	Object Oriented Programming in Python and File Handling and Exception Handling	20	19	3	1	CO5
Total		75	110	10	8	
Marks				30	80	

Course Objectives	i) To know the fundamentals of Python programming ii) To understand fundamental syntactic information about 'Python' iii) To develop various python programs
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Course Outcomes	CO1	CCN-302.1	Explain Basic constructs like operators, expressions and components of python programming as well as Editing and Debugging
	CO2	CCN-302.2	Write Python programs using Control statements, Loops
	CO3	CCN-302.3	Write python programs using Functions and arrays
	CO4	CCN-302.4	Develop Python programs using Data structures
	CO5	CCN-302.5	Develop Python application programs using OOP Concept, FILES, Exceptions

Learning Outcomes:

1. Introduction

- 1.1 History of Python.
- 1.2 List Python features
- 1.3 Explain Applications of Python
- 1.4 Describe Python Integrated Development and Learning Environment (IDLE)
- 1.5 Give the process of Running Python Scripts.
- 1.6 Explain Identifiers, Keywords, Indentation, Variables
- 1.7 Explain various datatypes (Int, float, Boolean, string, and list)
- 1.8 Explain declaration, initialization of variables.
- 1.9 Explain Input and Output statements.
- 1.10 Explain formatted input output.
- 1.11 State the usage of comments
- 1.12 Explain various Operators.
- 1.13 Explain Boolean values.
- 1.14 Explain Operator precedence rules.
- 1.15 State the purpose of modules.
- 1.16 Define functions.
- 1.17 List types of functions. (Built-in, User defined)
- 1.18 Explain Built-in Functions.
- 1.19 Give the Steps in Develop a simple python program and execution.

2. Control Flow and Loops

- 2.1 Explain various Control Flow constructs.
 - 2.1.1 If
 - 2.1.2 If-Else
 - 2.1.3 if-elif-else
- 2.2 Explain various Loop Statements.
 - 2.2.1 for Loop
 - 2.2.2 while loop
 - 2.2.3 break
 - 2.2.4 continue
 - 2.2.5 pass

3. Functions and Arrays

- 3.1 **Introduction** Function Arguments: Default arguments, Variable Length Arguments
- 1.2 Anonymous Functions
- 1.3 Return Statement
- 1.4 Scope of variables: Local Variables and Global Variables
- 1.5 Explain creation of modules.
- 1.6 Explain importing of modules.
- 1.7 Python Variable: Namespace and scoping
- 1.8 Python Packages
- 1.9 Explain Strings: String slices, immutability
- 1.10 Explain String functions and methods.

- 1.11 Explain about String module.
- 1.12 Explain about Python Arrays.
- 1.13 Explain accessing of elements in an Array.
- 1.14 Explain Array methods.

4. Data Structures

- 4.1 Explain Python Lists.
- 4.2 Describe Basic List Operations.
- 4.3 Explain List Slices.
- 4.4 Explain List methods.
- 4.5 Explain List loop
- 4.6 Explain mutability.
- 4.7 Explain aliasing.
- 4.8 Explain Cloning lists.
- 4.9 Explain List parameters.
- 4.10 Explain List comprehension.
- 4.11 Tuples.
 - 4.11.1 Explain Tuple assignment.
 - 4.11.2 Explain Tuple as return value.
 - 4.11.3 Explain Tuple Comprehension
- 4.12 Dictionaries
 - 4.12.1 Explain creation of dictionary/assignment.
 - 4.12.2 Explain Operations and methods.
 - 4.12.3 Explain Dictionary Comprehension.
- 4.13 Explain Sets.

5.0 Handling

Object Oriented Programming in Python and File Handling and Exception

- 5.1 Creating Classes
- 5.2 Creating Objects
- 5.3 Method Overloading and Overriding
- 5.4 Data Hiding
- 5.5 Data Abstraction
- 5.6 Opening files in different modes
- 5.7 Processing files
- 5.8 Closing a file
- 5.9 Exception Handling

COURSE CONTENT

UNIT – I:

Introduction: Introduction to Python and installation, data types: Int, float, Boolean, string, and list; variables, Arithmetic Operators, Comparison (Relational) Operators, Assignment Operators, Logical Operators, Bitwise Operators, Membership Operators, Identity Operators, Boolean values, expressions, statements, precedence of operators, comments; modules, functions--- function and its use, flow of execution, parameters and arguments.

UNIT – II:

Control Flow and Loops: Control Flow- if, if-elif-else, for, while, break, continue, pass

UNIT – III:

Functions and Arrays - Defining Functions, Calling Functions, Passing Arguments, Keyword Arguments, Default Arguments, Variable-length arguments, Anonymous Functions, Fruitful Functions (Function Returning Values), Scope of the Variables in a Function - Global and Local Variables, Modules: Creating modules, import statement, from Import statement, name spacing, Python packages, Introduction to PIP, Installing Packages via PIP, Using Python Packages - Strings: string slices, immutability, string functions and methods, string module; Python arrays, Access the Elements of an Array, array methods.

UNIT – IV:

Data Structures : Lists: list operations, list slices, list methods, list loop, mutability, aliasing, cloning lists, list parameters, list comprehension; Tuples: tuple assignment, tuple as return value, tuple comprehension; Dictionaries: operations and methods, comprehension-sets.

UNIT – V:

Object Oriented Programming OOP in Python: Classes, 'self variable', Methods, Constructor Method, Inheritance, Overriding Methods, Data hiding, File Handling: Open Files, File Processing and Closing a File Error and Exceptions: Difference between an error and Exception, Handling Exception, try except block, Raising Exceptions, User Defined Exceptions

REFERENCE BOOKS

1. Python Programming by K. Nageswara Rao, Shaikh Akbar - Scitech Publications (India) Pvt. Ltd.
2. Python Programming: A Modern Approach, Vamsi Kurama, Pearson
3. Learning Python, Mark Lutz, Orielly
4. Think Python, Allen Downey, Green Tea Press
5. Core Python Programming, W.Chun, Pearson.
6. Introduction to Python, Kenneth A. Lambert, Cengage

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.7
Unit test-2	From 3.8 to 5.9

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-303	Operating Systems	4	60	20	80

Time Schedule

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1.	Introduction to Operating system	12	16	2	1	CO1
2.	Process management	12	16	2	1	CO2, CO6
3.	Synchronization & Deadlocks	12	26	2	2	CO3, CO6
4.	Memory management	12	26	2	2	CO4, CO6
5.	Disk scheduling and File management	12	26	2	2	CO5, CO6
Total		60	110	10	8	
Marks				30	80	

Course Objectives	i) To know about the basics of Operating Systems ii) To familiarize with process management, Scheduling algorithms, Synchronization and deadlock techniques iii) To understand various Memory management techniques iv) To familiarize with File management
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Learning Outcomes:

1.0 Introduction to operating systems

- 1.1** Define an operating system
- 1.2** Discuss history of operating system
- 1.3** Discuss about various types of operating systems
- 1.4** Distinguish spooling and buffering
- 1.5** Explain the concepts multiprogramming and timesharing
- 1.6** Differentiate between distributed and real time systems
- 1.7** Describe multiprocessor systems
- 1.8** Describe the operating system components
- 1.9** Discuss operating system services
- 1.10** Define system call with an example
- 1.11** List and explain different types of system calls
- 1.12** Define single user, multi user operating system structure

2.0 Process management

- 2.1** Define process and process control block
- 2.2** Explain process state diagram
- 2.3** Describe process creation and termination
- 2.4** Discuss the relation between processes
- 2.5** Define Thread and describe multithreading
- 2.6** Explain scheduling concepts
- 2.7** Describe scheduling queues and schedulers
- 2.8** Explain CPU scheduling and scheduling criteria
- 2.9** Explain various scheduling algorithms
 - 2.9.1 FCFS
 - 2.9.2 SJF
 - 2.9.3 Round Robin
 - 2.9.4 Priority
 - 2.9.5 Multilevel Scheduling

3.0 Synchronization & Deadlocks

- 3.1** Describe semaphores
- 3.2** Explain inter process communication
- 3.3** Define Deadlock
- 3.4** State the necessary conditions for arising deadlocks
- 3.5** State various techniques for deadlock prevention
- 3.6** Discuss Deadlock avoidance and detection
- 3.7** Describe the process of recovering from deadlock

4.0 Memory management

- 4.1** Describe briefly address binding, dynamic loading, dynamic linking
- 4.2** Define overlays
- 4.3** Describe briefly on swapping
- 4.4** Explain single partition allocation
- 4.5** Explain multiple partition allocation
- 4.6** Explain the concept of fragmentation

- 4.7 Explain paging concept
- 4.8 Explain how logical address is translated into physical address
- 4.9 Explain segmentation and segmentation with paging
- 4.10 Define virtual memory techniques
- 4.11 Describe demand paging
- 4.12 Describe page replacement
- 4.13 Discuss on page replacement algorithms
 - 4.13.1 FIFO
 - 4.13.2 LRU
 - 4.13.3 Optimal
- 4.14 Explain the concept of thrashing
- 4.15 Explain working set model and page fault frequency
- 5.0 Disk scheduling and File management**
 - 5.1 List out various disk performance parameters
 - 5.2 Disk scheduling policies
 - 5.2.1 FIFO
 - 5.2.2 SSTF
 - 5.2.3 SCAN
 - 5.3 Define file management
 - 5.4 List and explain various file operations
 - 5.5 List and explain various access methods
 - 5.6 List and explain various allocation methods
 - 5.6 List and explain directory structure
 - 5.7 Explain disk organization and structure

COURSE CONTENT

1.0 Introduction to operating systems

Operating System –Evolution of operating system-Types of Operating Systems - Multi Programming and Time Sharing - Distributed and Real time Systems - spooling and buffering - Multi processor systems- Components of Operating Systems - operating System Services - system Calls - single User and Multi user operating System Structure.

2. Process management

Processes - Sequential Processes - Process State Diagram - Process Control Block - Process Creation and Termination - Relations between Processes - Threads and Multi-Threading - Scheduling Concepts • Schedulers - CPU scheduling and Scheduling criteria - scheduling algorithms.

3. Synchronization & Deadlocks

Inter Process Communications - semaphores – monitors

Deadlocks - principal of deadlock - deadlock prevention - deadlock detection - deadlock avoidance.

4. Memory management

Address binding -Dynamic Loading- dynamic linking-overlays-swapping• memory allocation-fragmentation-paging-segmentation- segmentation with paging-

Benefits of virtual memory - virtual memory techniques - demand paging - page replacements - page replacement algorithms – thrashing.

5.Disk scheduling and File management

Disk performance parameters - Disk scheduling policies –

Introduction to file systems - File management-File Operations - Access methods - Directory structure organization - File Protection.

REFERENCE BOOKS

1. Operating Systems -- Silberschatz and Galvin
2. Operating Systems -- William Stallings, PHI
3. Operating Systems -- Dietel and Dietel
4. Operating Systems -- Dhamdhare (TMH)
5. Advanced Operating Systems -- Tanenbaum

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.2
Unit test-2	From 3.3 to 5.7

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-304	Data Communication and Computer Networks	5	75	20	80

Time Schedule

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1.	Introduction to Data communication and Computer Networks	12	16	2	1	CO1
2.	Data Link Layer	20	26	2	2	CO2
3.	Network Layer	15	21	2	1 ½	CO3
4.	Transport & Session Layer	15	26	2	2	CO4
5.	Presentation & Application Layer	13	21	2	1 ½	CO5
Total		90	110	10	8	
Marks				30	80	

Course Objectives	i)To familiarize the concepts of data communication and computer networks ii)To understand the concepts of Error control and Flow control techniques during transmission in data link layer iii)To understand IP address and routing algorithms to find out the optimum route in the network iv)To know the concepts of traffic and its control which ensures quality of service(QoS) in Transport layer v)To familiarize the usage of Application layer protocols in Real world applications
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Course Outcomes	At the end of the course the student able to learn following:		
	CO1	CCN-304.1	Explain the features of data communication and Standard network reference models like ISO:OSI & TCP/IP
	CO2	CCN-304.2	Analyse the services and features of various protocols used in data link layer.
	CO3	CCN-304.3	Describe various routing protocols of Network layer.
	CO4	CCN-304.4	Explain various protocols that are used in Transport layer.
	CO5	CCN-304.5	Use various protocols and services of Application layer in Real World application

Learning Outcomes:

1.0 Introduction to Data Communication and Computer Networks

- 1.1. Define Data communication.
- 1.2. Explain various Data Transmission modes (Serial, Parallel, Synchronous, Asynchronous, Isochronous transmission)
- 1.3. Explain various Data transmission characteristics (Signalling rate, data rate, bit rate, baud rate)
- 1.4. Define computer network.
- 1.5. State the need of computer networks.
- 1.6. List the advantages and disadvantages of computer networks.
- 1.7. List the applications of computer networks
- 1.8. List and explain various Network Topologies(Mesh, Star, Bus, Tree, Ring and Hybrid)
- 1.9. Classification of Networks
 - 1.9.1. Based on Transmission Technologies(Point to-point, Multipoint, Broadcast)
 - 1.9.2. Based on Physical size(LAN, MAN, WAN)
 - 1.9.3. Based on Architecture (Peer to Peer, Client Server)
- 1.10. List and Explain various Networking devices(Repeater, Hub, Switch, Router, Bridge, Gateway)
- 1.11. Communication standards (Reference models)
 - 1.11.1. Explain OSI reference model.
 - 1.11.2. Explain TCP/IP Reference Model.
- 1.12. Transmission Media
 - 5.1.1. Guided Transmission media (Twisted pair cable, Coaxial cable, Optical Fiber cable)
 - 5.1.2. Unguided Transmission media(Radio waves, Micro waves, Infrared waves)
- 1.13. Define Analog and Digital Transmission

2.0 Data Link Layer.

- 2.1. List the Services provided to the Network Layer
- 2.2. Explain Fixed and variable sized framing
- 2.3. List different types of errors
- 2.4. Explain the following Error detection techniques
 - 2.4.1. Single parity

- 2.4.2. Two dimensional parity
- 2.4.3. Check sum
- 2.4.4. Cyclic Redundancy Check
- 2.5. Explain the following Error correction techniques
 - 2.5.1. Forward error correction.
 - 2.5.2. Backward Error correction
 - 2.5.3. Cyclic Redundancy check
 - 2.5.4. Hamming code
- 2.6. List and explain Flow control techniques
 - 2.6.1. Stop and wait
 - 2.6.2. Sliding window
- 2.7. Explain the following Medium Access control(MAC) Protocols
 - 2.7.1. ALOHA
 - 2.7.1.1. Pure ALOHA
 - 2.7.1.2. Slotted ALOHA
 - 2.7.2. CSMA
 - 2.7.3. CSMA/CA
 - 2.7.4. CSMA/CD
- 2.8. Explain IEEE Standards for the following LAN's
 - 2.8.1. Ethernet (IEEE 802.3)
 - 2.8.2. Token Bus (IEEE 802.4)
 - 2.8.3. Token Ring(IEEE 802.5)

3.0 Network Layer

- 3.1. List and Explain the services provided by Network layer
- 3.2. Circuit Switching Vs Packet Switching
- 3.3. Classify Routing Algorithms
 - 3.3.1. Explain Adaptive Routing Algorithms.
 - 3.3.1.1. Distance vector Routing
 - 3.3.1.2. Link state routing
 - 3.3.2. Explain Non Adaptive Routing Algorithms
 - 3.3.2.1. Flooding
 - 3.3.2.2. Shortest Path
- 3.4. Explain TCP/IP Addressing scheme
 - 3.4.1. Components of IP address
 - 3.4.2. IP address classes
 - 3.4.3. IP Subnetting
- 3.5. Explain the versions of IP
 - 3.5.1. IPv4
 - 3.5.2. IPv6

4.0 Transport Layer

- 4.1. List the explain the services provided by Transport layer
- 4.2. List and explain transport layer protocols
 - 4.2.1. TCP
 - 4.2.2. UDP
- 4.3. Explain flow control protocols in Transport layer
 - 4.3.1. Stop and wait
 - 4.3.2. Go Back N
 - 4.3.3. Selective repeat
- 4.4. Explain congestion control techniques/Algorithms in Transport layer
 - 4.4.1. Leaky Bucket

4.4.2.Token Bucket

- 4.5. List the services provided by session Layer.
- 4.6. List different protocols used in session layer.
- 4.7. Study the purpose of Session Layer.

5.0 Application Layer

- 5.1. List different services offered by Presentation layer.
- 5.2. List different protocols used in Presentation layer.
- 5.3. Study the purpose of PresentationLayer.
- 5.4. List and explain different services offered by application layer.
- 5.5. Explain different Application Layer protocols
 - 5.5.1. Hypertext Transfer protocol(HTTP)
 - 5.5.2. File Transfer Protocol(FTP)
 - 5.5.3. Teletype Network (TELNET)
 - 5.5.4. Simple Mail Transfer protocol(SMTP)
 - 5.5.5. Domain Name Service(DNS)
 - 5.5.6. Dynamic Host configuration protocol(DHCP)
- 5.6. Explain Network Management Protocol
 - 5.6.1. Simple Network Management Protocol (SNMP)
 - 5.6.2. Internet Control Message Point (ICMP)
- 5.7. Explain about Remote Monitoring (RMON)
- 5.8. Explain about World Wide Web (WWW)

COURSE CONTENT

1.0 Introduction to Data combination and Computer Networks

Data communication-Data Transmission modes (Serial, Parallel Synchronous, Asynchronous, Isochronous transmission)-Transmission characteristics (Signalling rate, data rate, bit rate, baud rate). Computer Networks-Need-Advantages-Disadvantages-Applications-Network topologies (Mesh, Star, Bus, Tree, Ring and Hybrid)-Network Classification(Based on Transmission Technologies,Scale, Architecture)-Network Devices.

The OSI reference model- TCP/IP Reference Model.Physical Layer: Transmission media- Analog Transmission- Digital transmission

2.0 Data Link Layer

Data Link Layer Design Issues-Services provided to the Network Layer-Error Detection and correction-Framing-Flow Control - Sliding Window Protocols- Multiple Access Protocols-IEEE Standard for LANs

3.0 Network Layer

Introduction to Network Layer – Services - Circuit Switching Vs Packet Switching-Types of Routing-routing algorithms-Network Protocols-IP- IPV4, IPV6

4.0 Transport Layer

The Transport Services–Internet Transport Protocols(TCP,UDP)-Flow control Protocols (Simple,Stop&wait) – Congestion Controls in Transport Layer (Leaky bucket,Token Bucket)-session layer services-protocols.

bApplication Layer

Presentation Layer-services-Protocols. Application Layer -HTTP-FTP-TELNET-SMTP-DNS-DHCP-SNMP-WWW

REFERENCE BOOKS

1. "Data Communication and Networking" by Behrouz A Forouzan - TMH
2. "Data and Computer Communication" by William Stallings-Pearson
3. "Internetworking with TCP/IP, Volume 1" by Douglas Comer-O'Reily
4. "TCP/IP Illustrated" by W Richard Stevens -Addison-Wesley
5. "Computer Networks: A Top – Down Approach" by FOROUZAN - TMH
6. Computer Networks -- Andrew S. Tanenbaum-Pearson Education

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.2
Unit test-2	From 3.3 to 5.8

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-305	DBMS	5	75	20	80

Time Schedule

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1.	Concepts of DBMS & RDBMS	18	29	3	2	CO1, CO3, CO4
2.	Concepts of SQL	22	26	2	2	CO2
3.	Basics of PL/ SQL	15	26	2	2	CO1, CO3
4.	Advance PL/SQL	10	16	2	1	CO1, CO4, CO5
5.	Concepts of NoSQL & MongoDB.	10	13	1	1	CO2, CO4
Total		75	110	10	8	
Marks				30	80	

Course Objectives	i) To know the fundamentals of DBMS ii) To familiarize insert, retrieve, update, delete data in database iii) To familiarize programming skills for insert, retrieve, update, delete data in database
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Course Outcomes	CO1	CCN-305.1	Describe fundamentals, types and Overall structure of DBMS
	CO2	CCN-305.2	Apply SQL commands to create, retrieve, update, delete data from the Relational data bases.
	CO3	CCN-305.3	Describe PL/SQL programming constructs, control statements and sub programs.
	CO4	CCN-305.4	Apply cursors, triggers and Exception handling concepts
	CO5	CCN-305.5	Use NOSQL database concepts and MongoDB data base concepts in designing database Schema.

Learning Outcomes:

1.0

Concepts of DBMS & RDBMS

1.1

Define Database Management System(DBMS)

- 1.2 List the advantages of DBMS
- 1.3 Explain Database Abstraction, Data Independence
- 1.4 Define Instances and Schemas
- 1.5 Explain Data Models.
- 1.6 Define Database languages DDL, DML, TCL
- 1.7 Explain Database Administrator, Users and Database System Architecture with diagram.
- 1.8 Define Entity, Entity sets, Relationship, Relationship sets, Super Key , Candidate Key and Primary Key, Foreign Key
- 1.9 Explain Mapping Cardinalities.
- 1.10 List the symbols used in ER model.
- 1.11 Know The Entity-Relationship Model.
- 1.12 Reduce the ER-diagrams to tables
- 1.13 Explain Generalization, Specialization &Aggregation.
- 1.14 Explain Functional Dependencies, Normalizations– 1 NF, 2 NF and 3NF

2.0 Concepts of SQL

- 2.1 Explain SQL and benefits of SQL.
- 2.2 Describe about Embedded SQL and Lexical conventions
- 2.3 Describe Naming of the Objects and parts and how to refer them.
- 2.4 Explain literals &different data types like character, number, long, date, raw and longraw etc.
- 2.5 Illustrate the comments within SQL Statement
- 2.6 Explain SQL Operators
- 2.7 Describe Data Definition Language commands CREATE, ALTER and DROP.
- 2.8 Explain integrity constraints through creating a table and altering table.
- 2.9 Describe Data Manipulation Language commands INSERT, UPDATE and DELETE
- 2.10 Explain SELECT statement with WHERE, ORDER BY, GROUP BY and HAVING clauses with examples
- 2.11 List and explain single row(Number, character, date and conversion) functions
- 2.12 List and Explain group functions
- 2.13 Explain Transaction Control Commands COMMIT, SAVEPOINT, ROLLBACK, GRANT, and REVOKE.
- 2.14 Explain Sub Queries with examples
- 2.15 Explain Joins (Equi Join, Non-Equi Joins, Inner Join, Outer Join, cross join and Self join) with syntax and examples.

3.0 PL/SQL

- 3.1 Explain PL/SQL Block structure.
- 3.2 List the features of PL/SQL
- 3.3 Explain the data types of PL/SQL
- 3.4 Declaration of variables
- 3.5 Explain PL/SQL tables and user defined records.
- 3.6 Explain Input/Output statements
- 3.7 Explain decision making statements and illustrate
- 3.8 Explain looping statements and illustrate
- 3.9 Define procedure and function
- 3.10 Describe the advantages of subprograms.
- 3.11 Explain handling procedures and functions with example programs.
- 3.12 Explain the parameter modes in PL/SQL with examples (in , out and in out)

4.0 Advanced PL/SQL

- 4.1 Define cursor.
- 4.2 Classify cursors

- 4.3 Explain implicit cursor with example
- 4.4 Explain explicit cursors with example
- 4.5 Define trigger
- 4.6 List Advantages of triggers
- 4.7 Explain database triggers.

5.0 Concepts of No SQL & Mongo DB.

5.1 No SQL

- 5.1.1 List features of NOSQL
- 5.1.2 Compare RDBMS and NoSQL
- 5.1.3 List the Advantages and Disadvantages of NoSQL
- 5.1.4 Know about the ACID and BASE system.
- 5.1.5 Compare ACID and BASE properties
- 5.1.6 NoSQL
 - 5.1.6.1 Key-value stores,
 - 5.1.6.2 Column-oriented,
 - 5.1.6.3 Graph oriented Databases
 - 5.1.6.4 Document oriented Databases.

5.2 MongoDB

- 5.2.1 What is MongoDB.
- 5.2.2 List the advantages of MongoDB
- 5.2.3 Explain the Creation, Dropping, Creation of Collection
- 5.2.4 Dropping of Collection of Database in MongoDB
- 5.2.5 Explain the Data types of MongoDB.
- 5.2.6 Explain Inserting Document, Query Document, Update Document, Deleting Document & Sorting Document.

COURSE CONTENT

1. Concepts of DBMS & RDBMS

Define DBMS – Purpose of DBMS - Data Abstraction – Data Models – Instances and Schemas – Data Independence – Data Definition Language - Data Manipulation Language – Database Administrator - Database Users – Database system Structure.

Entities – Relationships and Relationship sets – Mapping constraints – Entity – Relationship Diagram – Super key , Candidate key and Primary key - Reducing E- R Diagrams to tables – Generalization and Specialization – Aggregation – Functional Dependencies - Normal forms 1NF , 2 NF , 3 NF

2. Concepts of SQL

Benefits of SQL – Embedded SQL – Lexical conventions – Naming objects and parts – Referring objects and parts – Literals – Text – Integer – Number – Data types – Character data types – Number data type – Long data type – Raw and Long Raw data types – Pseudo columns – comments within SQL statements – comments on schema objects.

Operators – Unary and Binary operators – Precedence- Arithmetic operators – character operators – comparison operators – logical operators- set operators – other operators – DDL Commands – Integrity Constraints – DML Commands - functions – single row functions – numeric functions – character functions

– date functions – conversion functions – other functions- Group functions. Transaction control commands-Sub queries - Joins.

3. Basics of PL/SQL

Main features – architecture – advantage of PL/SQL – fundamentals – character set – Lexical units – Data types – data type conversion – Declaration -scope and visibility – assignments – expressions and comparisons – PL/SQL tables – user defined records.

Conditional control- IF statement – sequential control- GOTO and NULL statements. SQL support – national language support – Remote Access

Advantages of subprograms – procedures – Functions RETURN statement – forward declarations – actual versus formal parameters – positional and named notation - parameter modes

4. Advanced PL/SQL

Cursors – Implicit cursor – Explicit cursor – Triggers – Advantages - creating trigger – raising trigger -

5. NoSQL& Basics of MongoDB

Classification of Databases : RDBMS, OLAP, NoSQL.-Introduction to NoSQL- need for NoSQL – Comparison of RDBMS and NoSQL- Advantages and Disadvantages of NoSQL - BASE system – ACID System – Comparison of ACID and BASE properties – Classification of NoSQL as Key-value stores, Column-oriented, Graph and Document oriented Databases

Introduction to MongoDB - advantages of MongoDB - applications of MongoDB - Installation of MongoDB - Creation of Database - Dropping of Database - Creation of Collection - Dropping of Collection - Data types of MongoDB - different Commands of MongoDB - Inserting Document - Query Document - Updating Document – Deleting Documents - Sorting Documents

REFERENCE BOOKS

1. Database System Concepts --- Silberschatz, Henry F. Korth, S. Sudarshan
2. Oracle Database 11g :The Complete Reference - Kevin Loney
3. Understanding ORACLE -- James T. Peary & Joseph G. Laseer.
4. RDBMS with ORACLE -- Rolland.
5. ORACLE series books of ORACLE Press – TMH.
6. Starting out with Oracle – Covering Databases -- John Day & Craig Van
7. PL/SQL, Developer Tools & DBA -- Slyke, Dreamtech
8. www.nosql-database.org
9. www.mongodb.org

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.5
Unit test-2	From 3.6 to 5.2.5

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-306	Python Programming Lab	3	45	40	60

COURSE OBJECTIVES	Upon completion of the course the student shall able to learn - Basics of Python programming - Decision Making and Functions in Python - Object Oriented Programming using Python.	
CO No.	COURSE OUTCOMES	
CO 1	CCN-306.1	Execute Simple python programs
CO 2	CCN-306.2	Execute Python programs using expressions, operators
CO 3	CCN-306.3	Execute python programming using Functions, packages
CO 4	CCN-306.4	Demonstrate Python programs using Lists
CO 5	CCN-306.5	Develop Python programs using OOP Concepts and exceptions
CO 6	CCN-306.6	Demonstrate Debugging of Python Programs

LEARNING OUTCOMES

1. Write and execute simple python Program.
2. Write /execute simple 'Python' program: Develop minimum 2 programs using different data types (numbers, string, tuple, list, and dictionary).
3. Write /execute simple 'Python' program: Develop minimum 2 programs using Arithmetic Operators, exhibiting data type conversion.
4. (i) Write simple programs to convert U.S. dollars to Indian rupees.
(ii) Write simple programs to convert bits to Megabytes, Gigabytes and Terabytes.
5. Write simple programs to calculate the area and perimeter of the square, and the volume & perimeter of the cone.
6. Write program to: (i) determine whether a given number is odd or even. (ii) Find the greatest of the three numbers using conditional operators.
7. Write a program to: i) Find factorial of a given number. ii) Generate multiplication table up to 10 for numbers 1 to 5.
8. Write a program to: i) Find factorial of a given number. ii) Generate multiplication table up to 10 for numbers 1 to 5 using functions.
9. Write a program to: i) Find factorial of a given number using recursion. ii) Generate Fibonacci sequence up to 100 using recursion.
10. Write a program to: Create a list, add element to list, delete element from the lists.
11. Write a program to: Sort the list, reverse the list and counting elements in a list.
12. Write a program to: Create dictionary, add element to dictionary, delete element from the dictionary.

13. Write a program to: To calculate average, mean, median, and standard deviation of numbers in a list.
14. Write a program to: To print Factors of a given Number.
15. File Input/output: Write a program to: i) To create simple file and write "Hello World" in it.
ii) To open a file in write mode and append Hello world at the end of a file.
16. Write a program to :i) To open a file in read mode and write its contents to another file but replace every occurrence of character 'h' ii) To open a file in read mode and print the number of occurrences of a character 'a'.
17. Write a Program to: Add two complex number using classes and objects.
18. Write a Program to: Subtract two complex number using classes and objects.
19. Write a Program to: Create a package and accessing a package.

TIME SCHEDULE

Sl.	Major Topic	Periods	CO'S mapped
1.	Write and execute simple python Program.	3(2,1)	CO1,CO6
2.	Write /execute simple 'Python' program: Develop minimum 2 programs using different data types (numbers, string, tuple, list, dictionary).	4(2,1,1)	CO1,CO2,CO6
3.	Write /execute simple 'Python' program: Develop minimum 2 programs using Arithmetic Operators, exhibiting data type conversion.	4(2,1,1)	CO1,CO2,CO6
4.	(i) Write simple programs to convert U.S. dollars to Indian rupees. (ii) Write simple programs to convert bits to Megabytes, Gigabytes and Terabytes	3(1,1,1)	CO1,CO2,CO6
5.	Write simple programs to calculate the area and perimeter of the square, and the volume & perimeter of the cone.	3(1,1,1)	CO1,CO2,CO6
6.	Write program to: (i) Determine whether a given number is odd or even. (ii) Find the greatest of the three numbers using conditional operators.	3(1,1,1)	CO1,CO2,CO6
7..	Write a program to: i) Find factorial of a given number. ii) Generate multiplication table up to 10 for numbers 1 to 5.	2(1,1)	CO1,CO2,CO6
8.	Write a program to: i) Find factorial of a given number. ii) Generate multiplication table up to 10 for numbers 1 to 5 using functions.	2(1,1)	CO1,CO2,CO3
9.	Write a program to: i) Find factorial of a given number using recursion. ii) Generate Fibonacci sequence up to 100 using recursion.	2(1,1)	CO1,CO2,CO3
10.	Write a program to: To print Factors of a given Number.	2(1,1)	CO4,CO6

11.	Write a program to: Create a list, add element to list, delete element from the lists	3(1,1,1)	CO1,CO3,CO6
12.	Write a program to: Sort the list, reverse the list and counting elements in a list	3(1,1,1)	CO2,CO3,CO6
13.	Write a program to: Create dictionary, add element to dictionary, delete element from the dictionary.	3(1,1,1)	CO2,CO3,CO6
14.	Write a program to: To calculate average, mean, median, and standard deviation of numbers in a list.	3(1,1,1)	CO2,CO3,CO6
15.	File Input/output: Write a program to: i) To create simple file and write "Hello World" in it. ii) To open a file in write mode and append Hello world at the end of a file.	4(2,2)	CO1,CO6
16.	Write a program to :i) To open a file in read mode and write its contents to another file but replace every occurrence of character 'h' ii) To open a file in read mode and print the number of occurrences of a character 'a'.	4(2,2)	CO1,CO6
17.	Write a Program to: Add two complex number using classes and objects.	4(2,2)	CO5,CO6
18.	Write a Program to: Subtract two complex number using classes and objects	3(2,1)	CO5,CO6
19.	Write a Program to: Create a package and accessing a package.	2(1,1)	CO5,CO6
		60	

KEY COMPETENCIES

Sl.No	Name of the Experiment	Objectives	Key Competencies
1.	Write and execute simple python Program.	Write a simple python program to print Hello World! and debug and execute	1. Know the usage of Python IDLE 2. Edit and save the program 3. Check for the syntax errors and clear the errors 4. Run the program and check for the output.
2.	Write /execute simple 'Python' program: Develop minimum 2 programs using different data types (numbers, string, tuple, list, dictionary).	Write a Python program to identify different data types.	1. Identify different data types 2. Write basic python program using datatypes 3. Evaluate arithmetic expression 4. Run the program 5. Rectify the syntactical errors 6. Execute the program 7. Check the output for its correctness

3.	Write /execute simple 'Python' program: Develop minimum 2 programs using Arithmetic Operators, exhibiting data type conversion.	Write a Python program to identify arithmetic operators and data type conversion	1. Identify different arithmetic operators 2. Build arithmetic expressions 3. Identify the priorities of operators 4. Evaluate arithmetic expression 5. Run the program 6. Rectify the syntactical errors 7. Execute the program Check the output for its correctness
4.	(i) Write simple programs to convert U.S. dollars to Indian rupees. (ii) Write simple programs to convert bits to Megabytes, Gigabytes and Terabytes.	Write a Python program to identify arithmetic operators and data type conversion	1. Identify different arithmetic operators 2. Build arithmetic expressions 3. Identify the priorities of operators 4. Evaluate arithmetic expression 5. Run the program 6. Rectify the syntactical errors 7. Execute the program Check the output for its correctness
5.	Write simple programs to calculate the area and perimeter of the square, and the volume & perimeter of the cone.	Write a Python program to identify arithmetic operators and data type conversion	1. Identify different arithmetic operators 2. Build arithmetic expressions 3. Identify the priorities of operators 4. Evaluate arithmetic expression 5. Run the program 6. Rectify the syntactical errors 7. Execute the program Check the output for its correctness
6.	Write program to: (i) Determine whether a given number is odd or even. (ii) Find the greatest of the three numbers using conditional operators.	Write a Python program to identify conditional statements in Python.	1. Build a relational expression 2. Use the if statement for decision making 3. Rectify the syntax errors 4. Check the output for correctness
7.	Write a program to :i) Find factorial of a given number. ii) Generate multiplication table up to 10 for numbers 1 to 5.	Write a Python program to identify loops statements in Python.	1. Build the termination condition for looping 2. Use while statement with correct syntax 3. Check whether correct number of iterations are performed by the while loop 4. Rectify the syntax errors 5. Debug logical errors
8.	Write a program to : To print Factors of a given Number.	Write a Python program to make use of function.	1. Build the termination condition for looping 2. Use while statement with correct syntax 3. Check whether correct number of iterations are performed by the while loop 4. Rectify the syntax errors Debug logical errors

9.	Write a program to: i) Find factorial of a given number using recursion. ii) Generate Fibonacci sequence up to 100 using recursion.	Write a Python program to make use of recursion.	1. Build the application using recursion. 2. Build the terminating condition for recursion. 3. Rectify the syntax errors 4. Debug logical errors
10.	Write a program to : To print Factors of a given Number.	Write a Python program to identify loops statements in Python.	1. Build the termination condition for looping 5. Use while statement with correct syntax 6. Check whether correct number of iterations are performed by the while loop 7. Rectify the syntax errors 1. Debug logical errors
11.	Write a programs to: Create a list, add element to list, delete element from the lists.	Write a Python program to identify various lists and list manipulation methods in Python.	1. Create a one list with correct syntax 2. Create a list 3. Read elements from list 4. Add elements to list 5. Delete elements 6. Rectify the syntax errors 7. Debug logical errors 8. Check for the correctness of output for the given input
12.	Write a programs to: Sort the list, reverse the list and counting elements in a list.	Write a Python program to identify various lists and list manipulation methods in Python.	1. Create a one list with correct syntax 2. Create a list 3. Read elements from list 4. Add elements to list 5. Delete elements 6. Rectify the syntax errors 7. Debug logical errors 8. Check for the correctness of output for the given input
13.	Write a programs to: Create dictionary, add element to dictionary, delete element from the dictionary.	Write a Python program to identify various dictionary and dictionary manipulation methods in Python.	1. Create a one dictionary with correct syntax 2. Create a dictionary 3. Read elements from list 4. Add elements to dictionary 5. Delete elements from dictionary 6. Rectify the syntax errors 7. Debug logical errors 8. Check for the correctness of output for the given input

14	Write a program to: To calculate average, mean, median, and standard deviation of numbers in a list.	Write a Python program to identify various statistical functions.	1. Create a list 2. add elements to list 3. perform statistical functions on that list
15.	File Input/output: Write a program to :i) To create simple file and write "Hello World" in it. ii) To open a file in write mode and append Hello world at the end of a file.	Write a Python program to identify the steps to create a file and append to file.	1. Create a Python file 2. Add contents to file
16	Write a program to :i) To open a file in read mode and write its contents to another file but replace every occurrence of character 'h' ii) To open a file in read mode and print the number of occurrences of a character 'a'.	Write a Python program to identify the steps to open a file in read/write mode.	1. Open a Python file in write mode 2. Add contents to the file 3. Open a Python file in Read mode 4. Print the file
17.	Write a Program to: Add two complex number using classes and objects.	Write a Python program to identify the steps to create class and create an object in Python.	1. Create a class using Python 2. Create an object in Python 3. Debug the python program Check the correctness
18	Write a Program to: Subtract two complex number using classes and objects	Write a Python program to identify the steps to create class and create an object in Python.	1. Create a class using Python 2. Create an object in Python 3. Debug the python program Check the correctness
19	Write a Program to: Create a package and accessing a package.	Write a Python program to practice in creating packages and accessing packages	1. Create a package using Python 2. Access the package in Python 3. Debug the python program Check the correctness

Course Code	Course Title	No. of periods/week	Total No. of periods	Marks for FA	Marks for SA
CCN-307	Computer Hardware & Network Maintenance Lab	6	90	40	60

S No	Chapter/ Unit Title	No. of Periods	COs Mapped
1.	Computer Hardware	45	CO1,CO2,CO3
2.	Computer Networking	30	CO3.CO4.CO5
3.	Network Maintenance through server	15	CO4,CO5,CO6
	Total	90	
COURSE OBJECTIVES	1. Identify all the components of mother board. 2. Modify CCNOS settings as required 3. Troubleshoot desktop computer 4. Troubleshoot individual resources like keyboard, Monitor, Printers 5. Install drives, NIC cards, modems(internal, external) 6. Install network devices, design and develop network. 7. Understand IP address classes and sub netting 8. Prepare cross and straight Ethernet cables 9. Install and configure proxy server 10. Install any network operating system and Control/maintain the network and network resources using server administration and Troubleshoot the entire network		

Course Outcomes	Upon completion of the course the student shall be able		
	CO1	CCN-307.1	Assemble the PC with suitable components.
	CO2	CCN-307.2	Troubleshoot desktop system and individual peripheral devices.
	CO3	CCN-307.3	Demonstrate configuring computer network with subnetting
	CO4	CCN-307.4	Perform user and group management techniques through
	CO5	CCN-307.5	Troubleshoot the computer network.
	CO6	CCN-307.6	Configure any network device.

Learning out comes:

1. Identify various mother board Components
2. Perform various operations and modifications required for CCNOS setup.
3. Print the summary of your system Hardware and verify for correctness
4. Upgrading memory and verify the effect after upgrading.
5. Hard drive, optical drive installation.
6. How to recover lost data on hard drive.
7. Trouble shooting keyboard and monitor
8. Trouble shoot Printer Problems

9. Installation of Network card
10. Dis-assembling and assembling of working desktop.
11. Preparing the Ethernet cable for cross and direct connections using crimping tool and test using LAN tester.
12. Installation of a switch and connecting systems to a network switch.
13. Installation of a modem (internal, external or USB) and connecting to internet.
14. Using FTP for uploading and downloading files.
15. Installation and configuring the proxy server for internet access.
16. Setting of particular IP address to an existing terminal system
17. Installation of network operating system
18. Creating and managing user accounts through network server.
19. Configuration of DHCP and DNS
20. Exercise on File/Folder accessing rights for sharing
21. Exercise on remote desktop.
22. Exercise on setting up of VPN on network

OBJECTIVES AND KEY COMPETENCIES:

S.N o.	Name of the experiment	Objectives	Key Competencies
1	Exercise on Identification and familiarization of various components of mother board.	Identification and familiarization of various components of computer system.	❖ Identify and note down mother board, Components and Chips. ❖ Identify various Internal and External slots in the mother board and clean them with blower/ Brush. ❖ Practice Inserting and Removing RAM with care. ❖ Measure the Output voltages of SMPS.
2	Exercise on various operations and modifications required for CCNOS setup.	Perform various operations and modifications required for CCNOS setup.	❖ Identify location of CCNOS battery on mother board. ❖ Know how to replace CCNOS battery. ❖ Identify keyboard key for entering BIOS setup. ❖ Setup CCNOS settings ❖ Check the status of CCNOS settings after replacement.
3	Exercise on Print the summary of your system Hardware and verify for correctness	Print the summary of your system Hardware and verify for correctness	❖ Know how to open system summary window ❖ Check whether all the hardware peripherals are working properly or not. ❖ Know how to install device drivers ❖ Know how to enable and disable hardware peripherals. ❖ Print the hardware summary page.
4	Exercise on Upgrading memory and verify the effect after upgrading.	Upgrading memory and verify the effect after upgrading.	❖ Know the location of RAM slots ❖ Know how to insert or replace RAM chips ❖ Check the system properties for confirming the RAM up gradation.

5	Exercise on Hard drive, optical drive installation.	Hard drive, optical drive installation.	Hard drive: ❖ Identify the Hard drive slot. ❖ Know how to remove power supply and SATA cables from hard drive. ❖ Unscrew Hard drive from computer case ❖ Replace new Hard drive and fix it in computer case ❖ Know how to connect power supply cable and SATA cables to Hard drive ❖ Check for the working condition of new Hard Drive. Optical drive: ❖ Identify the Optical drive slot. ❖ Know how to remove power supply and SATA cables from Optical drive. ❖ Unscrew Optical drive from computer case ❖ Replace new Optical drive and fix it in computer case
			❖ Know how to connect power supply cable and SATA cables to Optical drive ❖ Check for the working condition of Optical drive.
6	Exercise on recovery of lost data on hard drive.	How to recover lost data on hard drive.	❖ Verify the available recovery tools of Operating system. ❖ Know how to recover lost data on Hard drive using Restore point. ❖ Know how to recover lost data on Hard drive using Recovery Image.
7	Exercise on Trouble shooting keyboard and monitor.	Trouble shooting keyboard and monitor.	❖ few keys do not work. ❖ keyboard does not work at all. ❖ Key continuous to repeat after being released. ❖ key produces wrong character. ❖ Power light (led) does not go on, no picture. ❖ Power LED light is on no picture power up. ❖ Power on but monitor display wrong character. ❖ Monitor flickers has wary lines. ❖ Screen goes blank 30 seconds or minute after the keyboard is left untouched
8	Exercise on Printer Problems	Printer Problems	Laser printer: ❖ Printer never leaves warm-up mode. ❖ Paper Jam message is displayed ❖ Printed messages are distorted ❖ RE-filling and replacing cartridge ❖ Replacing damaged drum with new one. ❖ Perform head cleaning DMP ❖ Print head moves back and forth but nothing prints. ❖ Print self-test works but printing from a computer application does not work etc.,

9	Exercise on Installation of Network card.	Installation of Network card.	❖ Identify the slot for placing NIC card ❖ Know how to place NIC card ❖ Install required NIC driver ❖ Check for working status of NIC card
10	Exercise on Dis-assembling and assembling of working desktop.	Dis-assembling and assembling of working desktop.	❖ Identify all the peripherals of Desktop computer. ❖ Check the working condition of system before dis-assembling it. ❖ Dis-assemble all the peripherals. ❖ Assemble all the peripherals. ❖ Check the working condition of system after assembling it.
11	Exercise on Preparing the Ethernet cable for cross and direct connections using	Preparing the Ethernet cable for cross and direct connections using crimping	❖ Know the color pattern of Ethernet cable for direct connection. ❖ Prepare UTP cable for direct connection using crimping tool. ❖ Check the working condition of cable using LAN tester.
	crimping tool and test using LAN tester.	tool and test using cable tester.	❖ Know the color pattern of Ethernet cable for cross connection. ❖ Prepare UTP cable for cross connection using crimping tool. ❖ Check the working condition of cable using LAN tester.
12	Exercise on Installation of a switch and connecting systems to a network switch.	Installation of a switch and connecting systems to a network switch	❖ Know the purpose of switch ❖ Run Ethernet cables from switch to individual nodes ❖ Connect Ethernet cables of nodes to switch. ❖ Check the network status of the connection in computer system.
13	Exercise on Installation of a modem (internal, external or USB) and connecting to internet.	Installation of a modem (internal, external or USB) and connecting to internet.	Internal modem ❖ Identify PCI slot for placing Internal modem ❖ Connect internal modem ❖ Install required modem driver ❖ Check for the working condition External modem ❖ Connect External modem ❖ Install required modem driver ❖ Check for the working condition USB modem ❖ Connect USB modem ❖ Install required modem driver ❖ Check for the working condition
14	Exercise on Using FTP for uploading and downloading files.	Using FTP for uploading and downloading files.	❖ Know about FTP protocol ❖ Know how to upload file using FTP ❖ Know how to download file using FTP

15	Exercise on Installation and configuring the proxy server for internet access	Installation and configuring the proxy server for internet access	❖ Know about proxy server. ❖ Know how to install proxy server. ❖ Know how to configure proxy server.
16	Exercise on Setting of particular IP address to an existing terminal system	Setting of particular IP address to an existing terminal system	❖ Know about IP addresses ❖ Know how to set IP addresses to the computer systems in a LAN
17	Exercise on Installation of network operating system	Installation of network operating system	❖ Know about different network operating systems. ❖ Install required Network operating systems ❖ Configure the system with the proper settings.
18	Exercise on Creating and managing user accounts through network server.	Creating and managing user accounts through network server.	❖ Know how to create user accounts ❖ Know how to modify user accounts ❖ Know how to delete user accounts
19	Exercise on Configuration of DHCP and DNS.	Configuration of DHCP and DNS.	❖ Know about static IP address, dynamic IP address ❖ Know about DHCP ❖ Configure the DHCP ❖ Know about DNS ❖ Configure the DNS
20	Exercise on File/Folder accessing rights for sharing	File/Folder accessing rights for sharing.	❖ Know the different accessing rights ❖ Know how to give access rights ❖ Know how to remove access rights ❖ Know how to share file/folders
21	Exercise on remote desktop.	Exercise on remote desktop.	❖ Know about remote login ❖ Know how to login to the remote desktop
22	Exercise on setting up of VPN on network	Exercise on setting up of VPN on network	❖ Know about VPN ❖ Know how to configure VPN service

Course Code	Course Title	No. of periods/week	Total No. of periods	Marks for FA	Marks for SA
CCN-308	DBMS Lab	4	60	40	60

Sno	UNIT TITLE	NO. OF PERIODS	COS
1	Concepts of DBMS & RDBMS	8	CO1
2	Concepts of SQL	16	CO2
3	Basics of PL/ SQL	12	CO3
4	Advance PL/SQL	16	CO4
5	Concepts of NoSQL & MongoDB.	8	CO5
		60	

COURSE OBJECTIVES	Upon completion of the course the student shall able to learn
	• Insert, update, delete and select data into/from Relation Database • Develop PL/SQL programs • Insert, update, delete and select data from MongoDB

LEARNING OUTCOMES

1. Know installation of Oracle
2. Exercise on creating tables.
3. Exercise on inserting records
4. Exercise on updating records
5. Exercise on modifying the structure of the table
6. Exercise on Select command
7. Exercise on querying the table using clauses like WHERE, ORDER BY, IN, AND, OR, NOT, IS NULL
8. Exercise on GROUP BY, HAVING
9. Exercise on Number functions, character functions, conversion functions and date functions, group functions
10. Exercise on set operators
11. Exercise on sub queries
12. Exercise on Joins
13. Exercise on various date and number format models
14. Exercise on creating tables with integrity constraints
15. Write programs using PL/SQL control statements
16. Exercise on Procedures
17. Exercise on Functions
18. Exercise on Cursors
19. Exercise on Triggers
20. Exercise on Installation of MongoDB
21. Exercise on Creation and Dropping of Database
22. Exercise on Creation and Dropping of Collections.
23. Exercise on Commands of MongoDB- Insert, update, find, delete and sorting of Documents.

Mini Project : Student has to develop a Mini project applying the skills acquired from the learning outcomes of this course.

Time Schedule:

Sl.No	Name of the Experiment	Periods
1	Know installation of Oracle	1
2	Exercise on creating tables.	1
3	Exercise on inserting records	2
4	Exercise on updating records	2
5	Exercise on modifying the structure of the table	1
6	Exercise on SELECT command	2
7	Exercise on querying the table using clauses like WHERE, ORDER, IN,AND, OR,NOT, IS NULL	18
8	Exercise on GROUP BY, HAVING	2
9	Exercise on Number functions, character functions, conversion functions and date functions, group functions	3
10	Exercise on set operators	2
11	Exercise on sub queries	3
12	Exercise on Joins	3
13	Exercise on various date and number format models	1
14	Exercise on creating tables with integrity constraints	2
15	Write programs using PL/SQLcontrol statements	6
16	Exercise on Procedures	2
17	Exercise on Functions	1
18	Exercise on Cursors	2
19	Exercise on Triggers	2
20	Exercise on Installation of MongoDB	1
21	Exercise on Creation and Dropping of Database	1
22	Exercise on Creation and Dropping of Collections	1
23	Exercises on commands of MongoDB	1
Total		60

KEY COMPETENCIES

Sl.No	Name of the Experiment	Objectives	Key Competencies
1	Know installation of Oracle	Perform the following: - To identify the version of Oracle being installed - To understand the RAM and HDD requirements for Oracle installation - To comprehend the installation steps correctly - Setting up of Oracle Administrative Password - Configuring the Oracle database after post- installation steps of Oracle viz configuring administrative rights for performing - To login to Oracle as administrator account and Oracle user account	❖ Observe Oracle version being installed ❖ Observe the RAM & HDD requirements ❖ Rectify for any Oracle installation errors ❖ Able to login as Administrator and as Oracle user account
2	Exercise on creating tables.	Perform the following: - To login with Oracle user account - To give correct syntax for table creation - To give correct data type for the required fields with appropriate size - To display the structure of the table	❖ Correct Table creation syntax errors ❖ Correct the wrong data types and inappropriate sizes for the respective fields ❖ Check for displaying the structure of the table
3	Exercise on inserting records	Perform the following: - Check for the required table present already - To insert the records correctly - To display the records correctly	❖ Correct syntax errors for Insertion of record ❖ Check for insertion of proper values for the required fields ❖ Verify the correct values pertaining to the record are inserted in the required table ❖ Check for displaying of the records correctly

Sl.No	Name of the Experiment	Objectives	Key Competencies
4	Exercise on updating records	Perform the following: i. Check for the required table present already ii. To update the records correctly iii. To display the updated records	❖ Correct syntax errors for updating of record ❖ Check for updating of propervalues for the required fields ❖ Check for displaying of the updated records correctly
5	Exercise on modifying the structure of the table	Perform the following: i. To identify the required table present in the system already ii. To add new column iii. To display the records correctly	❖ Correct syntax errors in modifying the structure of the table ❖ Check whether required field is newly added to the existing table ❖ Check for displaying of the modified table correctly
6	Exercise on SELECT command	Perform the following: i. To identify the required table present already ii. To display the records in the required table	❖ Check for syntax error in usage of Select command ❖ Check whether Select command is given correctly to display all the records
7	Exercise on querying the table using clauses like WHERE, ORDER, IN,AND, OR,NOT, IS NULL	Perform the following: i. To use the Select command ii. To use the clauses WHERE, ORDER, IN,AND, OR, NOT, IS NULL along with Select command on the given records in the table	❖ Check for syntax error in usage of Select command with appropriate clauses ❖ Check whether Select command along with appropriate clause is given correctly for the required condition ❖ Check the usage of clauses WHERE, ORDER, IN,AND, OR, NOT along with Select command appropriately
8	Exercise on GROUP BY, HAVING	Perform the following: i. To use the Select command To use the clauses GROUP BY, HAVING along with Select command on the given records in the table	❖ Check for syntax error in the usage GROUP BY, HAVING ❖ Check for usage of GROUP BY, HAVING ❖ Verify output values based on certain condition on few records
9	Exercise on Number functions, character functions, conversion functions and date functions, group functions	Perform the following: i. To use functions ii. To use set command along with WHERE condition, GROUP BY, HAVING	❖ Check for syntax error of various functions ❖ Check for usage of various functions ❖ Verify output values based on certain condition on few records

Sl.No	Name of the Experiment	Objectives	Key Competencies
10	Exercise on SET operators	Perform the following iii. To use set command iv. To use set command along with WHERE condition	❖ Check for syntax error in the usage of SET command ❖ Check for usage of SET command for updating values based on certain condition on few records
11	Exercise on sub queries	Perform the following i. To use Select command ii. To use appropriate Operators IN	❖ Check for the syntax error in usage of sub queries ❖ Check for the correctness of the usage of appropriate operators used
12	Exercise on Joins	Perform the following i. To create two tables ii. To use the common field if two tables are joined iii. To know different types of Join	❖ Check for the correctness of the syntax used for joining ❖ Check if the join is created between two tables ❖ Check if self-join is created
13	Exercise on various date and number format models	Perform the following: i. To use date formats correctly ii. To use number formats correctly	❖ Check for the syntax of the date formats ❖ Check for the syntax of the number formats
14	Exercise on creating tables with integrity constraints	Perform the following i. Create Primary key ii. Create Foreign key or referential integrity constraint iii. Create NOT NULL constraint iv. Create UNIQUE Key constraint v. Create CHECK constraint	❖ Check for the syntax errors in usage of all types of Integrity constraints ❖ Check whether different types of Integrity constraints are used
15	Write programs using PL/SQL control statements	Perform the following i. To use IF .. ELSE statements ii. To use iterative statements – Simple loop, While Loop, For Loop	❖ Check for the syntax of IF.. ELSE statements ❖ Check for the syntax of all iterative statements
16	Exercise on Procedures	Perform the following i. To know the concept of stored procedures ii. To declare procedures iii. The type of parameters IN, IN OUT, OUT iv. To call procedures from other procedures	❖ Check for proper declaration of procedures ❖ Check for syntax of parameters and its type ❖ Check for proper calling of procedures

Sl.No	Name of the Experiment	Objectives	Key Competencies
17	Exercise on Functions	Perform the following i. To know the concept of stored functions ii. To declare function with return data iii. To call functions from other functions	❖ Check for proper declaration of function ❖ Check for syntax of parameters and its data type ❖ Check for proper return data type from the functions ❖ Check for variable assignment to get the returned value from the function
18	Exercise on Cursors	Perform the following i. To know the concept cursors ii. To know the fetch data from database	❖ Check for the syntax of cursor ❖ Check for open cursor, fetch data, close cursor ❖ Check for the result
19	Exercise on Triggers	Perform the following i. To know the concept of triggers ii. Validation before and after insert, before and after update and , before and after delete data	❖ Check for the syntax of trigger ❖ Write a trigger which raises before insert data ❖ Raise trigger ❖ Repeat the procedure for remaining ❖ Check for the result
20	Exercise on Installation of MongoDB	Perform the following i. To download and install MongoDB	❖ Observe MongoDB version being installed ❖ Observe the RAM & HDD requirements ❖ Rectify for any MongoDB installation errors Able to login as Administrator
21	Exercise on Creation and Dropping of Database	Perform the following i. Create the Database ii. Drop the Database	❖ Know the use of create Database() and drop Database() ❖ Correct Database creation syntax errors ❖ Check for displaying the database name

Sl.No	Name of the Experiment	Objectives	Key Competencies
22	Exercise on Creation and Dropping of Collections	Perform the following i. Create the Collection ii. Drop the Collection	❖ Know the use of create Collection() and drop() ❖ Correct Database creation syntax errors ❖ Check for collection name ❖ Check for the collection dropped
23	Exercises on commands of MongoDB	Execute the following commands of MongoDB i. Insert the Document ii. update the Document iii. find the Document iv. Delete the Document v. sort the Documents	❖ Know the syntax of insert(), update(), find(), remove(), sort() functions. ❖ Correct syntax errors. ❖ Check out for different input values.

Course Code	Course Title	No. of periods/week	Total No. of periods	Marks for FA	Marks for SA
CCN-309	Multimedia Lab	3	45	40	60

S.NO	MAJOR TOPICS	NO. OF PERIODS	CO's mapped
1	MS Access - create Database, create table with and without constraints, Insert, delete, update records, implement queries, create relationship between two tables	10	CO1
2	Pagemaker - Creation of publication using tools, text, shapes, etc., Custom template, colours, text block, Objects, Styles, Page elements, Printing the documents	20	CO2
4	Photoshop – Different tools, Working with Layers, Working with painting tools, Colors, Brushes	15	CO3
	TOTAL	45	

Course Objectives	(i) Familiarize with the features of MS Access. (ii) Familiarize with the features of Adobe PageMaker (iii) Familiarize with the features of Adobe Photoshop
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CO No.	COURSE OUTCOMES
CO1	CCN-309.1 Working with databases, tables, manipulating records, queries and establishing relationship among tables.
CO2	CCN-309.2 Create and print publication files such as Textbooks, Visiting Cards, Invitation Cards, etc. using Adobe Pagemaker
CO3	CCN-309.3 Design and enhance the quality all types of picture files using Adobe Photoshop

Learning Outcomes:

Practice with MS-Access

1. To create Database
2. To Create table with and without constraints
3. To Insert, delete, update records
4. To implement queries
5. To create relationship between two table

Practice with Adobe Page Maker

6. Exercise on Installation, invoking and familiarizing Adobe Page Maker.
7. Exercise on Page Maker Tools.
8. Exercise on pallets and formatting pages
9. Exercise on text formatting
10. Exercise on Advanced text formatting
11. Exercise on Graphics tools
12. Exercise on object transformations.
13. Exercise on color options.
14. Exercise on graphics with layers using photoshop plug-ins
15. Exercise on import and export options.
16. Exercise on creating visiting card
17. Exercise on creating book cover page
18. Exercise on creating hotel menu card
19. Exercise on creating invitation card
20. Exercise on creating brochure

Practice with Adobe Photoshop

21. Exercise on Installation, invoking and familiarizing Adobe Photoshop
22. Exercise on Images
23. Exercise on Resizing& Cropping Images
24. Exercise on Working With Basic Selections
25. Exercise on Layers
26. Exercise on Painting In Photoshop
27. Exercise on Photo Retouching
28. Exercise on Colour Correction
29. Exercise on Quick Mask Mode
30. Exercise on Pen Tool
31. Exercise on Creating Special Effects
32. Exercise on Exporting Your Work
33. Exercise on Logo Creation

Mini Project : Student has to develop a Mini project applying the skills acquired from the learning outcomes of this course.

KEY COMPETENCIES

Exp No	Name of the Experiment	Objectives	Key Competencies
1	To create database	Open MS Access Create database save	Database creation
2	To create table	• Open MS Access • Create database • Create table • Use primary key	Table creation
3	To insert/delete/update records into table	7. Open msaccess 8. Create database 9. Create table 10. Insert/delete/update records	Able to insert/update/delete and delete and update records into the table
4	To implement queries	f. Open msaccess g. Create database Create table h. Use Select command	Able to display contents of the table based on the user requirement
5	Create relationships between tables	h. Create table i. Create one more tables j. Insert records k. Use relationship option	Able to link tables
6	Exercise on Installation, invoking and familiarizing Adobe Page Maker.	Installation, invoking and familiarizing Adobe Page Maker.	❖ Installing page maker ❖ Familiarize with pagemaker environment
7	Exercise on Page Maker Tools.	Page Maker Tools.	❖ Using Tool box, zero position, pointer tool, text tool, rotate tool, crop tool, oblique line tool,
			constrained line tool, box tool, rectangle frame, circle tool, circular frame, polygon tool, polygon frame, hand tool and zoom tool,
8	Exercise on pallets and formatting pages	pallets and formatting pages	❖ Use of paper size, page size, control pallet, color pallet, styles pallet, layers pallet, master page pallet, hyperlink pallet and measurement system, grids, rulers and guides, insert pages.

9	Exercise on text formatting	text formatting	❖ Know the purpose of master pages, placing a text, Formatting text(size, styles), Paragraph setting, tab setting, bullets, numbering, hyphenation setting, setting and creating styles ,rotating text and color to text and save the document.
10	Exercise on Advanced text formatting	Advanced text formatting	❖ use spell check, divide the text into columns, work with indexes and page nation, use the find feature and save the document.
11	Exercise on Graphics tools	Graphics tools	❖ Create a document to work with graphics with the help of line tool, box tool, ellipse tool, polygon tool, rounded corners, fill, stroke and to place various graphics(at least 2 for each graphic tool) and save the document.
12	Exercise on object transformations.	object transformations.	❖ Transform the objects such as transforming a rectangle, resizing an ellipse, inserting cropping an image.
13	Exercise on color options.	color options.	❖ Adding color to a graphic shape, creating own colors with RGB, editing, copying, removing and replacing colors, grouping and

			ungrouping objects, linking objects, masking objects and save the document.
14	Exercise on graphics with layers using photoshop plug-ins	graphics with layers using photoshop plug-ins	❖ Create a document of five pages containing text and graphics and work with layers, moving objects between layers using layer options, using stacking order, using photoshop plug-ins

15	Exercise on import and export options.	import and export options.	❖ Create a new document and import text from HTML, MSword, spreadsheet, photo CD, acquiring tif image, managing linked files, to use export options to export text and graphics to jpeg format and to print the document, to publish the document in internet and save it,
16	Exercise on creating visiting card	creating visiting card	❖ Create visiting card with text and graphics on both sides with proper formatting.
17	Exercise on creating book cover page	creating book cover page	❖ Create front and back cover page of a book with text and graphics with proper formatting.
18	Exercise on creating hotel menu card	creating hotel menu card	❖ Create a hotel menu card with text and graphics with proper formatting.
19	Exercise on creating invitation card	creating invitation card	❖ Create a invitation card with text and graphics with proper formatting for required no. of pages.
20	Exercise on creating brochure	creating brochure	❖ Create a brochure for the firms like real estate companies, hospitals, educational institutions etc.,
21	Exercise on Installation, invoking and familiarizing Adobe Photoshop	Installation, invoking and familiarizing Adobe Photoshop	❖ Exploring the Toolbox ❖ The New CS4 Applications Bar & the Options Bar ❖ Exploring Panels & Menus ❖ Creating & Viewing a New Document ❖ Customizing the Interface ❖ Setting Preferences

22	Exercise on Images	Working with Images	❖ Zooming & Panning an Image ❖ Working with Multiple Images, Rulers, Guides & Grids ❖ Undoing Steps with History ❖ Adjusting Color with the New Adjustments Panel ❖ The New Masks Panel & Vibrance Color Correction Command ❖ The New Note Tool & the Save for Web & Devices Interface ❖ The New Auto-Blend & Auto-Align Layers Commands ❖ The New 3D Commands
23	Exercise on RESIZING & CROPPING IMAGES	RESIZING & CROPPING IMAGES	❖ Understanding Pixels & Resolution ❖ The Image Size Command ❖ Interpolation Options ❖ Resizing for Print & Web ❖ Cropping & Straightening an Image ❖ Adjusting Canvas Size & Canvas Rotation
24	Exercise on WORKING WITH BASIC SELECTIONS	WORKING WITH BASIC SELECTIONS	❖ Selecting with the Elliptical Marquee Tool ❖ Using the Magic Wand & Free Transform Tool ❖ Selecting with the Regular & Polygonal Lasso Tools ❖ Combining Selections ❖ Using the Magnetic Lasso Tool ❖ Using the Quick Selection Tool & Refine Edge ❖ Modifying Selections
25	Exercise on LAYERS	Working on Layers	❖ Understanding the Background Layer

			❖ Creating, Selecting, Linking & Deleting Layers ❖ Locking & Merging Layers ❖ Copying Layers, Using Perspective & Layer Styles ❖ Filling & Grouping Layers ❖ Introduction to Blending Modes ❖ Blending Modes, Opacity & Fill ❖ Creating & Modifying Text
26	Exercise on PAINTING IN PHOTOSHOP	PAINTING IN PHOTOSHOP	❖ Using the Brush Tool ❖ Working with Colors & Swatches ❖ Creating & Using Gradients ❖ Creating & Working with Brushes ❖ Using the Pencil & Eraser Tools ❖ Painting with Selections
27	Exercise on PHOTO RETOUCHING	PHOTO RETOUCHING	❖ Using The Red Eye Tool ❖ The Clone Stamp Tool ❖ The Patch Tool & the Healing Brush Tool ❖ The Spot Healing Brush Tool ❖ The Color Replacement Tool ❖ The Toning & Focus Tools ❖ Painting with History
28	Exercise on COLOR CORRECTION	COLOR CORRECTION	❖ Using Color Spaces & Color Modes ❖ The Variations Command ❖ The Auto Commands ❖ Adjusting Levels ❖ Adjust Curves, Non-Destructively, with Adjustment Layers
29	Exercise on QUICK MASK MODE	Using QUICK MASK MODE	❖ Using Quick Mask Options ❖ Painting a Selection ❖ Saving & Removing a Selection from the Background
30	Exercise on PEN TOOL	Working with the PEN TOOL	❖ Understanding Paths & the Pen Tool ❖ Creating Straight & Curved Paths ❖ Creating Combo Paths ❖ Creating a Clipping Path
31	Exercise on CREATING SPECIAL EFFECTS	CREATING SPECIAL EFFECTS	❖ Getting Started with Photoshop Filters ❖ Smart Filters ❖ Creating Text Effects

			❖ Applying Gradients to Text
32	Exercise on Photo Shop Credits	EXPORTING YOUR WORK	❖ Saving with Different File Formats ❖ Saving for Web & Devices ❖ Printing Options ❖ Photo shop Credits
33	Exercise on Logo Creation	Logo Creation	❖ To apply all the tools ❖ Prepare college logo ❖ Prepare logo for industry

DIPLOMA IN COMMUNICATION AND COMPUTER NETWORKING ENGINEERING

SCHEME OF INSTRUCTIONS AND EXAMINATION

CURRICULUM-2023

(IV Semester)

Sub Code	Name of the Subject	InstructionPeriods/ Week		Total Periods Per Semester	Scheme Of Examinations			
		Theory	Practicals		Duration (hrs)	Sessional Marks	End Exam Marks	Total Marks
THEORY SUBJECTS								
CCN-401	Network Management through TCP/IP	5	-	75	3	20	80	100
CCN-402	Web Technologies	5	-	75	3	20	80	100
CCN-403	Digital Electronics and Computer Organisation	5	-	75	3	20	80	100
CCN-404	OOP through Java	5	-	75	3	20	80	100
CCN-405	Cryptography and Network Security	5	-	75	3	20	80	100
PRACTICAL SUBJECTS								
CCN-406	Web Technologies Lab	-	4	60	3	40	60	100
CCN-407	Java Programming Lab	-	4	60	3	40	60	100
CCN-408	Communication Skills	-	3	45	3	40	60	100
CCN-409	Computer Networks and Security Lab	-	3	45	3	40	60	100
	ACTIVITIES		3	45				
	Total	25	17	630	-	260	640	900

CCN-402 COMMON WITH CM-402

CCN-403 COMMON WITH AIM-304, CCN-404 COMMON WITH CM-404

CCN-406 COMMON WITH CM-406, CCN-407 COMMON WITH CM-407

CCN-408 COMMON WITH ALL BRANCHES, CCN-409 COMMON WITH CM-409

Course Code	Course Title	No. of Periods/week	Total No. of periods	Marks for FA	Marks for SA
CCN-401	Network Management through TCP/IP	5	75	20	80

Time Schedule

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1	Understand the Concept of Internetworking	10	13	1	1	CO1
2	Internet Addresses to Physical Addresses(ARP) Internet Protocol:	15	16	2	1	CO2
3	Connectionless Datagram Delivery	25	39	3	3	CO3
4	Internet Protocol: Routing IP Datagram	15	26	2	2	CO4
5	Internet Protocol: Error and Control Messages(IITP)	10	16	2	1	CO5
Total		120	110	10	8	
Marks				30	80	

Course Objectives	(i) To learn the concept of internetworking using TCP/IP. (ii) To learn about ARP Protocols. (iii) To understand the concept of Connectionless delivery of Datagram (iv) To learn about Routing (v) To learn about Error control in Network Management
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Course Outcomes	Upon completion of the course the student shall be able to	
	CO1	Internet Architecture, Internet Services
	CO2	Mapping of Internet Addresses to Physical Addresses
	CO3	Connectionless Datagram Delivery
	CO4	Routing in an Internet

	CO5	Error and Control Messages
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Learning Outcomes:

1. Understand the concept of internetworking

- 1.1 The motivation for internet working
- 1.2 The TCP/IP internet
- 1.3 Internet services
 - 1.3.1 Application - level Internet services
 - 1.3.2 Network – level Internet services
- 1.4 History of the internet
- 1.5 The internet Architecture Board
- 1.6 Internetworking concept
 - 1.6.1 Application - level Interconnection
 - 1.6.2 Network – level Interconnection
- 1.7 Explain Router
- 1.8 Internet Architecture
 - 1.8.1 Interconnection through Routers

2. Understand Mapping of Internet Addresses to Physical Addresses (ARP)

- 2.1 The Address Resolution Problem
 - 2.1.1 Resolution through Direct Mapping
 - 2.1.2 Resolution through Dynamic Binding
- 2.2 ARP Refinement
- 2.3 ARP implementation
- 2.4 ARP Encapsulation and identification
- 2.5 ARP Protocol format
- 2.6 Determining an Internet address at startup (RARP) Reverse Address Resolution Protocol.
- 2.7 Timing RARP Transactions
- 2.8 Primary & backup RARP server

3. Internet Datagram options

- 3.1 Explain circuit switched and packet switched networks
- 3.2 A virtual Network
- 3.3 Internet Architecture & philosophy
- 3.4 The concept of unreliable delivery
- 3.5 Connectionless Delivery system
- 3.6 Purpose of internet protocol
- 3.7 The internet Datagram
 - 3.7.1 Datagram Format
 - 3.7.2 Datagram type of service and datagram precedence
 - 3.7.3 Datagram Encapsulation
 - 3.7.4 Datagram size, Network MTU, Fragmentation
 - 3.7.5 Reassembly of fragment
 - 3.7.6 Fragment control
 - 3.7.7 Time to live (TTL)
 - 3.7.8 Other Datagram Header fields
- 3.8 Internet Datagram options
 - 3.8.1 Record Route Option
 - 3.8.2 Source Route Option

- 3.8.3 Time stamp Option
- 3.8.4 Processing Option During Fragmentation

4. Internet Protocol: Routing IP Datagram

- 4.1 Introduction
- 4.2 Routing in an Internet
- 4.3 Direct and indirect Delivery
 - 4.3.1 Datagram Delivery over a single network
 - 4.3.2 Indirect Delivery
- 4.4 Table driven IP Routing
- 4.5 Next- Hop Routing
- 4.6 Default – Router
- 4.7 Host – Specific Router
- 4.8 The IP Routing Algorithm
- 4.9 Routing with IP Addresses
- 4.10 Handling incoming Datagram
- 4.11 Establishing Routing Table

5. Internet Protocol: Error and Control Messages (IITP)

- 5.1 Introduction
- 5.2 The Internet Control Message Protocol
- 5.3 Error reporting vs Error correction
- 5.4 IITP Message Delivery
- 5.5 IITP Message Format
- 5.6 Testing Destination Reachability and status (Ping)
- 5.7 Echo Request and reply message format
- 5.8 Reports of unreachable destinations

COURSE CONTENT

1. Concept of Internetworking

Introduction–TCP/IPinternet–internetservices–applicationlevelservices–
networklevel–history–internet architecture board–application level
interconnection–network level interconnection–router–internet architecture–inter
connection through routers

2. Address Resolution Protocol

Address resolution problem–direct mapping–dynamic binding–ARP Refinement-
ARP Refinement- ARP implementation –ARP Encapsulation and identification-ARP
Protocol format-Reverse Address Resolution Protocol-Timing RARP
Transactions- Primary & backup RARP server Internet Protocol

3. Circuit switching and packet switching:

Virtual Network - Internet Architecture & philosophy-The concept to unreliable
delivery-Connectionless Delivery system -Purpose of internet protocol –The internet
Datagram–Format-type of service and datagram precedence – encapsulation -
datagram size, Network MTU, Fragmentation-Reassembly of fragment-Fragment
control-Time to live(TTL)-Other Datagram Header fields-Internet Datagram options-
Record Route Option - Source Route Option - Time stamp Option - Processing Option
During Fragmentation

4. Routing IP

Introduction-Routing in an Internet-Direct and indirect Delivery- Datagram Delivery over a single network - Indirect Delivery - Table driven IP Routing - Next-Hop Routing-Default-Router-Host-Specific Router-The IP Routing Algorithm - Routing with IP Addresses - Handling incoming Datagram – Establishing Routing Table

5. ITP

The Internet Control Message protocol-Error Reporting Vs. Error Correction-Message Delivery - Message format - Testing Destination Reachability and Status (Ping) - Echo Request and Reply Message format - Reports of Unreachable destinations

Reference Books

1. Internetworking with TCP/IP–Douglas E Comer

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.6
Unit test-2	From 3.7 to 5.8

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-402	Web Technologies	5	75	20	80

Time Schedule

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1.	Principles of Web Designing and HTML Introduction.	11	21	2	1 ½	CO1
2.	Understand various HTML tags and usage of style sheets.	14	21	2	1 ½	CO1,CO2
3.	Understand XML and Client side scripting using Java Script.	18	26	2	2	CO2
4.	JQuery	10	13	1	1	CO3
5.	Web servers and Server side scripting using PHP	22	29	3	2	CO4
Total		75	110	10	8	
Marks				30	80	

Course Objectives	i) Understand the basic elements of web page ii) Know the working with HTML, CSS iii) To familiarize the various Technologies like Java Script, JQuery, PHP. iv) To understand Database connectivity Using PHP
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Course Outcomes	CO1	Implement interactive web page(s) using HTML and CSS
	CO2	Know how to format and validate Web page elements using JavaScript and describe data in a web page using XML.
	CO3	To know the Usage of JQuery
	CO4	Build Dynamic web site using server side PHP Programming and database connectivity using PHP.

LEARNING OUTCOMES:

1. Principles of Web Designing and HTML Introduction.

1.1 Understand the principles of Web Designing

- 1.1.1 Basic web Terminology.
- 1.1.2 Describe Anatomy of web page.
- 1.1.3 Understand different Web page elements.
- 1.1.4 Navigate through web pages
- 1.1.5 Narrate steps in building web site
- 1.1.6 Narrate steps in launching
- 1.1.7 Narrate maintaining web site.

1.2 HTML Introduction

- 1.2.1 Introduction and Overview of HTML
- 1.2.2 Discuss the rules for designing a HTML document.
- 1.2.3 Explain the structure of HTML document.
- 1.2.4 Define HTML element and Attribute.
- 1.2.5 Study the basic tags in HTML <html>, <head>, <title>, <body>.
- 1.2.6 Study the header tags <h1> to <h6>
- 1.2.7 Discuss the Physical formatting tags , <i>, <u>, <strike>, <sub>, <sup>, <big>, <small>, <tt>
- 1.2.8 Discuss the Logical formatting tags <q>, , <cite>, <ins>, ,
- 1.2.9 Discuss the <marquee> with attributes.
- 1.2.10 List Character entities.
- 1.2.11 Explain the List tags like , , , <dl>, <menu> with attributes.

1.3 Describe the setting of tables.

- 1.3.1 Describe the tags <table>, <tr>, <td>, <th>, <tbody>, <thead>, <tfoot>

2. Understand various HTML tags and usage of style sheets.

- 2.1 Explain the link and imaging tags <a>, with attributes.

2.2 Explain <object> tag with attributes.

- 2.2.1 Explain the tags, <form>, <input>, <button>, <label>, <select>, <options>, <textarea>, <legend> with attributes.

2.3 Explain the tags, <frame>, <frameset>, <noframe>, <iframe> with attributes.

2.4 Illustrate about cascading style sheets

- 2.4.1 Understand the level of styles inline, internal and embedded style sheets.
- 2.4.2 Explain ID and Class selectors in CSS
- 2.4.3 Explain about Color and background properties
- 2.4.4 Explain about Box properties like Border, position, margin, padding of elements.

3. Understand XML and Java Script.

3.1 Understand XML

- 3.1.1 Describe how to organize data in the form of XML.
- 3.1.2 Explain the rules for designing XML document.
- 3.1.3 Understand the significance of Namespace.
- 3.1.4 List the various applications of XML.

3.2 Types of scripting-JavaScript

- 3.2.1 Differentiate between Client-side and Server-side scripting.
- 3.2.2 List Client side and server side scripting languages.
- 3.2.3 Describe the features of Java Script.
- 3.2.4 Placing JavaScript code in HTML.
- 3.2.5 Understand functions

- 3.2.5.1 Know how to define and call a function.
- 3.2.5.2 Know how to pass parameters.
- 3.2.5.3 Understand the purpose of getElementById method
- 3.2.5.4 Describe the global functions provided by JavaScript.
- 3.2.6 Form Handling in Java Script
- 3.2.7 Illustrate Arrays
 - 3.2.7.1 Understand single and multi-dimensional arrays.
 - 3.2.7.2 Design small programs using arrays.
- 3.2.8 Understand about various Objects provided by JavaScript
 - 3.2.8.1 **Math** object
 - 3.2.8.2 **String** object
 - 3.2.8.3 **Date** object
 - 3.2.8.4 **Boolean** and **Number** object
- 3.2.9 Describe events in java script.

4. JQuery

- 4.1 Define JQuery
- 4.2 List the features of JQuery
- 4.3 List JQuery plugins
- 4.4 Explain the steps for to include JQuery in Web Pages
- 4.5 Explain JQuery Syntax with example program
- 4.6 Describe the jQuery Selectors-Accessing HTML elements by using
- 4.7 Element Selectors
- 4.8 ID, Class Selectors
- 4.9 Explain the JQuery Document Ready Event
- 4.10 Describe the JQuery Event handling methods(Mouse Events, Keyboard Events, Form Events, DoCCNent/Window events)
- 4.11 Explain effects of JQuery(like hide, show, fadeIn, fadeout, fadeToggle ,fadeTo, slideDown, SlideUp, SlideToggle)
- 4.12 Explain Functions in JQuery like text(),html(), val(), attr(),css().

5. Web servers and Server side scripting using PHP.

5.1 Web servers:

- 5.1.1 Understand the architecture of a Web server.
- 5.1.2 List the various web servers.
- 5.1.3 Illustrate the various HTTP request types and their difference.
- 5.1.4 Compare the properties of IIS, and Apache.

5.2 Understand the Fundamentals of PHP

- 5.2.1 Explain how to combine HTML and PHP.
- 5.2.2 Explain how to access HTML, PHP documents from web servers.
- 5.2.3 List various Data types and explain them with examples.
- 5.2.4 Explain how to declare Variables and Constants.

5.3 List and explain string manipulation functions.

5.4 Understand Arrays

- 5.4.1 Explain types of arrays.
- 5.4.2 Design small programs using arrays.

5.5 Explain form handling in PHP

- 5.5.1 Access elements of form using \$_GET,\$_POST

- 5.6** Know how to access Mysql Database
 - 5.6.1 List and explain mysql database functions in PHP.
 - 5.6.2 Explain the steps of connecting to a Database.
 - 5.6.3 Know about retrieving data from a table.
 - 5.6.4 Know about inserting data into a table.
 - 5.6.5 Know about updating the data in a table.
 - 5.6.6 Know about deleting data from a table.
 - 5.6.7 Design some simple programs to insert, delete, update and retrieve data from database.
- 5.7** Cookies
 - 5.7.1 Define Cookie.
 - 5.7.2 Know how to create and delete a cookie.
 - 5.7.3 Know the purpose of cookie.
- 5.8** Sessions
 - 5.8.1 Define Session
 - 5.8.2 Understand how to create a session.
 - 5.8.3 Know how to destroy a session.
 - 5.8.4 Know the purpose of session.
 - 5.8.5 Differentiate Sessions and Cookies.
- 5.9** Passing data from one web page to other webpage using query string.

COURSE CONTENTS

1. Principles of Web Designing and HTML Introduction

Principles of Web Designing:

Anatomy of Web page, Format, Elements, Navigation, Building, Launching and maintaining web site

HTML:

Introduction to HTML, Format of web page, Tags and attributes, Formatting text, Adding images, Positioning. Lists, Colors, Tables.

2. HTML & CSS

Connecting to hyperlinks and Imaging, Forms, Frames, IFrame

CSS : Introduction, Inline styles, Embedded style sheets, Conflicting styles, Linking external Style sheets, Positioning elements, Backgrounds, Element dimensions

3. XML & JavaScript

XML: Introduction, Structuring Data, XML Namespaces, Applications of XML

JAVA SCRIPT

Introduction to Scripting, Client-Side versus Server-Side Scripting, JavaScript features, Functions – Function definitions, Use of getElementById, getElementByName, Global functions, Form handling.

Arrays – Declaring and allocating arrays, References and reference parameters, passing arrays to functions, sorting and Searching arrays, Multiple-Subscripted arrays

Objects – **Math** object, **String** object, **Date** object, **Boolean** and **Number** object.

4. JQuery

JQuery

Introduction to JQuery, Features of JQuery, Plugin used in JQuery, steps for to include JQuery in Web Page, JQuery Syntax, JQuery Selectors- Element, Selectors, ID, Class, Document Ready Event, JQuery Event handling methods, effects of JQuery, Functions in JQuery

5. Web servers and Server side scripting using PHP.

6. Web servers:

Introduction, HTTP Request Types, System Architecture, Client-Side versus Server-Side Scripting, Accessing Web Servers-IIS, Apache, Requesting HTML, PHP documents.

PHP

Fundamentals of PHP, Data types, String functions, Arrays, form handling, Databases, Cookies, Sessions, Passing data from one web page to other web page.

REFERENCE BOOKS

- 1) Principles of Web Design, Sklar, TMH
- 2) HTML complete reference, Powell, TMH
- 3) Basics of Web Site Design, NIIT – PHI
- 4) WWW Design with HTML, Xavier (TMH)
- 5) Internet & World Wide Web, Dietel and Dietel, Pearson education Asia.
- 6) Complete Reference PHP, Steven Holzer-McGraw Hill
- 7) JQueryCookbook, O'Reilly Media
- 8) www.w3schools.com
- 9) www.php.net

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.2.6
Unit test-2	From 3.2.7 to 5.9

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-403	Digital Electronics & Computer Organization	5	75	20	80

Time Schedule

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1.	Number systems, Logic Gates ,Boolean Algebra and basic Combinational circuits	15	16	2	1	CO1,CO3
2.	Flip-Flops & Counters	15	16	2	1	CO1,CO2
3.	Information representation & CPU Organization	15	26	2	2	CO1,CO2,CO3
4.	Memory Organization	15	26	2	2	CO2,CO4
5.	I/O Organization	15	26	2	2	CO3,CO4,CO5
Total		120	110	10	8	
Marks				30	80	

Course Objectives	i) To acquire the basic knowledge of Number systems, digital logic levels and apply of knowledge to understand digital logic circuits. ii) To prepare students to perform the analysis and design of various digital electronics circuits iii)To know about Processor organization, information Representation iv)To understand how memory and i/o is organized in an effective way
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Course Outcomes	At the end of the course the student able to learn following:		
	CO1	CCN-403.1	Describe fundamental Numbering concepts and techniques used in digital electronics, the switching algebra theorems and logic gates
	CO2	CCN-403.2	Analyse the operation of flip flops and counting circuits
	CO3	CCN-403.3	Explain the Basic computer organization techniques and information representation
	CO4	CCN-403.4	Explain Memory organization

	CO5	CCN-403.5	Describe Handling of I/O organization
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Learning Outcomes:

1.0 Circuits

1.1 Number systems

- 1.1.1 List the various number systems used in digital Computer.
- 1.1.2 Explain Decimal, Binary, octal, HexaDecimal number systems
- 1.1.3 Convert decimal number to other base conversion.
 - 1.1.3.1 Decimal to Binary
 - 1.1.3.2 Decimal to Octal
 - 1.1.3.3 Decimal to Hexadecimal
- 1.1.4 Convert binary number to other base conversion.
 - 1.1.4.1 Binary to Decimal
 - 1.1.4.2 Binary to octal
 - 1.1.4.3 Binary to Hexadecimal
- 1.1.5 Convert octal number to other base conversion.
 - 1.1.5.1 Octal to Decimal
 - 1.1.5.2 Octal to Binary
 - 1.1.5.3 Octal to Hexadecimal
- 1.1.6 Convert hexadecimal other base conversion.
 - 1.1.6.1 Hexadecimal to Decimal
 - 1.1.6.2 Hexadecimal to Binary
 - 1.1.6.3 Hexadecimal to Octal
- 1.1.7 Binary numbers representation.
 - 1.1.7.1 Define Binary numbers representation.
 - 1.1.7.2 List the types of Binary numbers representation.
 - 1.1.7.3 Explain Unsigned binary number representation.
 - 1.1.7.4 Explain Signed binary number representation.
- 1.1.8 Binary coded decimal (BCD) coding scheme.
 - 1.1.8.1 Define Binary coded decimal (BCD) coding scheme.
 - 1.1.8.2 List the types of Binary coded decimal (BCD)
 - 1.1.8.3 Draw and explain 8421 code.
 - 1.1.8.4 Draw and explain 2421 code.
 - 1.1.8.5 Draw and explain 8 4-2-1 code.

1.2 Boolean algebra

- 1.2.1 Explain AND, OR, NOT operations with truth tables.
- 1.2.2 Explain the working of EX-OR and EX-NOR operations with truth tables.
- 1.2.3 List the different postulates in Boolean algebra.
- 1.2.4 State De-Morgan's theorems.
- 1.2.5 Prove De-Morgan's theorems using truth tables.
- 1.2.6 Apply De-Morgan's theorems and other postulates of Boolean algebra to simplify the given Boolean expression.
- 1.2.7 Write Boolean expression for given truth table.
 - 1.2.7.1 Using Sum-Of-Products(SOP) method
 - 1.2.7.2 Using Product-Of-Sums(POS) method
- 1.2.8 Use K-map to simplify Boolean expression (up to 4 variables).
 - 1.2.8.1 Using Two variable K-Map
 - 1.2.8.2 Using Three variable K-Map
 - 1.2.8.3 Using Four variable K-Map

1.3 Logic Gates

- 1.3.1 Define Logic gate
- 1.3.2 List basic gates
- 1.3.3 Define OR gate
- 1.3.4 Explain OR gate with logic symbol and truth table.
- 1.3.5 Define AND gate
- 1.3.6 Explain AND gate with logic symbol and truth table.
- 1.3.7 Define NOT gate
- 1.3.8 Explain NOT gate with logic symbol and truth table.
- 1.3.9 What is universal gate? List universal gates
- 1.3.10 Define NOR gate
- 1.3.11 Explain NOR gate with logic symbol and truth table.
- 1.3.12 Define NAND gate
- 1.3.13 Explain NAND gate with logic symbol and truth table.
- 1.3.14 Define EX-OR and EX-NOR gates
- 1.3.15 Explain the working of EX-OR and EX-NOR gates with truth tables.
- 1.3.16 Implement AND, OR, NOT, EX-OR gates using NAND gates only
- 1.3.17 Implement AND, OR, NOT, EX-OR gates using NOR gate only.

1.4 Basic Combinational Circuits

- 1.4.1 Define the HalfAdder. Explain the function of Half Adder.
- 1.4.2 Draw Half-Adder circuit using an exclusive OR and an AND gate.
- 1.4.3 Draw a Half-Adder using only NAND gates or only NOR gates.
- 1.4.4 Define the Full Adder. Explain the function of Full Adder.
- 1.4.5 Construct Full Adder using two Half-Adder and an OR gate

2.0 Flip-Flops, Counters

2.1 FLIP-FLOPS

- 2.1.1 List the details of different logic families.
- 2.1.2 Define positive and negative logic levels.
- 2.1.3 Define Flip flop
- 2.1.4 Draw and explain the basic principle of operation of a Flip-flop.
- 2.1.5 Define Latch.
- 2.1.6 Explain the working of a NAND latch circuit with truth table and Timing diagram
- 2.1.7 Explain the working of a NOR latch circuit with truth table and Timing diagram
- 2.1.8 Differentiate between Latch and Flip-flop.
- 2.1.9 Explain with block diagram, waveforms and truth tables the working of RS Flip-flop.
- 2.1.10 Explain with block diagram, waveforms and truth tables the working of D Flip-flop.
- 2.1.11 Explain with block diagram, waveforms and truth tables the working of JK Flip-flop.
- 2.1.12 Explain with block diagram, waveforms and truth tables the working of T Flip-flop.
- 2.1.13 Draw and explain the need for a Master-Slave flip-flop.
- 2.1.14 Explain the working of a Master-Slave flip-flop using suitable circuit diagram and truth table.

2.2 Counters

- 2.2.1 Define Counter
- 2.2.2 List the types of counters.
- 2.2.3 Define Synchronous counter
- 2.2.4 Define Asynchronous counter

- 2.2.5 Distinguish between asynchronous and synchronous counters.
- 2.2.6 Draw and explain module-10 (decade) Asynchronous counter circuit diagram with waveforms and truth tables
- 2.2.7 Draw and explain module-8 synchronous counter circuit diagram with waveforms and truth tables
- 2.2.8 Draw and explain module-16 synchronous counter circuit diagram with waveforms and truth tables
- 2.2.9 List the advantages of synchronous counters
- 2.2.10 Programmable counter
 - 2.2.10.1 Draw and explain the need for a Programmable counter
 - 2.2.10.2 Explain how to design Programmable counter circuit diagram
- 2.2.11 List the applications of counter.

3.0 CPU Organization & Information representation and Arithmetic Operation

3.1 CPU Organization

- 3.1.1 Draw the functional block diagram of Digital computer and explain the function of each unit.
- 3.1.2 Define Register
- 3.1.3 State the purpose of
 - 3.1.3.1 Accumulator
 - 3.1.3.2 Program counter
 - 3.1.3.3 Instruction Register
 - 3.1.3.4 Memory Buffer Register
 - 3.1.3.5 Memory Address Register
- 3.1.4 Draw the block diagram of simple accumulator based CPU.
- 3.1.5 Explain the function of each unit
- 3.1.6 Define the terms micro operation, macro operation,
- 3.1.7 Define instruction cycle, fetch cycle and execution cycle.
- 3.1.8 What is stored program concept
- 3.1.9 Describe the sequential execution of a program stored in memory by the CPU

3.2 Information representation and Arithmetic Operation

- 3.2.1 Explain the basic types of information representation in a computer.
- 3.2.2 Define floating point representation and fixed point representation of numbers.
- 3.2.3 Illustrate the floating point and fixed point representations with example.
- 3.2.4 Distinguish between Fixed point and Floating point representations.
- 3.2.5 What is Instruction format
- 3.2.6 Define Opcode , Operand and address.
- 3.2.7 Explain different types of instructions with examples
 - 3.2.7.1 Zero address instructions
 - 3.2.7.2 One address instructions
 - 3.2.7.3 Two address instructions
 - 3.2.7.4 Three address instructions

- 3.3 List and explain various addressing modes.

4.0 Memory Organization

- 4.1 Distinguish between main and auxiliary memory.
- 4.2 State the need for memory hierarchy in a computer.
- 4.3 Explain memory hierarchy in a computer in detail

- 4.4 State the significance of various memory device characteristics: access time, access rate, alterability, permanence of storage, cycle time.
- 4.5 Discuss Associative Memory
- 4.6 Explain the principle of virtual memory organization in a computer system
- 4.7 Explain virtual address and physical address organization.
- 4.8 State the principle of locality of reference
- 4.9 Explain Cache memory organization.
- 4.10 Analyze the importance of the principle of memory interleaving in a computer.

5.0 I/O Organization

- 5.1 List the any five peripheral devices that can be connected to a computer.
- 5.2 Define Interface.
- 5.3 Explain the need for an interface.
- 5.4 List modes of data transfer.
- 5.5 Explain synchronous and asynchronous data transfer.
- 5.6 Compare synchronous and asynchronous data transfer.
- 5.7 Explain hand shaking procedure of data transfer.
- 5.8 Explain programmed I/O method of data transfer.
- 5.9 Explain interrupted initiated I/O.
- 5.10 Explain DMA controlled transfer.
- 5.11 Explain priority interrupt, polling, and daisy chaining priority.
- 5.12 Write about bus system
- 5.13 List the four bus systems.
- 5.14 Differentiate between i/o bus and memory bus

COURSE CONTENTS

1. Number systems, Boolean algebra and Logical Gates :List the various number systems used in digital Computer, Explain Decimal , Binary,octal, Hexa Decimal number systems, Convert decimal number to other base conversion, Convert binary number to other base conversion, Convert octal number to other base conversion, Convert hexadecimal other base conversion, Binary numbers representation, Signed binary arithmetic, Binary coded decimal (BCD) coding scheme, Character representation, AND, OR, NAND, NOT, NOR & EX-OR gates. Logical definitions – Symbols – truth tables. Boolean theorems, Boolean simplifications of Boolean expressions, Using De-Morgan's theorems, Formation and implementation of Logic expressions, Karnaugh's mapping, Applications involving developing of combinational logic circuits. Half-Adder, Full-adder.

2. FLIP FLOP: Different logic families, Basic principles of Flip Flop operation (with help of wave form & truth tables) of RS,T,D,JK and Master Slave JK flip flop.Counters: Basic Asynchronous, Synchronous.

3. CPU Organization & Information representation and Arithmetic Operation : functional block diagram of Digital computer -Simple accumulator based CPU and function of each unit.-Stored program concept Information representation and Arithmetic Operation- Basic types of information representation - floating point representation and fixed point representation of numbers, Operand, Opcode and address - zero address, one address, two address and three address instructions - different addressing modes.

4. Memory Organisation - Organization of Computer Memory system - Main and auxiliary memory -Need for memory hierarchy in a computer -Significance of various memory devices characteristics: access time, access rate, alterability, permanence of storage, cycle time - Associative Memory-Virtual memory organization in a computer system - Virtual address and physical address organization. -Principle and advantage of cache memory organization- Principle of memory interleaving in a computer

5. Input and output organization - Peripheral devices -Need for an Interface-Three modes of data transfer
 - Synchronous and asynchronous data transfer -Hand shaking procedure of data transfer -Programmed I/O
 method of data Transfer-Interrupted initiated I/O-DMA controlled transfer-Priority interrupt, polling, and
 daisy chaining priority-Bus systems

REFERENCE BOOKS

- | | | |
|--|----|-------------------|
| 1. Digital principles and applications | -- | Malvino and leach |
| 2. Digital Electronics | -- | Bignell - Thomson |
| 3. Modern Digital Electronics | -- | R.P. Jain |
| 4. Computer System Architecture | -- | Morris Mano. |

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 2.2
Unit test-2	From 2.3 to 5.14

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-404	OOP through Java	5	75	20	80

Time Schedule

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1.	Object oriented programming concepts and Basics of java, Overloading	13	13	1	1	CO1,CO2
2.	Concepts of inheritance, overriding, Interfaces and Packages	13	26	2	2	CO2
3.	I/O Streams and Collections.	15	21	2	1 ½	CO3
4.	Exception handling and Multi-threaded programming.	14	21	2	1 ½	CO4
5.	Applets, AWT and Event Handling	20	29	3	2	CO4,CO5
Total		75	110	10	8	
Marks				30	80	

Course Objectives	i) To now applying object oriented programming paradigm in problem solving on the platform of Sun Micro Systems. ii) Able to design multi-tasking application with the knowledge of multi-threading. iii) Familiarized to develop graphical user interface with event handling mechanism.
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Course Outcomes	At the end of the course the student able to learn following:		
	CO1	CCN-404.1	Know the object oriented programming concepts in problem solving. Use syntaxes and semantics of object oriented paradigm.

	CO2	CCN-404.2	Design optimized definition for an application with reusability features and packages in project development.
	CO3	CCN-404.3	Knows the usage of utilities in real time data structures.
	CO4	CCN-404.4	Demonstrate multithreading concepts to implement multitasking and multi programming applications.
	CO5	CCN-404.5	Demonstrate to design effective dynamic user interface for any front-end applications using Applets and events.

Learning Outcomes:

- 1.0 Object oriented programming concepts and Basics of java and over loading
- 1.1 Know about object-oriented programming
- 1.2 Compare procedure-oriented programming and object-oriented programming
- 1.3 List and explain features of object-oriented programming
- 1.4 Importance of Java in Internet programming.
- 1.5 Explain features of Java. Define Byte codes of Java, JVM.
- 1.6 How to write and executing a Java program. List different keywords and comment statements in Java.
- 1.7 Explain data types, scope and life time of variables.
- 1.8 Describe conversion and casting features.
- 1.9 Apply one-dimensional and two-dimensional arrays give example programs.
- 1.10 Illustrates usage of conditional and iteration statements of Java with an example program.
- 1.12 Describe usage of jump statements, break, and continue statements.
- 1.13 Describe how to create classes and objects.
- 1.14 Demonstrate Usage of new operator and methods.
- 1.15 Explain usage of constructors with an example program.
- 1.16 Apply method overloading and construction overloading in applications.
- 1.17 Describe usage of 'this' pointer with example.
- 1.18 Explain usage of static in variables, methods, and blocks.
- 1.19 Explain about string classes.
- 1.20 Usage of command-line arguments.

2.0 Concepts of inheritance, overriding, Interfaces and Packages

- 2.1 Explain implementation of inheritance with an example program.
- 2.2 Illustrate how to implement multilevel inheritance with an example program.
- 2.3 Explain method overriding and usage of super keyword.
- 2.4 Describe concept of Interfaces.
- 2.5 Define an Interface.
- 2.6 Differences between abstract classes and interface.
- 2.7 Explain how to implement interfaces with sample program.
- 2.8 Define a package.
- 2.9 Explain the concept of class path.

- 2.10** Describe concept of Access protection.
- 2.11** Illustrate the mechanism of importing packages.
- 2.12** Give simple application to design packages with sample programs.

3.0 I/O Streams and Collections.

- 3.1** List different types of I/O streams.
- 3.2** Explain how to read and write data through console input and output streams.
- 3.3** Explain how to use Data Input Stream and Data Output Stream to access primitivedata types.
- 3.4** Explain various file access operation by using File Streams.
- 3.5** Explain sample programs on above streams.
- 3.6** What is collection frame work and Hierarchy of collection frame work.
- 3.7** Discuss Array List, Linked List
 - 3.7.1 Constructors
 - 3.7.2 Methods
 - 3.7.3 Comparisons between above two classes.
 - 3.7.4 Explain Iterator and List Iterator interface methods
 - 3.7.5 Sample programs on Array List and LinkedList
- 3.8** Discuss List interface and HashSet and Hash Table class for
 - following3.8.1Constructors
 - 3.8.2 Methods
 - 3.8.3 Sample programs on above.
- 3.9** Discusses Map interface and HashMap class for the following
 - 3.9.1 Constructors
 - 3.9.2 Methods
 - 3.9.3 Sample programs on above.
- 3.10** Explain Enum set and Enum Map classes.
- 3.11.** Design simple programs using collections.

4.0 Exception handling and Multi-threaded programming.

- 4.1** Describe sources of errors.
- 4.2** Give advantages of Exception handling.
- 4.3** Types of exceptions
 - Checked
 - Unchecked
- 4.4** Apply following key words to handling exceptions through sample programs
 - Try
 - Catch
 - Finally
 - Throw
 - Throws
- 4.5** Explain concept of Multi-catch statements with example.
- 4.6** Explain how to write nested try in exception handling with example.
- 4.7** Describe built in exceptions.
- 4.8** Describe multithreading.
- 4.9** Explain Thread life cycle and states
- 4.10** Explain how to Creating single thread with example program.
- 4.11** Explain how to Creating multi thread with example program.

- 4.12 Illustrate thread priorities in multiple threads with an example.
- 4.13 Describe the concept of synchronization with example program.
- 5.0 Applets, AWT, Event Handling.**
- 5.1 Describe the basics of Applets – Life cycle of an applet.
- 5.2 Describe steps for design and execute sample applet program
- 5.3 Explain Graphics class methods
 - Update()
 - Paint()
 - Drawing Lines, Rectangle, circles, polygons
- 5.4 Working with Color Font classes.
- 5.5 Describe AWT classes
- 5.6 Explain how to design Frame window with example.
- 5.7 Describe Types of Events
- 5.8 List and explain sources of events.
- 5.9 List and explain different event classes.
- 5.10 List and explain event listener interfaces
- 5.11 Demonstrate event handling mechanism.
- 5.12 Demonstrate handling mouse events with sample program.
- 5.13 Demonstrate handling keyboard events with sample program.
- 5.14 Explain how to use AWT controls in applet programming.
 - Labels.
 - Buttons.
 - TextFields
 - Checkboxes.
 - Lists.
 - Choice
 - Scrollbars.

COURSE CONTENTS

1. Basics of java and overloading: object oriented programming-Importance of Java to Internet – Byte codes.Features of Java: OOPS concepts –Data types –type conversions – casting – Arrays. Usage of classes – objects – new – methods – constructors – method overloading, string classes – command line arguments-static members-this pointer

2. Concepts InheritanceOverridingInterfaces and Package:-Usages of Inheritance: inheritance super class, sub classes – Multi level inheritance – super keyword -overriding –Abstract classes-Interfaces- Packages.

3. Concepts of I/O Streams and Collections:I/O streams-Accessing data through console input and output-DataInputStream- DataInputStream –Collection Frame work-ArrayList-LinkedList-Iterator and List Interface-Hash table-Hash Map-Enum Set-Enum Map

4. Exception Handling and Multi-threading: – Exception handling: Source of errors – error handling – Exception handling-Multi catch statements- Define thread – life cycle of thread - Multi threading –Synchronization- Inter thread communication – Dead locks – Thread properties.

5. Applets, AWT and Event Handling: Basics of Applets – life cycle of an applet-Working with Graphics-color-fonts-AWT classes-Event classes-Listener interfaces-keyboard and Mouse events-AWT controls-Buttons-TextFields - CheckBox-List

REFERENCE BOOKS

1. The complete reference Java -- Patrick Naughten, Herbert Schildt
TMH Company Limited, New Delhi.
2. Programming in JAVA -- P. Radhakrishna, University Press
3. Programming in Java -- Muthu - Thomson

4. Java Foundations of Programming – NIIT, PHI

5. Programming with Java -- Balagurusamy, TMH

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.6
Unit test-2	From 3.7 to 5.14

Course code	Course Title	No. of Periods/ Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-405	Cryptography and Network Security	5	75	20	80

Time Schedule

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1.	Introduction to Network Security	10	13	1	1	CO1,CO2
2.	Classical Encryption Techniques	15	16	2	1	CO2
3.	Cryptographic integrity techniques.	15	26	2	2	CO3
4.	System security& VPN	20	29	3	2	CO4
5.	Firewalls and Ethical Issues	15	26	2	2	CO4,CO5
Total		75	110	10	8	
Marks				30	80	

Course Objectives	i) To study the concepts of System security and various Encryption Techniques. ii) Able to design Public key and Private key with the knowledge of message authentication iii) to Familiarize with Firewalls and security laws with Internet Governance.
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	At the end of the course the student will be able to:		
	CO1	CCN-405.1	Explain the concepts of Cryptography and Network Security
	CO2	CCN-405.2	Apply encryption algorithms and ciphers.
	CO3	CCN-405.3	Analysing cryptographic integrity techniques using public, private keys and digital signatures.
	CO4	CCN-405.4	Explain Securing system from intrusions, VPN, IP Whitelisting, SSL certificate
	CO5	CCN-405.5	Describe the Configuration of firewalls to reduce the effect of cybercrimes and ethical hacking.

Learning Outcomes:

1. Introduction to Network security

- 1.1. Define security and network security.
- 1.2. Describe OSI security architecture.
- 1.3. Discuss about different security goals.
- 1.4. Define cryptography.
- 1.5. Discuss about crypto system.
- 1.6. Discuss about authentication, Confidentiality, integrity w.r.t data.
- 1.7. Differentiate passive and active security threats.
- 1.8. List and explain categories of passive and active security attacks.
- 1.9. List and explain categories of security services.
- 1.10. List and explain categories of security mechanisms.
- 1.11. Explain the Model for network security and with block diagram.

2. Classical Encryption Techniques

- 2.1. Define the term encryption
- 2.2. Define the term Decryption
- 2.3. List the essential ingredients of a symmetric cipher.
- 2.4. Describe two basic functions used in encryption algorithms.
- 2.5. List keys required for two people to communicate via cipher.
- 2.6. Describe the general approaches to attack a cipher.
- 2.7. Describe the Caesar cipher.
- 2.8. Discuss the mono-alphabetic cipher.
- 2.9. Describe Playfair cipher. And Hill ciphers.
- 2.10. Discuss One-Time-Pad.
- 2.11. Differentiate mono and poly-alphabetic ciphers.
- 2.12. Discuss the problems with the one-time pad.
- 2.13. Explain a Transposition cipher.
- 2.14. Explain steganography.
- 2.15. Illustrate the ciphers with examples.

3. Cryptographic integrity techniques

- 3.1 List the principal elements of a public-key cryptosystem.
- 3.2 List the roles of the public and private keys.
- 3.3 Discuss about message authentication.
- 3.4 List and explain message authentication requirements. Explain message authentication functions.
- 3.5 Discuss about the Message authentication code.
- 3.7. Define digital signature.
- 3.8. List the properties of a digital signature.
- 3.9. List the digital signature requirements.
- 3.10. Give the step-by-step procedure of getting digital signature approval.
- 3.11. Differentiate between hash function and cryptography Hash function.
- 3.12. List the applications of cryptographic hash functions.

4. System security & VPN

- 4.1 Discuss about Intruders
 - 4.1.1. Explain intrusion detection
 - 4.1.2. Explain password management

4.2. Discuss about Virus Nature and its Classification

4.2.1. Define Macro Viruses and Virus Kits,

4.2.2. Define the term E-Mail Viruses

4.3. Discuss about Virus Counter measures:

4.3.1 Define Antivirus Approaches

4.3.2. Describe Advanced Antivirus Techniques

4.4 VPN

4.4.1 Need for VPN

4.4.2 List Types of VPN

4.4.3 List Protocols used in VPN

4.5 IP Whitelisting

4.6 SSL Certificate

4.6.1 Introduction to SSL Certification

4.6.2 How do SSL certificates work?

5. Firewalls and Ethical Issues

5.1 Define Firewall.

5.2 List types of firewalls.

5.3 Discuss about firewall characteristics

5.4 Analyze the importance of firewall

5.5 Discuss about cybercrime and computer crime,

5.6 Discuss the classification of computer crime based on the role that the computer plays in the criminal activity

5.6.1 Study of IT Act . Study of Cyber Laws..

5.7 Explain digital rights management

5.8 List the basic conditions that must be fulfilled to claim a copyright.

5.9 Describe the principal categories of users of digital rights management systems.

COURSE CONTENTS

- 1. Introduction to Network security:** Security, Need of Network security, security goals, cryptography, Attacks, Mechanisms and Services, The OSI Security Architecture: Security Services, Availability Services, Security Mechanisms and Security Attacks, A model for Network Security.
- 2. Classical Encryption Techniques:** Symmetric Cipher Model, Substitution Techniques : Caesar Cipher, Mono alphabetic Cipher, Playfair Cipher, Hill Cipher, Mono alphabetic cipher, One-Time Pad, Transposition Techniques, Steganography.
- 3. Cryptographic integrity techniques:** Principles of Public Key Cryptosystems, Authentication Requirements, Authentication Functions, Message Authentication Codes, Discuss ledger on HashFunctions and DigitalSignatures.
- 4. System security:** Intruders, Intrusion Detection Viruses: The Nature of Viruses, Viruses Classification, Virus Kits, Macro Viruses, E-Mail Viruses. Virus Counter measures: Antivirus Approaches, Advanced Antivirus Techniques. VPN, IP Whitelisting, SSL- certification.
- 5. Firewalls and Ethical Issues :** The Need for Firewalls, Firewall Characteristics, Types of Firewalls and their advantages. Legal and Ethical issues: Cybercrime and Computer Crime, Study of IT Act. Study of Cyber Laws Ethical Issues Related to

Computers and Information Systems

REFERENCEBOOKS:

1. Cryptography and Network Security: Principles and Practices, - William Stallings –Pearson Education.
2. Cryptography and Network Security– AtulKahate:McGrawHill
3. Network Security Essentials (Applications and Standards)- William Stallings, Pearson Education.
4. Cryptography and NetworkSecurity:2ndEdition-Behrouza.Forouzan.
5. Computer network top-down approach-James F. Kurose &KeithW.Ross,PearsonEducation.

Model Blue Print:

S.No.	Chapter/Unit title	No. of periods	Weightage Allocated	Marks Wise Distribution of Weightage				Question wise Distribution of Weightage				CO's Mapped
				R	U	Ap	An	R	U	Ap	An	
1	Introduction to Network Security	10	13	3	10			1	1			CO1, CO2
2	Classical Encryption Techniques	15	16	3	3	10		1	1	1		CO2
3	Cryptographic integrity techniques	15	26	3	13	10		1	2	1		CO3
4	System security	20	29	3	13	13		1	2	2		CO4
5	Firewall and Ethical Issues	15	26	3	13	10		1	2	1		CO5
	Total	75	110									

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.5
Unit test-2	From 3.6 to 5.9

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN- 406	Web Technologies Lab	4	60	40	60

S.No.	Chapter/Unit Title	No.of Periods	CO's Mapped
1.	Exercises on HTML, CSS&XML	15	CO1
2.	Exercises on Java Script, JQuery	20	CO2,CO3
3.	Exercises on PHP web applications and Database Applications	25	CO3,CO4
Total Periods		60	

Course Objectives	i) Understand the principles of creating an effective web page ii) To Know the working with HTML, CSS iii) To acquire knowledge and skills for creation of web site considering both client and server side iv) To familiarize the various Technologies like Java Script, JQuery, PHP. V)To understand Database connectivity Using PHP
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Course Outcomes	CO1	CCN-406.1	Implement interactive web page(s) using HTML, CSS and JavaScript.
	CO2	CCN-406.2	To know the Usage of JQuery
	CO3	CCN-406.3	Build Dynamic web site using server side PHP Programming
	CO4	CCN-406.4	To know database connectivity using PHP.
	CO5	CCN-406.5	Develop real world application with different web designing tools.

LEARNING OUTCOMES

HTML, CSS and XML:

1. Exercise on basic HTML tags.
2. Design a HTML page using suitable table tags and attributes.
3. Design a HTML page with a form containing various controls.
4. Design a HTML page on iframes.
5. Exercise on style.
6. Exercise on designing a XML document.

JavaScript, AJAX& JQuery:

7. Exercise on JavaScript functions.
8. Exercise on JavaScript arrays.
9. Write a program on mouse events using JQuery.
10. Design a webpage to apply the Effects of JQuery to HTML elements.
11. Exercise on changing background color using css() function in JQuery.
12. Write a JavaScript program using Responsive Slides JQuery plugin(download from responsiveslides.com)

PHP:

13. Install the following on local machine:
 - Apache Web server
 - MySQL
 - PHP and configure it to work with Apache Web server and MySQL.
14. Exercise on PHP arrays.
15. Design a form and access the elements of form using PHP.
16. Write PHP program to perform various operations on a database table using functions.
17. Write a PHP program to set a cookie.

KEY COMPETENCIES

Exp. No.	Name of the experiment	Objectives	Key Competencies
1.	Exercise on basic HTML tags	Create the HTML page with a title, heading, formatting and list tags in the body.	1) Identify the editor required for writing HTML 2) Add the tags with relevant content 3) Save the file 4) Open the file in a browser 5) Test the results

Exp. No.	Name of the experiment	Objectives	Key Competencies
2.	Design a HTML page using suitable table tags and attributes	Create the HTML page with a table and that table should have a header, body and footer.	1) Identify the tags for creating the table 2) Add header, body and footer to the table. 3) Put some content in each section of table 4) Save the file 5) Open the file in a browser 6) Test the results
3	Design a HTML page with a form containing various controls	Create the HTML page with a form and add some controls like textbox, label to the form.	1) Identify the tags to add a form and controls 2) Add the form and put some controls in it. 3) Save the file 4) Open the file in a browser 5) Test the results
4	Design a HTML page on frames	Create the HTML page with multiple iframes so that content in each frame will have different format and colors.	1) Identify the tags for creating multiple frames 2) Add some content to the frames and use different formats, colors for each frame. 3) Save the file 4) Open the file in a browser 5) Test the results
5	Design a style sheet to set the background color, position and dimensions of a HTML element	Create a style sheet which contains selectors to set the background color, position and dimensions of a HTML element.	6) Identify the editor required for creating CSS 7) Add selectors to set the background color, position and dimensions of an element. 8) Save the CSS file 9) Link the CSS file to a valid HTML page. 10) Save the HTML page 11) Open the HTML page in a browser 12) Test the results
6	Exercise on designing a XML document	Create a XML Document on Student data	1) Identify the editor required for creating XML 2) Add required elements for student data 3) Save the XML file as .xml extension 4) Open the XML document in browser 5) Test the results

Exp. No.	Name of the experiment	Objectives	Key Competencies
7	Exercise on JavaScript functions	Write a JavaScript program using function which performs sum of two numbers and function should call when button is clicked.	1) Create a HTMLfile 2) Write a JavaScript function which adds two numbers. 3) Add HTML button tag and assign a function to onclick attribute. 4) Save the HTMLfile. 5) Open the HTML page in a browser 6) Test the results Resolve the errors if any through debugging
8	Exercise on JavaScript arrays	Write JavaScript code to implement sorting like reading an array of 'n' numbers and sorting them in ascending order.	1) Create a HTML file 2) Add elements to read array and to sort. 3) Write the logic for sorting using iterative and conditional statements. 4) Save the HTML file. 5) Open the HTML page in a browser 6) Test the results Resolve the errors if any through debugging
9	Write a program on mouse events using JQuery	Write a JavaScript program using JQuery which displays different messages for mouse events like mouse enter, mouse leave, click, dblclick	1) Create a HTML file 2) Add a div tag with some content and border. 3) Write a JQuery functions which displays different messages when mouse enters in div tag, mouse leaves div tag and clicks on div tag. 4) Save the HTML file. 5) Open the HTML page in a browser 6) Test the results by moving mouse over the div tag. 7) Resolve the errors if any through debugging
10	Design a webpage to apply the Effects of JQuery	Write a JavaScript program using JQuery which performs effects like hide, show, slideup, fadeIn, fadeout, slideDown, SlideUp	1) Create a HTML file 2) Add a div tag with some content and border. 3) Add some buttons 4) Write a JQuery functions which performs some effect when click on respective button. 5) Save the HTML file. 6) Open the HTML page in a browser 7) Test the results by click on the button. Resolve the errors if any through debugging

Exp. No.	Name of the experiment	Objectives	Key Competencies
11	Exercise on changing background color using CSS properties in JQuery	Write a JavaScript program using JQuery which changes css properties like color, background-color, border etc.	1) Create a HTMLfile 2) Add a div tag with some content 3) Add some buttons 4) Write a JQuery functions which changes css properties like color, border when click on respective button. 5) Save the HTMLfile. 6) Open the HTML page in abrowser 7) Test theresults by click on the button. Resolve the errors if any through debugging
12	Write a JavaScript program using ResponsiveSlidesJquery plugin(download from responsiveslides.com)	Write a JavaScript program using JQuery which displays datepicker.	1) Create a HTMLfile 2) Add JQuery script tag. 3) Add slider plugin, which can be downloaded from http://responsiveslides.com 4) Add plugins file 5) Add images 6) Write JQuery code for display slideshow of images 7) Save the HTMLfile. 8) Open the HTML page in abrowser 9) Test theresults by click on the button. Resolve the errors if any through debugging
13	Install the following on local machine: • Apache Web server • MySQL • PHP and configure it to work with Apache Web server and MySQL.	Install a web server which supports PHP	1) Identify version compatible to system 2) Download the software 3) Install the server software 4) Configure the server 5) Write simple PHP program 6) Test the result
14	Exercise on PHP arrays	Write PHP program to implement searching like reading an array of 'n' numbers and finding smallest among them.	1) Create a PHPfile. 2) Add elements to read array and to find the smallest number. 3) Write the logic for sorting using iterative and conditional statements. 4) Save and Run the page. Test the result

Exp. No.	Name of the experiment	Objectives	Key Competencies
15	Design a form and access the elements of form using PHP	Write a php program which displays sum of two numbers submitted by the form	1) Create a HTML file 2) Add form with two textboxes for enter two numbers 3) Write a php program, which adds two numbers submitted by form and display the sum. 4) Place the files in server 5) Open the HTML file in browser 6) Test the results
16	Write PHP code to perform various operations on a database table using functions.	Write PHP code to perform retrieval, insertion, modification and deletion of data in a database table using functions	1) Understand the process of connecting to database and execute commands. 2) Create a PHP file. 3) Add required elements to the page. 4) Write the logic to retrieve, insert, update and delete data in the table using functions. 5) Save and Run the page. 6) Test the result
17	Write a PHP program to set a cookie.	Write PHP code to create a cookie and put some information in it.	1) Understand the significance of cookies. 2) Create a PHP file. 3) Write the logic to create and set a cookie 4) Save and Run the page. 5) Test the result.

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-407	Java Programming Lab	4	60	40	60

S.No.	Chapter/Unit Title	No.of Periods	CO's Mapped
1.	Basics, overloading, inheritance, overriding	16	CO1,CO2
2.	Streams, Interfaces and Packages and Collections.	10	CO2,CO3
3.	Exceptions and Multi-threaded programming.	14	CO3,CO4
4.	Applets and Event Handling	20	CO5
Total Periods		60	

Course Objectives	i) Design object oriented programming paradigm ii) Able to develop multi-tasking application with the knowledge of multi-threading. iii) Familiarized to develop graphical user interface with event handling mechanism.
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Course Outcomes	CO1	Perform object oriented programming model application design.
	CO2	Design optimized definition for an application with reusability features like inheritance and polymorphism analyse modular design for real time applications by using packages concept in projects.
	CO3	Apply IO on data and use collections
	CO4	Apply multi-threading concepts to implement multitasking and multi programming applications.
	CO5	Develop effective dynamic user interface for any front end applications using Applets and events.

Learning Outcomes

1. Exercise programs on conditional statements and loop statements
2. Exercise programs on Strings.
3. Exercise program to create class and objects and adding methods.
4. Exercise programs using constructors and construction over loading.
5. Exercise programs on command line arguments.
 - i) Input as command line arguments and perform operation on that data.
 - ii) Input as command line arguments and update manipulated data in Files.
6. Exercise programs using concept of overloading methods.
7. Exercise programs on inheritance.
8. Write a program using the concept of method overriding.
9. Exercise on packages.
 - i) Creation of packages
 - ii) Design module to importing packages from other packages.
10. Exercise programs on interfaces.
11. Exercise programs on I/O Streams
 - i) Reading data through Keyboard
 - ii) Perform Reading and Writing operations on files using File Streams.
12. Exercise programs on Collections.
 - i) Write a java program to search a student mark percentage based on pin number using Array list.
 - ii) Write a java program to create linked list to perform delete, insert, and update data in linked list with any application.
13. Exercise on exception handling.
 - i) Programs on try, catch and finally.
 - ii) Programs on multiple catch statements
 - iii) Programs on nested try statements.
14. Exercise on multithreading
 - i) Programs on creation of single and multiple threads.
 - ii) Programs on adding priorities to multiple threads.
15. Exercise on applets
 - i) Programs on Graphics and colors.
 - ii) Simple animations using threads and graphics.
16. Exercise on AWT controls

- i) Program to handle mouse events.
- ii) Program to handle keyboard events.
- iii) Programs to illustrate Text Fields and Button control.
- iv) Programs to illustrate Check Box and List control.
- v) Write an application program to illustrate multiple controls.

KEY COMPETENCIES

Exp. No.	Name of the experiment	Objectives	Key Competencies
1	Exercise programs on conditional statements and loop statements.	(a) Write program using if statement and switch (b) Write program using while, do and for constructs.	(a) Know the usage of IF and switch statements. (b) Compile the program and rectify the errors. (c) Observe the output.
2	Exercise programs on Strings.	(a) Write a programs to manipulate Strings (b) Write a programs to arrange array of strings in ascending order	(a) Create String objects (b) Use string class methods Observe the output.
3	Exercise program to create class and objects and adding methods.	(a) Write a program to create a class and create objects. (b) Write a program to create class adding methods and access class members.	(a) Create class. (b) Declare methods. (c) Create objects. (d) Write main method. (e) Access class members.
4	Exercise programs using constructors and construction over loading.	(a) Write a program using default constructor. (b) Write a program using parameterized constructor.	(a) Declare and define constructor. (b) Call default constructor. (c) Call parameterized constructor. (d) observe constructor overloading.
5	Exercise programs on command line arguments.	(a) Write a program to illustrate usage of command line arguments. (b) Write a program to read data as command line arguments and update it into Files.	(a) Use command line arguments. (b) Run the program. (c) Understand usage of Files. (c) Observe the output.
6	Exercise programs using concept of overloading methods.	(a) Write a program to illustrate method overloading. (b) Write a program to illustrate method overloading using constructors.	(a) Observe method overloading. (b) Overload constructor methods.
7	Exercise on inheritance.	(a) Write a program to illustrate single inheritance.	(a) Create base class. (b) Write base class constructor.

		(b)Write a program to illustrate multiple inheritance.	(c) Create derived class. (d) Use extends keyword. (e) Use super keyword. (f) Write derived class constructor.
8	Write a program using the concept of method overriding.	Write a program using the concept of method overriding.	(a) Use method overriding. (b) Use this keyword. (c) use super keyword
9	Exercise on importing packages.	Write a program to create and importing package.	(a) Create package. (b) Use of access specifiers. (b) Use package. (c) Use import keyword.
10	Exercise on interfaces.	Write a program to illustrate multiple inheritance using interfaces.	(a) Define interface. (b) Use extends keyword. (c) Use implements keyword. (d) Access interface variables.
11	Exercise programs on I/O Streams	(a) Write a program to give values to variables interactively through the keyboard. (b) Write program to read and write primitive data types. (c) Write programs to handle Files.	(a) Use different data types. (b) Use readLine() method. (c) Use println() method. (d)use File Streams Observe the output.
12	Exercise programs on Collections.	(a) Write a java program to search a student mark percentage based on pin number using Array list. (b)Write a java program to create linked list to perform delete, insert, and update data in linked list with any application.	(a) Define collection classes (b) use ArrayList, LinkedList (c) apply List and Iterator Interface
13	Exercise on exception handling	(a) Write a program to illustrate exception handling. (b) Write a program to illustrate exception handling using multiple catch statements. (c) Write a program to illustrate exception handling using nested try.	(a) Use try – catch. (b) Use multiple catch blocks. (c) Use finally statement. (d) use Nested try
14	Exercise on multithreading	(a) Write a program to create single a thread by extending the thread class. (b) Write a program to create a single thread by implementing the runnable interface. (c) Write a program to create multiple threads.	(a) Use extends, new. (b) Use run() and start() methods. (c) Observe thread execution. (d) Use implements runnable interface. (e) Use setPriority() and getPriority() methods. (f) use wait(),notify() methods

		(d) Write a program to illustrate thread priorities.	
15	Exercise on applets.	Write a program to create simple applet to display different shapes with colors. Write an applet program to design simple animation.	(a) Use <applet>...</applet> tag. (b) Add applet to html file. (c) Run the applet. (d) use graphics methods (e) use threads and graphics.
16	Exercise on AWT controls	(a) Write an applet program to handle key events. (b) Write an applet program to handle mouse events. (c) Write an applet program to illustrate Text Field and button control. (d) Write an applet program to illustrate Check box and List control. (e) Write an applet program to illustrate multiple controls.	(a) Use keyboard event methods (b) Use mouse event methods (c) Use Text Field class methods (d) Use button class methods (e) Use Check box and List class methods

Communication Skills

Course Code	Course Title	No. of Periods/Week	Total No. of Periods	Marks for FA	Marks for SA
CCN-408	Communication Skills	3	45	40	60

Course Objectives:	- to communicate effectively in diverse academic, professional and everyday situations
	- exhibit appropriate body language and etiquette at workplace
	- be employable through preparing appropriate job applications and attend interviews confidently with all necessary skills

CO No.	
CO1	Listen and comprehend the listening inputs related to different genres effectively
CO2	Communicate effectively in interpersonal interactions, interviews, group discussions and presentations
CO3	Acquire employability skills: job hunting, resume writing, attending interviews
CO4	Practise appropriate body language and professional etiquette

Course Delivery: Text book: **“English Communication Skills”**
by State Board of Technical Education and Training, AP

Sl No	Unit	Teaching Hours
1	Listening Skills	6
2	Workplace Etiquette	3
3	Introducing Oneself	3
4	Short presentation (JAM)	6
5	Group Discussion	6
6	Resume Writing and Cover Letter	3

7	Interview Skills	9
8	Presentation Skills	9
<i>Total</i>		45

Course Content:

UNIT 1: Listening Skills

6 periods

Pre – While- Post-listening activities- Listening to audio content (dialogues/ speech/ narrations) - answering the questions and fill in the blanks- vocabulary

UNIT 2: Workplace Etiquette

3 periods

Basics of Etiquette- politeness/ courtesy, good manners- features of work place etiquette- adaptability, positive attitude, body language.

UNIT 3: Introducing Oneself

3 periods

Speak about oneself - introduce oneself to a gathering / formal & informal situations- Know about others- filling in the grid- introducing oneself in interviews

UNIT 4: Short Presentation

6 periods

Dos and Don'ts in short presentation- speak for a minute without repetition, deviation & hesitation - the techniques to speak fluently – defining and describing objects, people, phenomena, events.- speaking on randomly chosen topics.

UNIT 5: Group Discussion

6 periods

Fundamentals of Group Discussion- Dos and Don'ts- filling the Grid- possible list of topics- practice sessions- sample videos-Group activity

UNIT 6: Resume Writing and Cover Letter

3 periods

Pre activity: answer the questions- jotting down biographical information- sample resumes- tips, Dos and Don'ts- model resumes- practice exercises on Resume writing

UNIT 7: Interview Skills

9 periods

Pre –while-post activities: - things to do at three stages – respond to notifications- know the information about the organisation-practice FAQs - preparation of good/ suitable C V, Body language, tips for success in interviews, model / mock interviews.

UNIT 8: Presentation Skills

9 periods

Preparatory work: observe pictures and answer questions- different kinds of presentations- PPTs, Flash cards, Posters, Charts. - tips to prepare aids, slide show, model PPTs, - checklist on pre, while and post presentations.

Mapping Course Outcomes with Programme Outcomes:

PO	1	2	3	4	5	6	7
CO	POs 1 to 5 are applications of Engineering Principles, can't be directly mapped to English Communication Skills					1,2,3,4	1,2,3,4

Unit wise Mapping of CO –PO

CO	Course Outcome	COs / Unit Mapped	POs mapping	Cognitive levels as per Bloom's Taxonomy R/U/A/An (Remembering / Understanding / Applying/ Analysing)
CO 1	Listen and comprehend listening inputs related to different genres effectively	Unit 1	6,7	R/U/A
CO2	Communicate effectively in interpersonal interactions, interviews, group discussions and presentations	Units 3,4,5,7,8	6,7	R/U/A/An
CO3	Acquire employability skills: job hunting, resume writing, attending interviews	Units 6,7	6,7	R/U/A/An

Course Code	Course title	No of periods/week	Total no of periods	Marks for FA	Marks for SA
CCN-409	Computer Networks and Security Lab	03	45	40	60

S No	Chapter/ Unit Title	No. of Periods	COs Mapped
1.	Computer Hardware	10	CO1,CO2,CO3
2.	Computer Networking	15	CO3.CO4.CO5
3.	SECURITY	20	CO4,CO5,CO6
	Total	45	

COURSE OBJECTIVES	1. Identify all the components of mother board. 2. Modify CCNOS settings as required 3. Install drives, NIC card, modem 4. Install network devices, design and develop network. 5. Understand ip address classes and subnetting 6. Prepare cross and straight Ethernet cables 7. Install and configure proxy server 8. To learn Different Cipher Techniques 9. To Implement the Symmetric key Algorithms 10. To Implement the Asymmetric key Algorithms 11. To use the network security tools and vulnerability assessment tools
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Course Outcomes	CO1	CCN-409.1	Assemble the PC with suitable components.
	CO2	CCN-409.2	Install network devices, design and develop network Install any network device and configure
	CO3	CCN-409.3	Develop the cipher techniques for encryption
	CO4	CCN-409.4	Implement symmetric key Algorithms
	CO5	CCN-409.5	Demonstrate Asymmetric key Algorithm

Learning Outcomes:

Computer Hardware

1. Identification of various Hardware components on Motherboard
2. Using various options of CCNOS setup
3. Print the summary of your system Hardware and verify for correctness
4. Hard drive, optical drive installation.
5. How to recover lost data on hard drive.

Computer Networking

6. Preparing the Ethernet cable for cross and direct connections using crimping tool and test using LAN tester.
7. Installation of a switch and connecting systems to a network switch.
8. Installation of a modem (internal, external or USB) and connecting to internet.
9. Using FTP for uploading and downloading files.
10. Installation and configuring the proxy server for internet access.
11. Setting of IP address to an existing terminal

CYBERSECURITY CONCEPTS:

SYMMETRIC KEY ENCRYPTION TECHNIQUES

12. perform encryption and decryption by using Caesar Cipher technique
13. Exercise encryption and decryption by using Playfair Cipher technique
14. Exercise encryption and decryption by using Hill Cipher technique
15. perform encryption and decryption by using Vigenere Cipher

ASYMMETRIC KEY ENCRYPTION TECHNIQUES

16. Perform encryption and decryption using RSA public and private key.
17. To perform the validation of the digital document using Digital signature standard encryption and decryption
18. To perform the procedure of installation process of antivirus to detect threats.
19. To ensure security basic firewalls can be enabled in the system.

KEY COMPETENCIES

S.N o.	Name of the experiment	Objectives	Key Competencies
1	Exercise on Identification and familiarization of	Identification and familiarization of various	❖ Identify and note down mother board , Components and Chips.

	various components of computer system.	components of computer system.	❖ Identify various Internal and External slots in the mother board and clean them with blower/ Brush. ❖ Practice Inserting and Removing RAM with care. ❖ Measure the Output voltages of SMPS.
2	Exercise on various operations and modifications required for CCNOS setup.	Perform various operations and modifications required for CCNOS setup.	❖ Identify location of CCNOS battery on mother board. ❖ Know how to replace CCNOS battery. ❖ Identify keyboard key for entering BIOS setup. ❖ Setup CCNOS settings ❖ Check the status of CCNOS settings after replacement.
3	Exercise on Print the summary of your system Hardware and verify for correctness	Print the summary of your system Hardware and verify for correctness	❖ Know how to open system summary window ❖ Check whether all the hardware peripherals are working properly or not. ❖ Know how to install device drivers ❖ Know how to enable and disable hardware peripherals. ❖ Print the hardware summary page.
4	Exercise on Hard drive, optical drive installation.	Hard drive, optical drive installation.	Hard drive: ❖ Identify the Hard drive slot. ❖ Know how to remove power supply and SATA cables from Hard drive. ❖ Unscrew Hard drive from computer case ❖ Replace new Hard drive and fix it in computer case ❖ Know how to connect power supply cable and SATA cables to Hard drive ❖ Check for the working condition of new Hard Drive. Optical drive: ❖ Identify the Optical drive slot. ❖ Know how to remove power supply and SATA cables from Optical drive. ❖ Unscrew Optical drive from computer case ❖ Replace new Optical drive and fix it in computer case ❖ Know how to connect power supply cable and SATA cables to Optical drive ❖ Check for the working condition of Optical drive. SSD DRIVES: ❖ Identify SSD slots

			❖ How to install SSD drives
5	Exercise on recovery of lost data on hard drive.	How to recover lost data on hard drive.	❖ Verify the available recovery tools of Operating system. ❖ Know how to recover lost data on Hard drive using Restore point. ❖ Know how to recover lost data on Hard drive using Recovery Image.
6	Exercise on Installation of Network card.	Installation of Network card.	❖ Identify the slot for placing NIC card ❖ Know how to place NIC card ❖ Install required NIC driver ❖ Check for working status of NIC card
7	Exercise on Dis-assembling and assembling of working desktop.	Dis-assembling and assembling of working desktop.	❖ Identify all the peripherals of Desktop computer. ❖ Check the working condition of system before dis-assembling it. ❖ Dis-assemble all the peripherals. ❖ Assemble all the peripherals. ❖ Check the working condition of system after assembling it.
8	Exercise on Preparing the Ethernet cable for cross and direct connections using crimping tool and test using LAN tester.	Preparing the Ethernet cable for cross and direct connections using crimping tool and test using cable tester.	❖ Know the color pattern of Ethernet cable for direct connection. ❖ Prepare UTP cable for direct connection using crimping tool. ❖ Check the working condition of cable using LAN tester. ❖ Know the color pattern of Ethernet cable for cross connection. ❖ Prepare UTP cable for cross connection using crimping tool. ❖ Check the working condition of cable using LAN tester.
9	Exercise on Installation of a modem (internal, external or USB) and connecting to internet.	Installation of a modem (internal, external or USB) and connecting to internet.	Internal modem ❖ Identify PCI slot for placing Internal modem ❖ Connect internal modem ❖ Install required modem driver ❖ Check for the working condition External modem ❖ Connect External modem ❖ Install required modem driver ❖ Check for the working condition USB modem ❖ Connect USB modem ❖ Install required modem driver Check for the working condition

10	Exercise on Using FTP for uploading and downloading files.	Using FTP for uploading and downloading files.	❖ Know about FTP protocol ❖ Know how to upload file using FTP ❖ Know how to download file using FTP
11	Exercise on Installation and configuring the proxy server for internet access	Installation and configuring the proxy server for internet access	❖ Know about proxy server. ❖ Know how to install proxy server. ❖ Know how to configure proxy server.
12	Exercise on Setting of particular IP address to an existing terminal system	Setting of particular IP address to an existing terminal system.	❖ Know about IP addresses ❖ Know how to set IP addresses to the computer systems in a LAN
13	To implement Transportation and Substitution using Caesar Cipher Technique	Learn to implement the Caesar Cipher Transportation Technique on information	❖ Compile program ❖ Input key value ❖ Input text to be encrypted ❖ Rectify the syntax errors ❖ We will get Encrypted text as output - Check the output for correctness
14	To implement Transportation and Substitution using Playfair Cipher Technique	Learn to implement the Playfair Cipher Transportation Technique on information	❖ Compile program ❖ Input key value ❖ Input text to be encrypted ❖ Rectify the syntax errors ❖ We will get Encrypted text as output ❖ Check the output for correctness
15	To implement Transportation and Substitution using Hill Cipher Technique	Learn to implement the Hill Cipher Transportation Technique on information	❖ Input the plain text and key from the user. ❖ Split the plain text into groups of length three. ❖ Arrange the keyword in a 3*3 matrix. ❖ the two matrices to obtain the cipher text of lengththree. ❖ Combine all these groups to get the complete cipher text.
16	To implement Vigenere Cipher Technique	Learn to implement the Vigenere Cipher Technique on information	❖ Arrange the alphabets in row and column of a 26*26 matrix. ❖ Circulate the alphabets in each row to position left such that the first letter is attached to last. ❖ Repeat this process for all 26 rows and construct the final key matrix.

			❖ The keyword and the plain text is read from the user. ❖ The characters in the keyword are repeated sequentially so as to match with that of the plaintext. Pick the first letter of the plain text and that of the keyword as the row indices and column indices respectively. ❖ The junction character where these two meet forms the cipher character. ❖ Repeat the above steps to generate the entire cipher text.
17	To implement Encryption and Decryptions using RSA algorithm	Learn to implement the RSA Public Key Encryption Algorithm	❖ By using RSA Public Key & Private key for Encryption and Decryption of the message ❖ By using Public key message will be ciphered ❖ By using Private key message will be deciphered ❖ Both the keys are Asymmetric
18	Implementation of Digital Signature Standard	Learn the Digital signature Implementation and its usage	❖ Read the 256-bit key values. ❖ Divide into five equal-sized blocks named A, B, C, D and E. ❖ The blocks B, C and D are passed to the function F. ❖ The resultant value is permuted with block E. ❖ The block A is shifted right by 's' times and permuted with the result of step-4. ❖ Then it is permuted with a weight value and then with some other key pair and taken as the first block. ❖ Block A is taken as the second block and the block B is shifted by 's' times and taken as the third block. ❖ blocks C and D are taken as the block D and E for the final output.
19	Study of any Antivirus Installation & Configurations Study/Demo Study of Standard Vulnerabilities.	Learn to install the Antivirus Software in Computer System and know the configuration Setting	❖ Choose the Appropriate Antivirus Software to install in the Computer System ❖ Do the Appropriate settings to configure the Antivirus software in the System. ❖ Observing System Performance While using the Antivirus software ❖ Observe the Viruses/Threats when attacked to Computer System
20	Setting firewall with Windows OS, its importance and Problems.	Study the implementation of firewall Settings in Computer system and learn how it manage the System	❖ Choose The Appropriate System settings to implement Firewall ❖ Observe the System performance while having firewall

Software requirements	❖ Linux /Windows Operating system ❖ C compiler ❖ Java Compiler
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**DIPLOMA IN COMMUNICATION AND COMPUTER NETWORKING ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION**

CURRICULUM-2023

(V Semester)

Sub Code	Name of the Subject	Instruction Periods/Week		Total Periods Per Semester	Scheme Of Examinations			
		Theory	Practicals		Duration (hrs)	Sessional Marks	End Exam Marks	Total Marks
THEORY SUBJECTS								
CCN-501	Industrial Management and Entrepreneurship	5	-	75	3	20	80	100
CCN-502	Wireless Networks	5	-	75	3	20	80	100
CCN-503	Software Engineering	5	-	75	3	20	80	100
CCN-504	Internet Of Things	4	-		3	20	80	100
CCN-505	High Speed Networks	5	-	75	3	20	80	100
PRACTICAL SUBJECTS								
CCN-506	Wireless Networks Lab	-	3	45	3	40	60	100
CCN-507	Advanced Computer Networks Lab	-	3	45	3	40	60	100
CCN-508	Life Skills	-	3	45	3	40	60	100
CCN-509	Project work	-	6	90	3	40	60	100
	ACTIVITIES		3	45				
	Total	24	18	630	-	260	640	900

Note:CCN-501 Common to all branches.

CCN-503 common with CM-401

CCN-504 common with CM-504

CCN-508 common with all branches

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-501	Industrial Management & Entrepreneurship	5	75	20	80

TIME SCHEDULE

Chapter No.	Chapter/Unit Title	No. of Periods	Marks	No. of Short Answer Questions	No. of Essay Type Questions	CO's Mapped
1.	Principles of Management.	10	16	2	1	CO1
2	Organization Structure & Organizational Behaviour	18	26	2	2	CO2
3.	Production Management.	18	26	2	2	CO3
4.	Engineering Ethics & Safety and Labour Codes	19	26	2	2	CO4
5.	Entrepreneurship & Start-ups.	10	16	2	1	CO5
Total		75	110	10	8	

Course Objectives and Course Outcomes

COURSE OBJECTIVES		Upon completion of the course the student shall be able to (i) Familiarize the concepts of management, and organization structures. (ii) Exposure to organizational behavioral concepts, basics of production management in industries. (iii) Exposure to Engineering Ethics, Industrial Safety, Labour codes and entrepreneurial start-ups programmes.	
COURSE OUTCOMES	CO1	CCN-501.1	Understand the principles of management as applied to industry.
	CO2	CCN-501.2	Explain types of the industrial organization structures and the behaviour of an individual in an organization, motivational and leadership styles.
	CO3	CCN-501.3	Explain the different aspects of production management.
	CO4	CCN-501.4	Explain Engineering Ethics, Industrial Safety and industrial Labour Codes.
	CO5	CCN-501.5	Explain Entrepreneurial development programmes and Start-ups.

CO and PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	1							1		
CO2		1								2
CO3	3								2	2
CO4						3		1	2	
CO5							2	1	2	

3: High, 2: Moderate, 1: Low Note:

The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

(i) Assignments (ii) Tutorials (iii) Seminars (iv) Guest Lectures (v) Group Discussions
(vi) Quiz (vii) Industry Visits (viii) Tech-Fest (ix) Mini Projects (x) Library Visits.

Learning Outcome

Understand the principles of management as applied to industry.

1.0 Principles of Management

- 1.1 Define industry, commerce (Trade) and business.
- 1.2 Know the need for management.
- 1.3 Understand functions of Management.
- 1.4 List the principle of scientific management by F.W.Taylor
- 1.5 List the principle of modern management by Henry Foyal.
- 1.6 Differentiate management, administration and organization
- 1.7 Differentiate Lower, Middle and Top level management
- 1.8 Explain the importance of Managerial skills (Technical, Human, Conceptual)
- 1.9 Know the objectives of Management Information Systems.
- 1.10 Know the Characteristics of Management Information Systems.

2.0 Organization Structure & Organizational Behaviour

- 2.1 Define organization structure.
- 2.2 Explain line, staff and line & staff organization structures with advantages, disadvantages and applications.
- 2.3 List various Motivation theories.
- 2.4 Explain Maslow's Hierarchy of needs.
- 2.5 Explain Different leadership styles.
- 2.6 Explain Trait theory of leadership
- 2.7 Explain Behavioral theory of Leadership.
- 2.8 Explain the Responsibility of human resource management.
- 2.9 Understand the process of recruitment, selection and training
- 2.10 State the Objectives of Job Analysis.

3.0 Production Management

- 3.1 Define Production, Planning and Control.

- 3.2 Explain Briefly Mass production, Batch production and Job order production.
- 3.3 Define the terms Routing, Scheduling and Dispatching.
- 3.4 List applications of network diagrams in production planning and control.
- 3.5 Draw PERT and CPM Network Diagrams – Simple Problems.
- 3.6 Know the functions of Materials Management.
- 3.7 Explain ABC analysis of Inventory.
- 3.8 Explain concept of Economic ordering quantity.
- 3.9 Explain meaning of Supply chain management.
- 3.10 Write processes of Supply Chain Management
- 3.11 List the Functions of Purchase Department.
- 3.12 Write functions of Stores Department.

- 4.0 Engineering Ethics & Safety and Labour Codes
- 4.1 Definition of Engineering Ethics.
- 4.2 Understand Core qualities of Professional Engineers.
- 4.3 Explain Different types of Ethics in Engineering.
- 4.4 State the meaning of Intellectual Property Rights
- 4.5 List common types of Intellectual Property Rights.
- 4.6 List Activities of Corporate Social Responsibility (CSR).
- 4.7 State the need of Human values in engineering fields.
- 4.8 Comprehend the importance of safety at Workplace.
- 4.9 List Different hazards in the industry.
- 4.10 State the causes of accidents costs of accidents and their prevention.
- 4.11 List Salient features of Code on Wages, 2019.
- 4.12 List Salient features of Industrial Relations Code, 2020,
- 4.13 List Salient features of Code on Social Security, 2020
- 4.14 List Salient features of Occupational Safety, Health and Working Conditions Code, 2020.

5. Entrepreneurship & Start-ups

- 5.1 Define the word Entrepreneur and Entrepreneurship.
- 5.2 Explain various self - employment schemes
- 5.3 List the Financial assistance programmes provided by the Governments.
- 5.4 Explain the concept of TQM and ISO 9000 series and BIS 14000 Series.
- 5.5 List the Advantages and Drawbacks of ISO 9000 series of standards.
- 5.6 Explain the Concept of Incubation center's.
- 5.7 Explain Startup and its stages.
- 5.8 Explain Break Even Analysis to make or buy the products.
- 5.9 State the Importance of Branding.
- 5.10 State the significance of Business name, logo and tag line.
- 5.11 Explain the Concepts of Digital Marketing.
- 5.12 Know the Role of E-commerce and Social Media.

Course Content

1.0 Principles of Management

Introduction: Industry, Commerce and Trade; Definition of management; Functions of management; Principles of Scientific Management: F.W. Taylor - Principles of Modern Management: Henry Fayol; Administration organization and management; Levels of management - Managerial skills - Management Information Systems: Objectives and Characteristics.

2.0 Organization Structure & Organizational Behaviour

Organization Types: Line, Staff and Line & Staff Organizations – Maslow's motivational theory; Leadership Styles – Trait theory of leadership – Behavioural theory of Leadership.

Job Analysis - Responsibility of human resource management - Selection procedure – Training of Workers: Apprentice Training – On job training.

3.0 Production Management.

Production, Planning and Control – Types of Production - Routing, Scheduling and Dispatching - PERT and CPM Network Diagrams – Applications - –Calculate Project Duration and identify the critical path of the Project – Simple Problems; Functions of Materials Management - ABC analysis of Inventory. - Economic ordering quantity- Meaning of Supply Chain Management – Processes of Supply Chain Management - Functions of Purchase Department - Purchasing Procedure -Functions of Stores Department – Bin Card.

4.0 Engineering Ethics & Safety and Labour Codes

Engineering Ethics: Definition – Classification of Engineering Ethics - Personal and Business ethics –Value based ethics - Environmental ethics – Meaning of Intellectual Property Rights – Common types of Intellectual Property Rights - Activities of Corporate Social Responsibility (CSR).

Human values : Morals – Values –Character- Caring –Courage - Cooperation – Commitments –Empathy – Honesty- Integrity - Respect for others – Sharing–Service learning.

Industrial Safety: The importance of safety at Workplace -Hazard and accident - Different hazards in the industry -The causes of accidents and prevention of accidents - Direct and indirect cost of accidents.

Industrial Labour Codes: Meaning of Employer and Employee - Objectives of Industrial Labour Codes – Salient features of Code on Wages, 2019 - Salient features of Industrial Relations Code, 2020 - Salient features of Code on Social Security, 2020 - Salient features of Occupational Safety, Health and Working Conditions Code, 2020.

5.0 Entrepreneurship & Start-ups

Entrepreneur - Entrepreneurship - Role of Entrepreneur - Qualities of an entrepreneur- Requirements of an entrepreneur - Expectations of Entrepreneurship - Self-employment schemes - Institutional support - Concept of TQM –Pillars of TQM- Importance of ISO 9000 certification - Concepts of ISO 9000 Series and BIS 14000 Series- Advantages and Drawbacks of ISO 9000 series of standards - List the beneficiaries of ISO 9000. Financial assistance programmes - Concept of Incubation center's – Start-up and its stages — Make or Buy Decision – Break Even Analysis - Branding – Business name, logo and tag line – Concepts of Digital Marketing - Role of E-commerce and Social Media.

REFERENCE BOOKS

1. Industrial Engineering and Management - by O.PKhanna
2. Production Management - by Buffa.
3. Engineering Economics and Management Science - by Banga& Sharma.
4. Personnel Management - by Flippo.
5. Production and Operations Management – S.N.Chary
6. Converging Technologies for Smart Environments and Integrated Ecosystems IERC Book Open Access 2013 pages-54-76.

Table specifying the scope of syllabus to be covered for Unit Test-I & Unit Test-II

Unit Test	Learning Outcomes to be covered
Unit Test – I	From 1.1 to 3.12
Unit Test – II	From 4.1 to 5.12

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-502	Wireless Networks	5	75	20	80

Time Schedule

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1.	Introduction to Wireless Networks	10	13	1	1	CO1
2.	Wireless Network Architectures	15	16	2	1	CO1,CO2
3.	Wireless Communications	20	26	2	2	CO1,CO3
4.	Wireless LAN & PAN	20	29	3	2	CO4
5.	Advanced Wireless Technologies	10	26	2	2	CO1,CO5
Total		75	110	10	8	
Marks				30	80	

Course Objectives	i)To Understand concepts and principles of wireless networking ii)To familiarize Design of wireless LAN & WAN iii) To know Advanced wireless technologies
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Course Outcomes	At the end of course student able to learn the following:		
	CCN-502.1	CO1	Describe the features and applications of wireless communication networks and wireless devices.
	CCN-502.2	CO2	Explain Wireless Physical and Logical Architectures
	CCN-502.3	CO3	Justify the usage of choosing proper wireless communication in real world applications.
	CCN-502.4	CO4	Analyse Wireless LAN & PAN standards
	CCN-502.5	CO5	Apply Advanced Wireless Technologies

Learning Outcomes:

1. Introduction to Wireless Networks

- 1.1. Evolution of wireless Communication
- 1.2. Define wireless networking
- 1.3. State the need of wireless Networks
- 1.4. List and explain types of wireless networks
- 1.5. Describe Wireless networking standards
- 1.6. List Advantages and Disadvantages of Wireless Networks
- 1.7. List Applications of Wireless Networks
- 1.8. Differentiate between wireless networks and wired network
- 1.9. What is WI-FI wireless network?
- 1.10. List the benefits of a Wi-Fi wireless network
- 1.11. How does WI-FI wireless network is used?
- 1.12. Explain how to deploy a wireless network.
- 1.13. List and explain various Wireless communication networks
- 1.14. Define Wireless Access point(WAP)
- 1.15. Why we use a WAP to set up a wireless network?
- 1.16. List and explain Common types of access point configurations
- 1.17. History Bluetooth Technologies
- 1.18. List and explain Steps to connect Bluetooth devices
- 1.19. List applications of Bluetooth devices
- 1.20. Differentiate between Bluetooth and Wi-Fi
- 1.21. Explain How to Set Up a Small Wireless Network for an organization
- 1.22. What is Multi-factor authentication?

2. Wireless Network Architectures

- 2.1. Taxonomy and terminology
 - 2.1.1. Where do wireless networks fit?
 - 2.1.2. Classification
- 2.2. Example wireless network architectures and operational issues
 - 2.2.1. Cellular telephone networks
 - 2.2.2. Wireless local area networks
 - 2.2.3. Wireless personal arean networks
 - 2.2.4. Mobile ad hoc networks
- 2.3. Explain about Wireless network Logical Architecture.
- 2.4. Explain about Wireless network Physical Architecture
- 2.5. LWAPP Protocol
- 2.6. Wireless LAN Devices

3. Wireless Communications

3.1 Radio Communication basics

- 3.1.1 The RF Spectrum
 - 3.1.2 Spread Spectrum Transmission
 - 3.1.3 Wireless Multiplexing and Multiple Access Techniques.
 - 3.1.4 RF Signal Propagation and Reception
 - 3.1.5 NFC

3.2 Infrared Communication Basics

3.2.1 IR Spectrum

3.2.2 IR Signal Propagation and Reception

3.3 Satellite Communication Basics

3.3.1 SATELLITE COMMUNICATION – INTRODUCTION

3.3.2 Need of Satellite Communication

3.3.3 How a Satellite Works?

3.3.4 Pros and Cons of Satellite Communication

3.3.5 Applications of Satellite Communication.....

4. Wireless PAN & LAN

4.1. Wireless PAN

4.1.1. Define PAN

4.1.2. List various wireless PAN standards

4.1.3. Bluetooth

4.1.3.1. List versions of Bluetooth

4.1.3.2. State the importance of Bluetooth

4.1.3.3. Explain Bluetooth Protocol stack

4.1.3.4. Describe Bluetooth connection states

4.1.3.5. Describe Wireless USB

4.1.4. ZigBee

4.1.4.1. State the importance of zigbee

4.1.4.2. Explain ZigBee Super frame Structure in detail

4.1.5. Implementation of Wireless PAN

4.1.5.1. Compare Technical Attribute of PAN Technologies

4.1.5.2. Describe how to choose Wireless PAN Technology

4.2. Wireless LAN

4.2.1. Wireless LAN Standards

4.2.1.1. List and explain WLAN standards

4.2.1.2. Describe WLAN operations

4.2.1.3. Describe 802.11 Network Components

4.2.2. The 802.11 MAC Layer

4.2.2.1. Wireless media access

4.2.2.2. 802.11 CSMA/CA

4.2.2.3. List and explain 802.11 station services

4.2.2.4. List 802.11 distribution system services

4.2.3. The 802.11 PHY Layer

4.2.3.1. Describe 802.11a PHY Layer

4.2.3.2. List RF Band channels used in 802.11a OFDM PHY

4.2.3.3. List modulation methods used in 802.11a OFDM PHY

4.2.3.4. List modulation methods used in 802.11b DSSS PHY

4.2.3.5. Explain 802.11b/g Mixed Mode Interoperability Mechanisms

4.2.3.6. List Data rates at PHY layer

4.2.3.7. List Data rates at MAC layer

4.2.4. Implementing WLAN

4.2.4.1. List and explain Key steps in the planning and implementation of a wireless LAN

4.2.4.2. List Requirements to implement WLAN

4.2.4.3. List Factors Affecting WLAN Operating Range

- 4.2.5. WLAN Security
 - 4.2.5.1. What is Hacking Threat
 - 4.2.5.2. List and explain Wireless LAN Security Threats
 - 4.2.5.3. List various Wireless LAN Security Measures
 - 4.2.5.4. Explain Wired Equivalent Privacy Encryption
 - 4.2.5.5. Explain Wi-Fi Protected Access

5. Advanced Wireless Technologies

- 5.1. Introduction to 4G
 - 5.1.1. List 4G Features,
 - 5.1.2. List Challenges in 4G
 - 5.1.3. List Applications in 4G
 - 5.1.4. Explain Multicarrier Modulation in 4G.
 - 5.1.5. List and explain Smart antenna techniques
 - 5.1.6. The Specifications required for IMT-Advanced Standard (4G)
- 5.2. Introduction to 5G
 - 5.2.1. State the need of 5G.
 - 5.2.2. List Underlying technologies makeup 5G
 - 5.2.3. List features of 5G.
 - 5.2.4. List application areas of 5G.
 - 5.2.5. Give the 5G Road map.
 - 5.2.6. Explain the 5G communications systems architecture.
 - 5.2.7. List the explain security issues and challenges in 5G communication systems.
- 5.3. Light Fidelity(Li-Fi)
 - 5.3.1. What is Light Fidelity (Li-Fi)
 - 5.3.2. History of Li-Fi.
 - 5.3.3. List standards in Li-Fi.
 - 5.3.4. List applications of Li-Fi
 - 5.3.5. Describe Li-Fi Technology details
 - 5.3.6. Differentiate between WiFi (Wireless Fidelity) and LiFi (Light Fidelity)

COURSE CONTENT

1. Introduction To Wireless Networks: Evolution of wireless Communication - wireless networking-need of wireless Networks-Usage of wireless network -Types of wireless networks-Wireless networking standards-Advantages and Disadvantages of Wireless Networks-Applications of Wireless Networks-Wireless networks vs wired network-benefits of a Wi-Fi wireless network-deployment a wireless network-Various Wireless communication networks-Wireless Access point-History of Bluetooth Technologies-connecting Bluetooth devices.

2.Wireless Network Architectures:

Wireless Network Logical Architecture: Need of OSI Network Model - Layers of OSI Network Model -Network Layer Technologies - Data Link Layer Technologies -Explain Physical Layer Technologies

Wireless Network Physical Architecture :List and explain various Wireless Network Topologies - Wireless PAN Devices - Wireless LAN Devices

3.Wireless Communications :

Radio Communication Basics-The RF Spectrum - Spread Spectrum Transmission - Wireless Multiplexing and Multiple Access Techniques- RF Signal Propagation and Reception - Near Field Communications - Infrared Communication Basics- Infrared Propagation and Reception- Satellite Communication Introduction

4. Wireless LAN & PAN:

Wireless PAN: ZigBee - Implementation of Wireless PAN

Wireless LAN: Wireless LAN Standards - The 802.11 MAC Layer - The 802.11 PHY Layer

- Other WLAN Standards - Implementing WLAN - WLAN Security

5. Advanced Wireless Technologies: Introduction to 4G - Introduction to 5G - State the purpose of Li-Fi - Optical communication: Li-Fi.

REFERENCE BOOKS

- 1) Wireless Networking Technology From Principles to Successful implementation by Steve Rackley, Elsevier, Linacre House, Newnes Publications, Great Britain (Unit 1, 2, 3 & 4 units from this book)
- 2) Fundamentals 5G Mobile Networks Jonathan Rodriguez, Wiley Publications (unit 5 from this book)
- 3) WIRELESS COMMUNICATIONS AND NETWORKS Second Edition William Stallings Pearson Prentice Hall publishers
- 4) Advanced wireless Networks 4G, Savoglisis, Wiley Publications
- 5) https://www.4gon.co.uk/solutions/introduction_to_4g.php (Unit 5: comparisons of 4G Systems, specifications)
- 6) https://www.tutorialspoint.com/satellite_communication/satellite_communication_tutorial.pdf

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.2.2
Unit test-2	From 3.3.1 to 5.3.6

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-503	Software Engineering	5	75	20	80

Time Schedule

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1.	Basics of Software Engineering Designs & Life Cycle Models	10	16	2	1	CO1
2.	Software Project Management	18	16	2	1	CO2
3.	Requirement Analysis & Specifications	10	13	1	1	CO3
4.	Software Design, Coding	22	39	3	3	CO4
5.	Software testing, Debugging, Reliability, Quality Management & Maintenance	15	26	2	2	CO5
Total		75	110	10	8	
Marks				30	80	

Course Objectives	i)To know the fundamentals of software engineering & life cycle modes ii)To familiarize project managements iii)To design software projects with the help of software engineering principles and UML models
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Course Outcomes	At the end of the course the student able to learn following:		
	CO1	CCN-503.1	Explain Software life cycle models and basics of software engineering.
	CO2	CCN-503.2	Describe Software Project Management
	CO3	CCN-503.3	Prepare SRS document
	CO4	CCN-503.4	Apply Design ,coding techniques.
	CO5	CCN-503.5	Apply Testing Techniques ,Quality and reliability metrics

Learning Outcomes:

1.0 Basics of Software Engineering Designs & Life Cycle Models

1.1 Know the Evolution and Impact of the Software Engineering

1.1.1 Evolution of an Art to an Engineering Discipline

1.1.2 A Solution to the Software Crisis?

1.2 Know the difference between Programs and Software Products

1.3 Understand the evolution of Software Engineering Design

- 1.3.1 Early Computer Programming
- 1.3.2 High Level Language Programming
- 1.3.3 Control Flow-Based Design
- 1.3.4 Data Structure-Oriented Design
- 1.3.5 Data Flow-Oriented Design
- 1.3.6 Object Oriented Design
- 1.3.7 Other Developments
- 1.4 Explain the Software Life Cycle Models
 - 1.4.1 Classical Waterfall Model
 - 1.4.2 Iterative Water fall Model
 - 1.4.3 Prototyping Model
 - 1.4.4 Evolutionary Model
 - 1.4.5 Spiral Model
 - 1.4.6 AGILE Model
 - 1.4.7 Comparison of Different Life Cycle Models

2.0 Software Project Management

- 2.1 Software Project Manager
 - 2.1.1 Job Responsibilities of a Software Project Manager
 - 2.1.2 Skills Necessary for Software Project Management
- 2.2 Know about Software Project Planning
- 2.3 The SPMP Document
- 2.4 Metrics for Project Size Estimation
 - 2.4.1 Lines of Code
 - 2.4.2 Function Point Metric
- 2.5 Project Estimation Techniques
 - 2.5.1 Empirical Estimation Technique
 - 2.5.2 Heuristic Technique
- 2.6 Staffing Level Estimations
 - 2.6.1 Nordens Work
 - 2.6.2 Putnam's Work
- 2.7 Scheduling
 - 2.7.1 Work Break Down Structure
 - 2.7.2 Activity Networks
 - 2.7.3 Gantt Charts
 - 2.7.4 PERT Charts
- 2.8 Learn how to do Staffing
- 2.9 Who is a Good Software Engineer?
- 2.10 Risk Management
 - 2.10.1 Risk Identification
 - 2.10.2 Risk Assessment
 - 2.10.3 Risk Containment

3.0 Requirement Analysis & Specifications

- 3.1 Requirements Gathering and Analysis
- 3.2 Software Requirement Specifications

- 3.2.1 List Contents of the SRS Document
- 3.2.2 Explain Functional Requirements
- 3.2.3 Describe Procedure to identify the Functional Requirements
- 3.3 How to Document the Functional Requirements
- 3.4 Explain requirements Traceability
- 3.5 List Characteristics of a Good SRS Document
- 3.6 Give Examples of Bad SRS Document
- 3.7 Explain Organization of the SRS Document

4.0 Software Design, Coding

- 4.1 What is a good Software Design?
- 4.2 Define and Classify Cohesion and Coupling
 - 4.2.1 Classification of Cohesiveness
 - 4.2.2 Classification of Coupling
- 4.3 Approaches of Software Design
 - 4.3.1 Function-Oriented Design
 - 4.3.2 Object-Oriented Design
 - 4.3.3 Function-Oriented vs Object-Oriented Design
- 4.4 User Interface Design
 - 4.4.1 List the Characteristics of a good User Interface
 - 4.4.2 Understand the Basic Concepts
 - 4.4.2.1 User Guidance and Online Help
 - 4.4.2.2 Mode Based vs Modeless Interface
 - 4.4.2.3 Graphical User Interface (GUI) vs Text-Based User Interface
 - 4.4.3 Types of User Interface
 - 4.4.3.1 Command Language Based Interface
 - 4.4.3.2 Menu Based Interface
 - 4.4.3.3 Direct Manipulation Interfaces
 - 4.4.4 Component Based GUI Development Window System and Types of Widgets.
- 4.5 Unified Modeling Language
 - 4.5.1 List the goals of UML
 - 4.5.2 Role of UML in Object oriented Design
 - 4.5.3 List and explain Building blocks of UML
 - 4.5.4 List different symbols used in UML notation
 - 4.5.5 Classify and list standard UML diagrams
 - 4.5.6 Know the purpose of Class diagram and draw simple class diagrams
 - 4.5.7 Use case diagram
 - 4.5.7.1 Define the term Use case
 - 4.5.7.2 Know the purposes of Use case diagram
 - 4.5.7.3 Learn to draw the Use case diagram
 - 4.5.8 Interaction diagram
 - 4.5.8.1 State the purpose of Interaction diagram
 - 4.5.8.2 Interaction diagrams
 - 4.5.8.3 List interaction diagrams(sequence & collaboration)
 - 4.5.8.4 learn to draw the Interaction diagrams
- 4.6 Understand the concept of Software Coding

- 4.6.1 Coding Standards and Guidelines - Code Review- Code Walk- Throughs - Code Inspection
- 4.6.2 Clean Room Testing - Software Documentation- Software Testing

5.0 Testing, Debugging, Reliability, Quality Management & Maintenance

- 5.1 Understand Testing
 - 5.1.1 What is Testing?
 - 5.1.2 Differentiate Verification and Validation
 - 5.1.3 List 3 Designs of Test Cases
 - 5.1.4 Compare Testing in the Large vs Testing in the Small
 - 5.1.5 Explain Unit Testing
 - 5.1.6 Explain Black box Testing and White Box Testing.
 - 5.1.7 Explain Open source software testing tools : Selenium, Bugzilla
- 5.2 Debugging
 - 5.2.1 Explain Debugging Approaches.
 - 5.2.2 List the Debugging Guidelines.
- 5.3 Explain Program Analysis Tools (Static Analysis Tools& Dynamic Analysis)
- 5.4 List and Explain Integration Testing
- 5.5 Explain System Testing
- 5.6 Explain Performance Testing.
- 5.7 Understand the concept of Software Reliability
 - 5.7.1 Differentiate Hardware Reliability and Software Reliability
 - 5.7.2 List the different Reliability Metrics
 - 5.7.3 Understand the Reliability Growth Modeling
- 5.8 State the importance of Statistical Testing
- 5.9 Explain Software Quality Management systems
- 5.10 Define SEI Capability Maturity Model

COURSE CONTENT

1. Basics of Software Engineering Designs & Life Cycle Models - Life Cycle Models.

2. Software Project Management- Responsibilities of a Software Project

Manager- Project planning – Metrics-Project Estimation Techniques- Staffing Level Estimation – Scheduling – Risk Management

3. Requirement Analysis and Specification: Requirement Gathering and Analysis - SRS document

4. Software Design , Coding : Good software design, Cohesion and Coupling, Software Design Approaches, User interface Design, Software Coding and

Goals of UML - Role of UML in Object oriented Design - Building blocks of UML : Things, Relationships, and Diagrams - Symbols used in UML notation - Classify and list standard UML diagrams - Class diagram, purposes of class diagram, draw the class diagram - Use case diagram, define the term Use case, purposes of Use case

diagram, draw the Use case diagram - Interaction diagram, purposes of Interaction diagram, the types of interaction diagrams : Sequence diagram and Collaboration diagram, draw the Interaction diagrams.

5. Software Testing, Debugging ,Reliability, Quality Management and maintenance – Testing, Debugging software Reliability- Statistical Testing, Software Quality, Software Quality Management System, SEI capability Maturity Model

REFERENCE BOOKS

1. Fundamentals of Software Engineering – Rajib Mall (PHI) Second Edition.
2. Software Engineering - Jawadekar (TMH)
3. Software Engineering Concepts - Fairley (TMH)
4. Pankaj Jalote international approach to software engineering “:2nd edition
Narosal publishing house 1997
5. <http://www.tutorialspoint.com/uml/>
6. The Unified Modelling Language User guide...Grady Booch

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.6
Unit test-2	From 4.1 to 5.7

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-504	Internet of Things	4	60	20	80

Time Schedule

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1.	Introduction of IOT	12	16	2	1	CO1
2.	Data Protocols	15	26	2	2	CO2
3.	Communication Technologies	15	26	2	2	CO3
4.	Wireless Sensor Networks	15	29	3	2	CO4
5.	Role of IOT	10	13	1	1	CO5
Total		60	110	10	8	
Marks				30	80	

Course Objectives	i)To assess the vision of IoT. ii)To classify Real World IoT applications in various Domains. iii)To understand design methodology for IoT platforms.
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Course Outcomes	At the end of course student able to learn the following :		
	CCN-504.1	CO1	Understand the basic concepts like usage of sensors ,components and frequently used technologies of IoT from a global context
	CCN-504.2	CO2	Illustrate the application of Data protocols of IoT
	CCN-504.3	CO3	Understand various communication technologies of IOT
	CCN-504.4	CO4	Illustrate the use of sensor networks in applications of various domains
	CCN-504.5	CO5	Illustrate applications of IOT

Learning Outcomes:

1. Introduction of IOT

1.1. Introduction:

- 1.1.1. Define IOT
- 1.1.2. List Features
- 1.1.3. Evolution of Connected Devices
- 1.1.4. List Advantages
- 1.1.5. List Disadvantages
- 1.1.6. List the components of IoT
- 1.1.7. List Applications IOT
- 1.1.8. List various connecting technologies
- 1.1.9. Sensors
 - 1.1.9.1. Need of sensor
 - 1.1.9.2. Features of Sensors
 - 1.1.9.3. Classify Sensors based on output, on data types
- 1.1.10. Define actuator
- 1.1.11. List types of Actuators
- 1.1.12. Explain IoT Solutions Architecture
- 1.1.13. List IOT challenges
- 1.1.14. Explain IoT Ecosystem

1.2 Various Connectivity Technologies in IOT:

- 1.2.1 6LoWPANs Technologies
 - 1.2.1.1 Features
 - 1.2.1.2 Applications
 - 1.2.1.3 Advantages
 - 1.2.1.4 Addressing
 - 1.2.1.5 List and explain different packet formats
 - 1.2.1.6 Explain 6LoWPAN protocol stack architecture
- 1.2.2 Describe Routing protocols (RPL, CORPL, LOADng)
- 1.2.3 RFID Technologies
 - 1.2.3.1 What is RFID
 - 1.2.3.2 List features
 - 1.2.3.3 Applications
 - 1.2.3.4 Advantages
 - 1.2.3.5 Explain Working principle
 - 1.2.3.6 Applications

1. Data Protocols

1.1. Message Queue Telemetry Transport(MQTT)

- 1.1.1. Define MQTT
- 1.1.2. List features of MQTT
- 1.1.3. Explain MQTT
- 1.1.4. List components
- 1.1.5. List Methods
- 1.1.6. List Applications
- 1.1.7. Define Secure MQTT
- 1.1.8. Explain Secure MQTT

1.2. Constrained Application Protocol (CoAP)

- 1.2.1. Define CoAP
- 1.2.2. List features
- 1.2.3. Explain CoAP
- 1.2.4. List CoAP message types
- 1.2.5. Explain CoAP message types
- 1.2.6. Differentiate between COAP and MQTT

1.3. Extensible Messaging and Presence Protocol(XMPP)

- 1.3.1. List Features of XMPP
- 1.3.2. History of XMPP
- 1.3.3. Explain XMPP
- 1.3.4. Describe core XMPP Technologies
- 1.3.5. List applications of XMPP

1.4. Advanced Message Queuing Protocol (AMQP)

- 1.4.1. List Features of AMQP
- 1.4.2. Explain AMQP in detail
- 1.4.3. List applications of XMPP

2. Communication Technologies

2.1. IEEE 802.15.4

- 2.1.1. List features of IEEE 802.15.4
- 2.1.2. Why 802.15.4 is preferred in IoT communication?
- 2.1.3. Explain IEEE 802.15.4
- 2.1.4. List IEEE 802.15.4 Variants
- 2.1.5. List and explain IEEE 802.15.4 Types

2.2. ZIGBEE

- 2.2.1. What is ZIGBEE
- 2.2.2. List features
- 2.2.3. List components

- 2.2.4. List different topologies
- 2.2.5. List types
- 2.2.6. List Applications
- 2.2.7. Explain different topologies of ZIGBEE
- 2.2.8. Explain ZIGBEE types
- 2.3. Near field communication(NFC)
 - 2.3.1. What is NFC
 - 2.3.2. List types
 - 2.3.3. List applications
 - 2.3.4. Explain working principle
 - 2.3.5. Describe modes of operation of NFC
- 2.4. Bluetooth
 - 2.4.1. What is the purpose of Bluetooth
 - 2.4.2. List features
 - 2.4.3. List functions
 - 2.4.4. List applications
 - 2.4.5. Explain Bluetooth technology in detail
 - 2.4.6. Describe Pico Net

4 Wireless Sensor Networks (WSN)

- 4.1. State the importance of Wireless Sensor Network
- 4.2. List Applications
- 4.3. List characteristics of Sensor Networks
- 4.4. What are the Requirements of WSN
- 4.5. List and explain WSN Network Topologies
- 4.6. Sensor networks
 - 4.6.1. List and types of sensor networks
 - 4.6.2. Single Source Single Object Detection
 - 4.6.3. Single Source Multiple Object Detection
 - 4.6.4. Multiple Source Single Object Detection
 - 4.6.5. Multiple Source Multiple Object Detection
- 4.7. Describe Challenges in Wireless Sensor Networks
- 4.8. Define Node in WSN
- 4.9. Explain node Behaviour in WSNs
- 4.10. List and explain Security Services in WSN
- 4.11. Explain Information theoretic self-management in WSN
- 4.12. Applications of WSN
- 4.13. State importance of Wireless Multimedia Sensor Networks
- 4.14. Explain Wireless Multimedia Sensor Networks(WMSN)
- 4.15. State importance of Stationary Wireless Sensor Networks
- 4.16. Explain Stationary Wireless Sensor Networks
- 4.17. State importance of Mobile Wireless Sensor Networks
- 4.18. Explain Mobile Wireless Sensor Networks
- 4.19. What is Machine to Machine Communications(M2M)
- 4.20. List applications and features of M2M
- 4.21. List and explain M2M sensor nodes

5. Role of IOT

5.1. Explain Role of IOT in automation of the following applications

5.1.1.State the importance of automation in IOT.

5.1.2.List automation applications of IOT

5.1.3.List advantages of IOT in automation

5.1.4.List disadvantages of IOT in automation

5.1.5.What Is The Impact Of Iot On Industrial Automation?

5.1.6.List Types Of Industrial Automation

5.1.7.APPLICATIONS OF IOT

5.1.7.1. Health care

5.1.7.2. Smart Home

5.1.7.3. Smart Cities

5.1.7.4. Smart class rooms

5.1.7.5. Smart Energy

5.1.7.6. Smart Transportation and Mobility

5.1.7.7. Smart Factory

COURSE CONTENT

UNIT 1: Introduction of IOT

INTRODUCTION to IOT – Definition – Applications – Technologies – Sensor features –Types – Actuator list – Components – Challenges

Connectivity technologies - 6LoWPAN –Features – Addressing –Routing

RFID – features – working principle – Applications

UNIT 2: Data Protocols

MQTT – Definition – features – components – applications – MQTT – SMQTT

CoAP- Definition – message types

XMPP – features – core technologies – applications

AMQP- Features-applications

UNIT 3 : Communication Technologies

IEEE 802.15.4 – features – variants – types

ZIGBEE –features – components – technologies – types – applications

NFC – types –modes – applications

Bluetooth - purpose –features - Technologies- applications

UNIT 4: Wireless Sensor Networks

Wireless Sensor Networks- Applications -Types-Challenges-node Behaviour-Information theoretic self-management-Applications-WMSN-.

Stationary Wireless Sensor Networks-Mobile Wireless Sensor Networks-M 2 M-applications -features-sensor nodes-

UNIT 5: Role Of IOT

Role of IOT in automation of applications - Health care -Smart Home-Smart Cities

REFERENCE BOOKS

- 1) <https://onlinecourses-archive.nptel.ac.in/>
- 2) Vijay Madiseti, Arshdeep Bahga, "Internet of Things: A Hands-On Approach", Orient Blackswan Pvt., Ltd., New Delhi, 2015.
- 3) Waltenegus Dargie, Christian Poellabauer, "Fundamentals of Wireless Sensor Networks: Theory and Practice", A John Wiley and Sons, Ltd., Publication, 2010.
- 4) Jeeva Jose, "Internet of Things", (ISBN: 978-93-86173-591) KBP House, 1st edition, 2018.
- 5) Interconnecting Smart Objects with IP: The Next Internet, Jean-Philippe Vasseur, Adam Dunkels, Morgan Kuffmann
- 6) Designing the Internet of Things, Adrian McEwen (Author), Hakim Cassimally
- 7) Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems, Dr. Ovidiu Vermesan, Dr. Peter Friess, River Publishers
- 8) Internet of Things (A Hands-on-Approach), Vijay Madiseti, Arshdeep Bahga
- 9) 6LoWPAN: The Wireless Embedded Internet, Zach Shelby, Carsten Bormann, Wiley
- 10) Building the internet of things with ipv6 and mipv6, The Evolving World of M2M Communications, Daniel Minoli John Wiley & Sons
- 11) Recent research/white papers

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.2
Unit test-2	From 3.3 to 5.13

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-505	High Speed Networks	5	75	20	80

Time Schedule

S.No.	Chapter/Unit title	No. of Periods	Weightage of Marks	Short Type	Essay Type	COs mapped
1.	Introduction to High Speed Networks	16	16	2	1	CO1
2.	Congestion control and traffic management	16	19	3	1	CO2
3.	QOS in IP Networks	15	23	1	2	CO3
4.	Optical Networks.	14	26	2	2	CO4
5.	Logical Network design in High speed networking	14	26	2	2	CO5
Total		75	110	10	8	
Marks				30	80	

Course Objectives	i) To know the basic concepts, the architectures, the protocols, the advantages and limitations, and the recent development of various high-speed networking technologies. ii) Design TCP and ATM Congestion Control for High speed networks with respect to some protocol design issues iii) Evaluate the QOS related performance measurements of High speed networks iv) Know the significance of optical networks v) Familiarize with logical network design
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Course Outcomes	At the end of the course the student will be able to:		
	CO1	CCN-505.1	Describe fundamental concepts of protocols and architecture used for high-speed networks.
	CO2	CCN-505.2	Analyze Congestion control and traffic management techniques
	CO3	CCN-505.3	Apply the techniques of QOS in IP networks to ensure equality
	CO4	CCN-505.4	Explain features of Optical networks
	CO5	CCN-505.5	Use various techniques of logical network design

Learning Outcomes

1. Introduction to High Speed Networks

- 1.1. Introduction to frame relay network.
 - 1.1.1 Define frame relay network.
 - 1.1.2 Give the header format of framerelay network.
 - 1.1.3 Explain congestion control mechanisms in Framerelay network
- 1.2. State the importance of x.25
- 1.3. Explain X.25 network
- 1.4. Distinguish between Framerelay and X.25.
- 1.5. Explain Asynchronous transfer mode (ATM) technology.
 - 1.5.1 Define ATM
 - 1.5.2 State the need of ATM Technology.
 - 1.5.3 Cell structure format of ATM
 - 1.5.3.1. Explain Generic flow control
 - 1.5.3.2. Explain Header error control
 - 1.5.3.3. Explain Effect of error in cell header
- 1.6. Describe ATM protocol architecture.
- 1.7. Explain ATM logical connection in networks.
 - 1.7.1. State the importance of VPS (Virtual Private Service)
 - 1.7.2. .Call establishment using VPS (Virtual Private Service)
- 1.8. Explain ATM service categories.
 - 1.8.1.Real time services
 - 1.8.2.Non-real time services
- 1.9. Explain different types of High speed LAN's
 - 1.9.1.Fast Ethernet
 - 1.9.2.Gigabit Ethernet
 - 1.9.3.Fiber channel
- 1.10. Explain various services of IEEE 802.11
 - 1.10.1. WI-Fi and Bluetooth
 - 1.10.2. Wireless LAN's
 - 1.10.3. MANET-Mobile Ad-hoc networking
 - 1.10.4. LI-Fi
- 1.11. State the purpose of IEEE 802.16 (Wi-Max-Worldwide Interoperability for Microwave Access)
- 1.12. Comparison between Wi-Fi and WiMax

2.0 Congestion Control and Traffic Management

- 2.1. Define Congestion and its impact in networks.
- 2.2. Explain various Congestion control mechanisms in high speed networks.
 - 2.2.1. Implicit and Explicit congestions
 - 2.2.2. Frame relay discard eligibility using DE bit(Discard Eligibility bit)

- 2.2.3. Frame relay error checking using CRC
- 2.3. Explain congestion control techniques in packet switching networks
 - 2.3.1. Open loop congestion control policies
(Retransmission, window, discarding and acknowledgement)
 - 2.3.2. Closed loop congestion control (Backpressure, Choke packet, Implicit signalling, Explicit signalling)
- 2.4. Explain the following concepts in TCP
 - 2.4.1. TCP Traffic
 - 2.4.2. TCP load balancing
- 2.5. Explain the following TCP/IP internal gateway routing protocols
 - 2.5.1. RIP (Routing Information Protocol)
 - 2.5.2. OSPF (Open Shortest Path First)
 - 2.5.3. IGRP (Internal Gateway Routing Protocol)
 - 2.5.4. EIGRP (Extended Internal Gateway Routing Protocol)
- 2.6. Explain the following TCP/IP exterior gateway routing protocols
 - 2.6.1. BGP (Border Gateway Routing Protocol)
 - 2.6.2. EGP (Exterior Gateway Routing Protocol)

3.0 Quality of service in IP networks

- 3.1. Explain Integrated service Architecture (ISA) in IP networks
 - 3.1.1. ISA implementation in router
 - 3.1.2. ISA functions
- 3.2. Explain the following ISA components
 - 3.2.1. Explain ISA components Background functions
 - 3.2.1.1. RSVP (Resource Reservation Protocol)
 - 3.2.1.2. Admission Protocol
 - 3.2.1.3. Management agent
 - 3.2.1.4. Routing protocol
 - 3.2.2. Explain ISA components Forward functions
 - 3.2.2.1. Classifier and route selection
 - 3.2.2.2. Packet Scheduler
- 3.3. Explain ISA services
 - 3.3.1. Guarantees service
 - 3.3.2. Controlled load
 - 3.3.3. Best effort
- 3.4. Explain the following Queuing discipline techniques to ensure QOS in IP
 - 3.4.1. FIFO (First In First Out)
 - 3.4.2. FQ (Fair Queuing)
 - 3.4.3. BRfq (Bit Round Fair Queuing)
- 3.5. Comparison of FIFO, FQ and BRfq
- 3.6. Explain weighted fair queuing technique to ensure QOS in IP
- 3.7. Distinguish FIFO and WFQ

- 3.8. Explain the RED(Random Early Detection) algorithm
- 3.9. List Goals of RED
- 3.10. Explain RSVP(Resource Reservation Protocol)for QOS support
- 3.11. Explain MPLS(Multi Protocol Label Switching) for QOS support

4.0 Optical Networks

- 4.1** Introduction to Optical networks
 - 4.1.1 Define Optical Network
 - 4.1.2 State the Role of optical network
 - 4.1.3 Describe Structure of optical networks
 - 4.1.4 List various optical networks
- 4.2** Explain Optical network components
 - 4.2.1 Couplers
 - 4.2.2 Isolators
 - 4.2.3 Circulators
 - 4.2.4 Multiplexers&Filters
 - 4.2.5 Optical amplifiers
 - 4.2.6 Switches
 - 4.2.7 Wavelength converters
- 4.3** Explain SONET (Synchronous Optical network) architecture
- 4.4** List the applications of SONET
- 4.5** Optical multiplexing techniques
 - 4.5.1 Define Multiplexing
 - 4.5.2 State the usage of Multiplexing in optical networks
 - 4.5.3 Explain OTDM(Optical Time Division Multiplexing)
 - 4.5.4 Describe WDM(Wavelength Division Multiplexing)
- 4.6** Explain the following optical network switching technologies
 - 4.6.1 Optical Burst Switching(OBS)
 - 4.6.2 Optical Packet Switching(OPS)
 - 4.6.3 Optical Circuit Switching(OCS)
- 4.7** List the applications of optical networks

5.0 Logical Network Design in High Speed Networking

- 5.1 State the need of logical network design
- 5.2 Explain the following logical network design topologies
 - 5.2.1. Hierarchical Network Design
 - 5.2.2. Redundant Network Design Topologies
 - 5.2.3. Modular Network Design
 - 5.2.4. Designing a Campus Network Design Topology
 - 5.2.5. Designing the Enterprise Edge Topology
 - 5.2.6. Secure Network Design Topologies
- 5.3 What are the guidelines for assigning network layer addresses?

- 5.4 Explain Hierarchical Model for Assigning the Addresses
- 5.5 List the guidelines for designing a Model for Naming
- 5.6 Switching and routing protocols
 - 5.6.1 Define switching and routing
 - 5.6.2 State the need of switching and routing in high speed networks
- 5.7 Explain Spanning Tree protocol in high speed network switching
- 5.8 Explain the following Routing protocols
 - 5.8.1.IP Routing
 - 5.8.2.Novell NetWare Routing

COURSE CONTENTS: HIGH SPEED NETWORKS

1. Introduction to High Speed Networks:

Introduction-frame relay networks –ATM protocol architecture-ATM logical connection –ATM cells-ATM service categories -AAL- high speed LANS: The emergence of high speed LANS-Ethernets-fibre channel-wireless LANS, Local broadband and Adhoc networks. Wireless LANS-IEEE802.11

2. Congestion Control and Traffic Management

Congestion control in data networks and internets- Congestion control mechanisms- congestion control techniques in packet switching networks - TCP traffic - TCP load balancing- TCP/IP internal gateway routing protocols-TCP/IP exterior gateway routing protocols-Interior routing protocols.

3. QOS in IP Networks:

Integrated service architecture- ISA components - ISA services - Queuing discipline techniques to ensure QOS in IP - random early detection – RSVP (Resource Reservation Protocol) - MPLS (Multi Protocol Label Switching)

4. Optical Networks:

Optical networks - Role of optical network - Structure of optical networks - Optical network components – SONET (Synchronous Optical network) - Optical multiplexing techniques – optical network switching technologies-applications of optical networks.

5. Logical Network Design in High Speed Networking:

Designing Network Design: Hierarchical Network Design, Redundant Network Design Topologies, Modular Network Design, Designing a Campus Network Design Topology, Designing the Enterprise Edge Topology, Secure Network Design Topologies

Designing Models for Addressing and Naming: Guidelines for Assigning Network Layer Addresses, Using a Hierarchical Model for Assigning Addresses, Designing a Model for Naming.

Selecting Switching and Routing Protocols:

Selecting Bridging & Switching Protocols, Spanning Tree Protocol Enhancements–

Selecting Routing Protocols: Characterizing Routing protocols, IP Routing, Novell NetWare Routing,

Text Books:

1. William Stallings, "HIGH SPEED NETWORKS AND INTERNET", Pearson Education, Second Edition, 2002.
2. "Data Communication and Networking" by Behrouz A Forouzan
3. "Data and Computer Communication" by William Stallings

References:

1. Kaven Pahlavan And Prashant Krishnamoorthy, "Principles Of Wireless Network", Prentice Hall Of India
2. Adrian Farrel, "The Internet And Its Protocols", Elsevier Publications, 2011 Download
3. Larry L. Peterson and Bruce S. Davie, "Computer Networks", Third edition, Elsevier Publications, 2003

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.5
Unit test-2	From 3.6 to 5.8

Course Code	Course title	No. of periods/week	Total no. of periods	Marks for FA	Marks for SA
CCN-506	Wireless Networks Lab	3	45	40	60

Guidelines:

1. Number of periods allotted per week is 4 periods.
2. Out of 4 Periods one period is allotted for explaining syntax and usage of commands and three periods are for practicing.
3. In theoretical part of each week the syntax of tcl commands, usage of tools, and prerequisite theory for respective exercise is to be taught to the students.
4. All of the exercises of this course are to be run on simulators.
5. The simulators like NS2/NS3/Packet Tracer/GNS3 or any relevant simulators can be used.
6. The lecturer/Instructor has to demonstrate the respective exercise during the theoretical part.
7. Windows/Linux Operating system can be used.

Software Requirements:

Operating System: Windows 7 or above / Ubuntu / Linux

Simulator: NS2/NS3/GNS3/Packet Tracer/ ANY RELEVANT SIMULATOR

Hardware requirements:

Processor: P-IV or above ,AM:2GB,HDD:512GB

Note: All the experiments can be done in the physical networking infrastructure.

S No	Chapter/Unit Title	No. of Periods	Cos Mapped
1.	WLAN	11	CO1
2.	Wifi, Bluetooth and ZigBee	11	CO2
3.	4G and 5G	23	CO3
	Total	45	

COURSE OBJECTIVES	1. Introduce the concept of wireless networks simulation to the students 2. Involve students in analytical studies of Computer Networks through network simulation 3. Configuration and use of Wireless technologies using simulators
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Course Outcomes	At the end of the course the student will be able to		
	CO1	CCN-506.1	Practice the establishment and configuration of standard protocols Of WLAN to test the data transfer parameters.
	CO2	CCN-506.2	Use standard protocols of Short Range Wireless Networks to test The data transfer parameters.
	CO3	CCN-506.3	Demonstrate the configuration of standard protocols of Mobile Wireless Network Technologies to test the data transfer parameters.

Learning Outcomes:

Wireless LAN:

1. Create WLAN and configure
2. Simulate CSMA/CA protocol on WLAN
3. Configuration of wireless networks using CISCO AP
4. Simulate ICMP and DHCP on WLAN
5. Create public and private networks simultaneously using Gateway

Wi-Fi, Bluetooth and ZigBee:

6. Perform data transfer using Bluetooth Technology
7. Simulate Bluetooth Network and draw the Bluetooth parameter values
8. Simulate ZigBee network and draw the ZigBee parameter values
9. Create Wi-Fi Network and perform data exchange using RTC/CTS
10. Perform data transfer using RFID
11. Simulate ZigBee with Energy Model and find the rx Power and tx Power

Mobile 4G and 5G Technologies

12. Perform data transfer using GPRS Mobile Handset on Wireless Network
13. Perform data transfer using UMTS Mobile Handset on Wireless Network
14. Calculate SINR including Beam Tilt for Downlink and Uplink on Wireless Network
15. Find the Frequency Reuse for Co-Channel cells and Cell Cluster
16. Simulate MIMO technology in wireless networks
17. Simulate 4G LTE Network
18. Simulate 5G Network

The competencies and key competencies to be achieved by the student

S.N o.	Name of the experiment	Objectives	Key Competencies
1	Exercise on creation of WLAN and configuration.	❖ Identification and familiarization of various components of Simulator ❖ Know WLAN design and configuration	❖ Installation of simulator ❖ Identify devices, cables, ports and connection types in the simulator ❖ Practice connecting devices in the WLAN with different connection types and ports care. ❖ Configure all PCs, Devices and Ports with suitable IPs ❖ Check proper connections established or not and connectivity
2	Exercise on simulation of CDMA/CA on WLAN	❖ WLAN design and configuration ❖ Simulation of CDMA/CA	❖ Practice connecting devices in the WLAN with different connection types and ports. ❖ Configure all PCs ,Devices and Ports with suitable IPs ❖ Check proper connections established or not and connectivity ❖ Simulation of CDMA/CA ❖ Calculate Throughput of the WLAN
3	Exercise on configuration of Wireless Network using CISCOAP	❖ WLAN design and configuration ❖ CISCO AP configuration in the WLAN	❖ Practice connecting devices in the WLAN with different connection types and ports. ❖ Configure all PCs, Devices and Ports with suitable IPs ❖ Check proper connections established or not and connectivity ❖ Connecting CISCOAP into the WLAN ❖ Configuration of CISCOAP ❖ Check communication between CISCOAP and WLAN devices
4	Exercise on simulate ICMP and DHCP on WLAN	❖ WLAN design and configuration ❖ Configure DHCP in WLAN	❖ Practice connecting devices in the WLAN with different connection types and ports care. ❖ Check proper connections established or not And connectivity ❖ DHCP configuration and run ❖ Check IP addresses of the WLAN devices ❖ Check ICMP packets transmission between devices

5	Exercise on creating public and private networks simultaneously with gateway	❖ WLAN design and configuration ❖ Configure Private, Public network and Gateway	❖ Practice connecting devices in the WLAN with different connection types and ports care. ❖ Check proper connections established or not and connectivity ❖ Configuration of Private Network ❖ Configuration of Public Network ❖ Configuration of Gateway ❖ Check Connectivity between Public and private using Gateway
6	Exercise on perform data transfer using Bluetooth	❖ WLAN design and configuration ❖ Configure Bluetooth and perform data transfer	❖ WLAN Design and configuration ❖ Adding Blue tooth in the WLAN ❖ Configure Bluetooth in the WLAN ❖ Practice data transfer from master To slave and vice versa
7	Exercise on simulate Bluetooth network and draw parameter values	❖ WPAN design and configuration ❖ Perform data transfer and know the throughput of the network	❖ WPAN design and configuration ❖ Practice data transfer from master to slave and vice versa ❖ Know the simulation Bluetooth parameters ❖ Draw the Bluetooth parameters ❖ Know/calculate the throughput of the Data transfer
8	Exercise on simulate ZigBee network and draw parameter values	❖ Design and configure WPAN with ZigBee ❖ Perform data transfer and know the throughput of the network	❖ Design and configure WPAN with ZigBee ❖ Practice data transfer from source to destination ❖ Know the simulation ZigBee parameters ❖ Draw the ZigBee Parameter ❖ Know/calculate the throughput of the data transfer
9	Exercise on create Wi-Fi network and perform data exchange using RTC/CTS	❖ Design and configure Wi-Fi network ❖ Perform data exchange using RTC/CTS	❖ Design and Configure Wi-Fi Network ❖ Practice data exchange using RTC/CTS ❖ Know the simulation Wi-Fi parameters ❖ Draw the Wi-Fi parameters ❖ Know/calculate the throughput of the data transfer
10	Exercise on Perform data transfer using RFID	❖ Design and configure Wireless network with RFID ❖ Perform data transfer using RFID technology	❖ Design and Configure Wireless Network ❖ Add RFID card/tag and RFI reader ❖ Practice data transfer from RFID reader to RFID card/tag ❖ Draw the RFID parameters ❖ Know/calculate the throughput of the data transfer
11	Exercise on Simulate ZigBee with Energy Model and find the rx Power and tx Power	❖ Design and configure WPAN with ZigBee ❖ Perform data transfer and know the throughput of the network	❖ Design and configure WPAN with ZigBee ❖ Add Energy Model to ZigBee ❖ Practice data transfer from source to destination ❖ Know the simulation ZigBee parameters ❖ Draw the rx Power and tx Power Parameter values ❖ Know/calculate the throughput of the data transfer

12	Exercise on Perform data transfer using GPRS Mobile Handset on Wireless Network	❖ Design and configure Wireless Network ❖ Perform data transfer using GPRS Mobile Handset	❖ Design and configure Wireless Network ❖ Add GPRS Mobile Handset ❖ Practice Data Transfer using GPRS mobile handset ❖ Know the simulation parameters of GPRS mobile data transfer ❖ Know/calculate the throughput of the data transfer
13	Exercise on perform data transfer using UMTS mobile handset on Wireless Network	❖ Design and configure Wireless Network ❖ Perform data transfer using UMTS Mobile Handset	❖ Design and configure Wireless Network ❖ Add UMTS Mobile Handset ❖ Practice Data Transfer using UMTS mobile handset ❖ Know the simulation parameters of UMTS mobile data transfer ❖ Know/calculate the throughput of the Data transfer
14	Exercise on calculating SINR for Downlink and uplink in the wireless network	❖ Use predefined wireless network ❖ Calculate SINR for downlink and uplink	❖ Use the predefined network ❖ Add mobile in the wireless network ❖ Practice downlink operation ❖ Practice uplink operation ❖ Draw parameter values ❖ Calculate SINR for downlink ❖ Calculate SINR for uplink
15	Exercise to find the frequency reused for co-channel cells and cell clusters	❖ Use predefined wireless cellular network ❖ Find the frequency reused for co-channel cells and cell clusters	❖ Use the predefined network/design new wireless cellular network suitable for scenario ❖ Select cells in the cellular network ❖ Practice with different cells in the cellular network ❖ Identify the co-channel cells ❖ Identify the cell clusters
16	Exercise on simulate MIMO technology in Wireless network	❖ Design and configure wireless network ❖ Configure MIMO technology	❖ Design and configure Wireless network ❖ Configure MIMO technology ❖ Practice data transfer using MIMO technology ❖ Know/calculate the throughput of the Wireless network
17	Exercise on simulate 4G mobile Network	❖ Design and configure wireless mobile network	❖ Design and configure Wireless network ❖ Configure 4G technology ❖ Practice data transfer using 4G technology ❖ Know/calculate the throughput of the wireless network
18	Exercise on simulate 5G mobile Network	❖ Design and configure wireless mobile network	❖ Design and configure Wireless network ❖ Configure 5G technology ❖ Practice data transfer using 5G technology ❖ Know/calculate the throughput of the wireless network

References:

- Introduction to Network Simulator NS2, 2 nd Edition; Teerawat Issariyakul, Ekram Hossain
 - Manual of NS3 <https://www.nsnam.org/docs/manual/ns-3-manual.pdf>
 - Tutorials of NS3 <https://www.nsnam.org/docs/tutorial/ns-3-tutorial.pdf>
 - Advanced Network Technologies Virtual Lab <http://vlabs.iitkgp.ernet.in>
 - Follow the steps to configure ns2 simulator
<https://www.youtube.com/watch?v=D3A2ZNZI1Q>
- Packet tracer experiments <https://www.youtube.com/watch?v=frUQMHXhnvs>
- Fading Channels and Mobile Communications Virtual Laboratory
<http://vlabs.iitkgp.ac.in/fcmc/index.html>

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-507	Advanced Computer Networks Lab	3	45	40	60

Guide Lines:

1. Number of periods allotted per week is 3 periods.
2. Out of 3 Periods one period is allotted for explaining syntax and usage of commands and two periods are for practicing.
3. In theoretical part of each week the syntax of tcl commands, usage of tools, and prerequisite theory for respective exercise is to be taught to the students.
4. Most of the exercises of this course are to be run on simulators (using respective commands) and very few are to be developed through programming language.
5. The simulators like NS2/NS3/Packet Tracer/GNS3 or any other open-source simulators can be used.
6. The lecturer/Instructor has to demonstrate the respective exercise during the theoretical part.
7. Windows/Linux Operating system can be used.
8. For certain programming exercises Java/C/Python can be used.

NOTE: If full-fledged physical networking infrastructure is available then utilize it instead of simulators.

Sl.No.	Chapter/Unit Title	No. of Periods	CO's Mapped
1	Basics of Local area networks	12	C01
2	Concepts of Flow Control protocols	12	C02
3	Concepts of Access Control	6	C03
4	Concepts of Routing Algorithms	6	C04
5	Concepts of client server, wireless Networks	9	C05
Total number of periods		45	

Course Objectives	• Introduce the concept of network simulation to the students • Involve students in analytical studies of Computer Networks through network simulation • To understand client server model
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Course outcomes	At the end of the course the student able to :		
	C01	CCN-507.1	Demonstrate LAN topologies bus, ring, star
	C02	CCN-507.2	Perform Flow control protocols over noiseless channel and noisy channel
	C03	CCN-507.3	Use Access control like CSMA/CD and CSMA/CA
	C04	CCN-507.4	Practice Routing algorithms used in TCP (distance vector) and UDP (link state routing)
	C05	CCN-507.5	Implement Client server networking model, wireless networks like wireless sensor network and MANET

Learning Outcomes:

1. Exercise on Knowing installation of NS2/NS3/Packet Tracer/GNS3
2. Exercise on Bus topology
3. Exercise on Ring topology
4. Exercise on Star topology
5. Exercise on Stop and Wait sliding window protocol
6. Exercise on Go back N protocol
7. Exercise on selective repeat protocols
8. Exercise on high level data link control protocol
9. Exercise on performance analysis of CSMA/CD
10. Exercise on performance analysis of CSMA/CA
11. Exercise on Distance vector routing algorithm
12. Exercise on Link state routing algorithm
13. Exercise on socket programming
14. Exercise on implementing a wireless sensor network
15. Exercise on implementing a mobile adhoc network

Key Competencies

Sl.No.	Name of Experiment	Objectives	Key Competencies
1	Exercise on Knowing installation of NS2/NS3/ Packet Tracer/GNS3	• Learn the basic concepts about open-source network simulator NS-2/NS-3, • Learn to download, install and work with NS-2/NS-3 • Identifying and solving typical errors encountered during installation of NS-2/NS-3	• Observe the NS2/NS-3 simulator running • Initialize and terminate aspects of network simulator object • Define the network topology: nodes, links, queues, mobility of nodes, if any • Define the network traffic: creating agents and their applications • Set “trace” for Network Animator (NAM) • Trace
2	Exercise on Bus topology	• Learn about Local Area Network (LAN); Bus topology and protocols uses in LAN • Practice simulating a LAN using ns2/NS-3	• Observe the NS2/NS-3 simulator running • Initialize and terminate aspects of network simulator object • Define the network topology: nodes, links, queues, mobility of nodes, lan if any • Define the network traffic: creating agents and their applications • Set trace for Network Animator (NAM) • Trace

3	Exercise on Ring topology	• Learn about Local Area Network (LAN); Ring topology and protocols uses in LAN • Practice simulating a LAN using ns2/NS-3	• Observe the NS2/NS-3 simulator running • Initialize and terminate aspects of network simulator object • Define the network topology: nodes, links, queues, mobility of nodes,lan if any • Define the network traffic: creating agents and their applications • Set trace for Network Animator (NAM) • Trace
4	Exercise on Star topology	• Learn about Local Area Network (LAN); Star topology and protocols uses in LAN • Practice simulating a LAN using ns2/NS-3	• Observe the NS2/NS-3 simulator running • Initialize and terminate aspects of network simulator object • Define the network topology: nodes, links, queues, mobility of nodes,lan if any • Define the network traffic: creating agents and their applications • Set trace for Network Animator (NAM) [optional] • Trace
5	Exercise on Stop and Wait sliding window protocol	• Learn about flow control protocols for noiseless channel; Stop and Wait protocol • Practice simulating a flow control using ns2/NS-3	• Observe the NS2/NS-3 simulator running • Initialize and terminate aspects of network simulator object • Define the network topology: nodes, links, queues, mobility of nodes, if any • Define the network traffic: creating agents and their applications • Set trace for Network Animator (NAM) • Trace
6	Exercise on Go back N protocol	• Learn about flow control protocols for noisy channel; Go back N protocol • Practice simulating a flow control using ns2/NS-3	• Observe the NS2/NS-3 simulator running • Initialize and terminate aspects of network simulator object • Define the network topology: nodes, links, queues, mobility of nodes, if any

			• Define the network traffic: creating agents and their applications • Set trace for Network Animator (NAM) • Trace
7	Exercise on selective repeat protocols	• Learn about flow control protocols for noisy channel; selective repeat protocols • Practice simulating a flow control using ns2/NS-3	• Observe the NS2/NS-3 simulator running • Initialize and terminate aspects of network simulator object • Define the network topology: nodes, links, queues, mobility of nodes, if any • Define the network traffic: creating agents and their applications • Set trace for Network Animator (NAM) [optional] • Trace
8	Exercise on study of high level data link control protocol	• Learn about data link layer protocol; high level data link control protocol • Practice simulating a flow control using ns2/NS-3	• Observe the NS2/NS-3 simulator running • Initialize and terminate aspects of network simulator object • Define the network topology: nodes, links, queues, mobility of nodes, if any • Define the network traffic: creating agents and their applications • Set trace for Network Animator (NAM) • Trace
9	Exercise on performance analysis of CSMA/CD	• Learn about Access control protocol; CSMA/CD • Practice simulating a access control using ns2/NS-3	• Observe the NS2/NS-3 simulator running • Initialize and terminate aspects of network simulator object • Define the network topology: nodes, links, queues, mobility of nodes, if any • Define the network traffic: creating agents and their applications • Set trace for Network Animator (NAM) [optional] • Trace
10	Exercise on performance analysis of CSMA/CA	• Learn about Access control protocol; CSMA/CA	• Observe the NS2/NS-3 simulator running

		Practice simulating a access control using ns2/NS-3	- Initialize and terminate aspects of network simulator object - Define the network topology: nodes, links, queues, mobility of nodes, if any - Define the network traffic: creating agents and their applications - Set trace for Network Animator (NAM) - Trace
11	Exercise on Distance vector routing algorithm	- Learn about routing algorithms; Distance vector routing algorithm - Practice simulating a routing algorithm using ns2/NS-3	- Observe the NS2/NS-3 simulator running - Initialize and terminate aspects of network simulator object - Define the network topology: nodes, links, queues, mobility of nodes, if any - Define the network traffic: creating agents and their applications - Set trace for Network Animator (NAM) - Trace
12	Exercise on Link state routing algorithm	- Learn about routing algorithms; Link state routing algorithm; Dijkstra algorithm. - Practice simulating a routing algorithm using ns2/NS-3	- Observe the NS2/NS-3 simulator running - Initialize and terminate aspects of network simulator object - Define the network topology: nodes, links, queues, mobility of nodes, if any - Define the network traffic: creating agents and their applications - Set trace for Network Animator (NAM) [optional] - Trace
13	Exercise on socket programming	- Understand basics of .net package in java or any respective package depending language used - Develop socket programming in client server model	Practice Communication through sockets in client server model in simplex, half duplex, full duplex modes
14	Exercise on implementing a wireless sensor network	Understand basic concepts of Wireless Sensor Networks (WSNs), types, applications of WSN	- Observe the NS2/NS-3 simulator running - Initialize and terminate aspects of network simulator object

			• Define the network topology: nodes, links, queues, mobility of nodes, if any • Define the network traffic: creating agents and their applications • Set trace for Network Animator (NAM) • Trace
15	Exercise on implementing a mobile adhoc network	• Understand about the basics of Mobile Ad-hoc Networks (MANETs) and different routing protocols • Setup a network with wireless nodes using ns2/NS-3	• Observe the NS2/NS-3 simulator running • Initialize and terminate aspects of network simulator object • Define the network topology: nodes, links, queues, mobility of nodes, if any • Define the network traffic: creating agents and their applications • Set trace for Network Animator (NAM) • Trace

Software requirements	• NS2/NS3 Simulator in Linux /Windows Operating system • Sample configuration steps for Windows 10 64 bit Operating System • Ubuntu app for windows • Xming server for Windows
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References:

- Introduction to Network Simulator NS2, 2nd Edition; TeerawatIssariyakul , Ekram Hossain
- Advanced Network Technologies Virtual Lab <http://vlabs.iitkgp.ernet.in> › ant
- Follow the steps to configure ns2 simulator <https://www.youtube.com/watch?v=D3A2ZNZI1Q>

Course Code	Course Title	No. of Periods/Week	Total No. of Periods	Marks for FA	Marks for SA
CCN-508	Life Skills	3	45	40	60

Course Objectives:	Understand the relevance of life skills in both personal and professional lives
	Practice life skills complementarily in life-management to lead a happy and successful life

	Course Outcomes:
CO1	exhibit right attitude and be adaptable in adverse and diverse situations
CO2	set appropriate goals and achieve them through proper planning, time management and self-motivation
CO3	solve diverse real-life and professional problems with critical thinking and creativity for a stress-free life
CO4	be an ideal team player and manifest as a leader

Course Delivery:

Text book: **"Life Skills"** – by State Board of Technical Education and Training, AP

Sl no	Unit	Teaching Hours
1	Attitude	4
2	Adaptability	4
3	Goal Setting	4
4	Motivation	4
5	Time Management	4
6	Critical Thinking	4
7	Creativity	4
8	Problem Solving	5
9	Team work	4
10	Leadership	4
11	Stress Management	4
	Total	45

Course Content:

UNIT 1: Attitude *matters!*

Preparatory activity-Role play; Generating word bank; Types of attitude. Read the passage and answer the related questions, read the story and discuss issues raised; Express opinions on the given topic and fill the grid with relevant words.

UNIT 2: Adaptability... *makes life easy!*

Pair work-Study the given pictures and understand adaptability -read the anecdote and discuss, read the story and answer the questions, role play

UNIT 3: Goal Setting... *life without a goal is a rudderless boat!*

Short term goals and long term goals-SMART features, observe the pictures and answer questions- matching- read the passage and answer questions-filling the grid.

UNIT 4: Motivation... *triggers success!*

Types of motivation-difference between motivation and inspiration- matching different personalities with traits - dialogue followed by questions - writing a paragraph based on the passage.

UNIT 5: Time Management ... *the need of the hour!*

Effective Time Management- Time quadrant - Group task on management of time- Time wasters-fill in the grid, read the story and answer the questions- prioritising tasks.

UNIT 6: Critical Thinking... *Logic is the key!*

Preparatory activity-read the passage and answer the questions- differentiate between facts and assumptions- components of critical thinking- complete the sets of analogies- choose the odd one out- true or false statements- decide which of the conclusions are logical.

UNIT 7: Creativity.... *The essential YOU!!*

Definition- Pre-activity-read the anecdote and answer the questions- matching celebrities with their fields of specialisation- think of creative uses of objects- think creatively in the given situations.

UNIT 8: Problem Solving... *there is always a way out!*

Preparatory activity-read the story and answer the questions- discuss the given problem and come out with three alternative solutions- group activity to select the best solution among available alternatives- discuss the problem and plan to analyse it.

UNIT 9: Team Work... *Together we are better!*

Advantages of team work- Characteristics of a team player- Activity-Observe the pictures and classify them into two groups- team game - read the story and answer the questions- fill in the grid.

UNIT 10 : Leadership... *the making of a leader!*

Characteristics of effective leadership- styles of leadership- Activity-read the dialogue and answer the questions- identify the people in the picture and describe them- discuss leadership qualities of the given leaders- filling the grid- read the quotes and write the name of the leader.

UNIT 11: Stress Management ... *live life to the full* !!

Types of stress- Strategies for Stress Management- Activity-read the passage and answer the questions, read the situation and write a paragraph about how to manage stress.

Mapping COs with POs

POs	1	2	3	4	5	6	7
COs	POs 1 to 5 are applications of Engineering Principles, can't directly be mapped with Life Skills					1,2,3,4	1,2,3,4

Unit wise Mapping of COs- POs

CO	Course Outcome	CO Unit Mapped	PO mapped	Cognitive levels as per Bloom's Taxonomy R/U/Ap/An/Ev/Cr(Remembering / Understanding/ Applying/Analysing/ Evaluating/ Creating)
CO 1	To exhibit right attitude and be adaptable to adverse and diverse situations	All Units (1 to 11)	6,7	U/Ap/ An
CO2	To set appropriate goals and achieve them through proper planning, time management and self-motivation	Units 3,4,5	6,7	U/Ap/An
CO3	To solve diverse real-life and professional problems with critical thinking and creativity for a stress-free life	Units 6,7,8,11	6,7	U/Ap/An/ Ev/ Cr.
CO4	To be an ideal team player and manifest as a leader	Units 9,10	6,7	U/Ap/An/ Ev

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CCN-509	PROJECT WORK	6	90	40	60

Course Objectives	i)To inculcate team spirit among students ii)To apply software life cycle models iii)To design, develop, test and deploy project
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Course Outcomes	At the end of course student able to		
	CO1	CCN-509.1	Identify the hardware, software problems and their feasibility
	CO2	CCN-509.2	Prepare SRS document based on gathered and analysed requirements
	CO3	CCN-509.3	Design the plan document based on SRS
	CO4	CCN-509.4	Code and test the software based on the design document
	CO5	CCN-509.5	Practice software maintenance skills and maintaining quality and reliability
	CO6	CCN-509.6	Calculate software metrics like cost, loc, scheduling, manpower and other resources.

LEARNING OUTCOMES

1. Identify different works to be carried out in the Project
2. Collect data relevant to the project work
3. Carryout need survey and identify the problem(project)
4. Select the most efficient software life cycle from the available choices based on preliminary investigation
5. Estimate the cost of project, technological need, computer skills, materials and other equipment
6. Prepare the plan and schedule of starting time and sequence of operations to be carried out at various stages of the project work in detail
7. Prepare SRS document
8. Design the required elements of the project work as per standard models such as UML
9. Develop the working software modules required for the project work
10. Prepare critical activities at various stages of the project work
11. Test ,Debug, verify and validate the project
12. Record the results
13. Preparation of project report (and user manual if necessary) to enable the client to maintain the project

THE PROJECT CAN BE CHOSEN FROM THE FOLLOWING DOMAINS:

1. SOFTWARE PROJECTS

- a. Web site designing
- b. Banking
- c. Income tax calculation package
- d. Examinations cell.
- e. Student database management
- f. Library management
- g. Stores Management
- h. Staff data management
- i. Payrolls
- j. Inventory Control
- k. Hostel management
- l. Tourism package
- m. Institution management software
- n. Anti-Virus software development.
- o. Folder-locking.
- p. Terminate stay resident systems.

2. HARDWARE and NETWORKING PROJECTS

- a. QLAN establishing
- b. Using interfacing devices
- c. Voice synthesizer
- d. Voice recognizer
- e. Printer sharer
- f. ADD ON cards or any relevant

3. SOFTWARE AND HARDWARE PROJECTS

- a. Using interfaces, microcontrollers. Microprocessors and PCs
- b. Inter-cum
- c. Assembling computer along with peripherals.
- d. Traffic light controller
- e. Stepper motor related
- f. Lift controllers
- g. Level controllers
- h. Temperature controllers

4. To develop above projects and deploy in cloud platform

5. To develop IOT based applications

6. To maintain the software products based on the ever changing needs of and quality measures required by the clients

Evaluation Scheme for the Project Work

S. No.	Tasks	Max. Marks Allotted for each task INTERNAL /EXTERNAL (40+60=100)
1.	Feasibility study of the problem	4/6
2.	Requirement Analysis of the problem, SRS document preparation	4/8
3.	Designing the problem	6/10
4.	Implementation	8/10
5.	Testing and verification	10 /16
6.	Project report preparation and presentation	8/10
	Total:	40/60 (100)

**DIPLOMA IN COMMUNICATION AND COMPUTER NETWORKING ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION**

CURRICULUM-2023

(VI Semester)

CCN-601 Industrial Training

Course Code	Course title	No of periods/week	Duration	Marks for FA	Marks for SA
CCN-601	INDUSTRIAL TRAINING (Online Certificate courses /Industry)	42	6 months	240	60

LEARNING OUTCOMES (In Industry): The student shall be able to display the following skill sets

1. Apply knowledge and skill already learnt in the institution.
2. Acquire the required skills of analysis, design and development, testing, verification and Validation.
3. Acquire skills of deployment and distribution of the product.
4. Involve in product design, development, and quality testing and maintenance production by Exhibiting the strength, teamwork spirit and self-confidence
5. Prepare product documents like user manual and installation guide and operational manuals.
6. Perform the activities of deploying product at customer site and training the end user.
7. Maintaining the system at user site (Post product services)

S No	Unit Title	Duration	COs Mapped
1	Application of Knowledge acquired.	1 month	CO1
2	Skill Acquirement.	2 months	CO2
3	Participate in product development.	2 months	CO3
4	Preform onsite service.	1 month	CO4
	Total	6 months	

Course Outcomes	At the end of course student able to:		
	CO1	CCN-601.1	Apply knowledge and skill already learnt in the institution.
	CO2	CCN-601.2	Acquire the required skills of analysis, design and development, testing, verification and validation, deployment and distribution of the product.
	CO3	CCN-601.3	Involve in product design, development, quality testing and maintenance production by exhibiting the strength, teamwork spirit and self-confidence
	CO4	CCN-601.4	Prepare product document, gain the skills in deploying product at customer site , training the end user, maintaining the system.

LEARNING OUTCOMES – SCHEM OF EVALUATION (Two Online Certificate courses):

TRAINING MODULE NO.	TOPIC	LEARNING OUTCOMES (In-house training)	MARKS
First 3 Months/12 weeks	1) Registration at Nptel/Swayam/Moocs/course era/lectera/caltech/NASSCOM/AP SKILL DEVELOPMENT/ oxford/ hackerrank/udemy... etc.,	i) Learning ii) Mini Application development iii) Report preparation iv) 1 st Assessment	120
Next 3 Months/12 Weeks	1) Registration at Nptel/Swayam/Moocs/course era/lectera/caltech/NASSCOM/AP SKILL DEVELOPMENT/ oxford/ hackerrank/udemy... etc.,	i) Learning ii) Mini Application development iii) Report preparation iv) 2 st Assessment	120
External Evaluation	Seminar on two reports/viva	Evaluation by GUIDE/Co - Examiner, HOD and External Examiner	60
		TOTAL	300

Scheme of evaluation(Training at Industry)

Sl. No.	Subject	Duration	Scheme of evaluation		
			Item	Nature	Max. Marks
1	Industrial Training	6 months	1.First Assessment at Industry (After 12 Weeks)	Assessment of learning outcomes by both the faculty and training mentor of the industry	120
			2.Second Assessment at the Industry (After 20 weeks))	Assessment of learning outcomes by both the faculty and training mentor of the industry	120
			Final Summative assessment at institution level	Training Report	20
				Demonstration of any one of the skills listed in learning outcomes	30
				Viva Voce	10
TOTAL MARKS					300

The industrial training shall carry 300 marks and pass marks are 50%.A candidate failing to secure the minimum marks should complete it at his own expenses.

During Industrial training the candidate shall put in a minimum of 90% attendance.

Weightage of marks for Assessment of Learning Outcomes during first and second assessment (at industry)

Sl.No	Learning Outcome	Max Marks Allotted For first assessment	Max Marks Allotted For second assessment
1	Apply knowledge and skill already learnt in the institution.	50	10
2	Acquire the required skills of analysis, design and development , testing, verification and validation , deployment and distribution of the product.	70	30
3	Involve in product design, development, quality testing and maintenance production by exhibiting the strength, teamwork spirit and self-confidence	-	40
4	Prepare product document, gain the skills in deploying product at customer site , training the end user, maintaining the system.	-	40
	Total	120	120

During assessment the performance of the students shall be assessed in those skills in which the student has been trained and be awarded the marks as per the weightage assigned as above. In case the student has undergone training in a few skill sets then the total marks obtained shall be raised to 120 marks for the given assessment i.e. either assessment 1 or 2. However the performance of the student shall be assessed at the most skill sets listed above but not less than three skill sets.

Illustration for First assessment.

If the student has undergone training in only in 2 skill sets (namely 1 for 50 marks, and 2 for 40 marks) out of 3 (namely 1 for 50 marks, 2for 40 marks and 3 for 30 marks) in First assessment and marks awarded during assessment is 60 out of 90 marks, then the marks of 60 shall be enhanced to 120 proportionately as $(60/90)*120=80$.

Illustration for second assessment.

If the student has undergone training in only in 5 skill sets (namely 1 for 10 marks, 2 for 20 marks , 3 - for 10 marks, 4 for 25 marks, 5 For 15 marks) out of 7 (namely 1 for 10 marks, 2 for 20 marks , 3 For 10 marks, 4 for 25 marks, 5 For 15 marks, 6for 25 marks and 7for 15 marks) in Second assessment and marks awarded during assessment is 65 out of 80 marks, then the marks of 65 shall be enhanced to 120 proportionately as $(65/80)*120=97.5$ =rounded to 98.

GUIDELINES FOR INDUSTRIAL TRAINING OF DIPLOMA IN COMMUNICATION AND NETWORKING ENGINEERING PROGRAMME

1. Duration of the training: 6 months.
2. Eligibility: The As per SBTET norms
3. Training Area: Students can be trained in either in In-house/Industry in the areas of
4. Application Software Development / system software Development / firmware development/ Mobile application development/ Database applications / Web development/ IoT application development / smart technologies / Hardware interfacing/ Networking .
5. The candidate shall put a minimum of 90% attendance during Industrial Training.
6. If the student fails to secure 90% attendance during industrial training, the student shall reappear for 6 months industrial training.
7. Formative assessment at industry level shall be carried out by the Mentor from of the industry, where the student is undergoing training and the faculty in charge (Guide) from the concerned section in the institution.
8. The Industrial training shall carry 300 marks and pass marks is 50% in assessments at industry (first and second assessment) and final summative assessment at institution level put together i.e. 150 marks out of 300 marks.
9. If the student fails to secure 50% marks in final summative assessment at institution level, the student should reappear for final summative assessment in the subsequent board examination.
10. Final summative assessment at institution level is done by a committee including1. Head of the section (of concerned discipline ONLY), 2.External examiner from an industry and 3. Faculty member who assessed the student during Industrial Training as members.

Guidelines and responsibilities of the faculty members who are assessing the students

performance during industrial training

**DEPARTMENT OF TECHNICAL EDUCATION
NAME OF THE INSTITUTION
INDUSTRIAL TRAINING FIRST ASSESSMENT**

PIN:

NAME OF THE STUDENT:

Name of the Industry:

<i>Skill Set Sl.No</i>	<i>SKILL SET</i>	<i>Max Marks Allotted For each parameter</i>	<i>Marks obtained</i>
1	Apply knowledge and skill already learnt in the institution.	50	
2	Acquire the required skills of analysis, design and development , testing, verification and validation.	40	
3	Acquire the required skills of deployment and distribution of the product.	30	
	<i>Total</i>	<i>120</i>	

(Marks in words:)

Signature of the Training In-charge (Mentor)

Signature of the visiting staff (Guide)

Name:

Name:

Designation:

Designation:

**DEPARTMENT OF TECHNICAL EDUCATION
NAME OF THE INSTITUTION
INDUSTRIAL TRAINING SECOND ASSESSMENT**

PIN:

NAME OF THE STUDENT:

Name of the Industry:

<i>Skill Set Sl.No</i>	<i>SKILL SET</i>	<i>Max Marks Allotted For each parameter</i>	<i>Marks obtained</i>
1	Apply knowledge and skill already learnt in the institution.	10	
2	Acquire the required skills of analysis, design and development, testing, verification and validation.	20	
3	Acquire the required skills of deployment and distribution of the product.	10	
4	Involve in product design, development, quality testing and maintenance production by exhibiting the strength, teamwork spirit and self-confidence	25	
5	Prepare product documents like user manual and installation guide and operational manuals.	15	
6	Perform the activities of deploying product at customer site and training the end user.	25	
7	Maintaining the system at user site (Post product services)	15	
		120	

(Marks in words:)

Signature of the Training In-charge (Mentor)

Signature of the visiting staff (Guide)

Name:

Name:

Designation:

Designation: